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DES SCIENCES

FALCUTÉ DES SCIENCES ET TECHNOLOGIE (FST)

RAPPORT DE TRAVAUX DIRIGÉS I : INTRODUCTION À CISCO PACKET TRACER ET TOPOLOGIES RÉSEAUX

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Niveau : L3

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Introduction

Dans le cadre du développement de compétences en administration du système d'exploitation Linux, la maîtrise de l'interface en ligne de commande et la compréhension de la gestion des environnements graphiques constituent des éléments majeurs. Ce rapport présente les travaux réalisés au cours du TD3, autours des thèmes suivants, Commandes de base, utilitaires, gestion des paquets et bureaux KDE Plasma et XFCE.

Ce TD a pour objectif principale d'acquérir une autonomie pratique dans la gestion quotidienne d'un système Linux. De la manipulation de bas niveau jusqu'à la configuration de l'interface utilisateur.

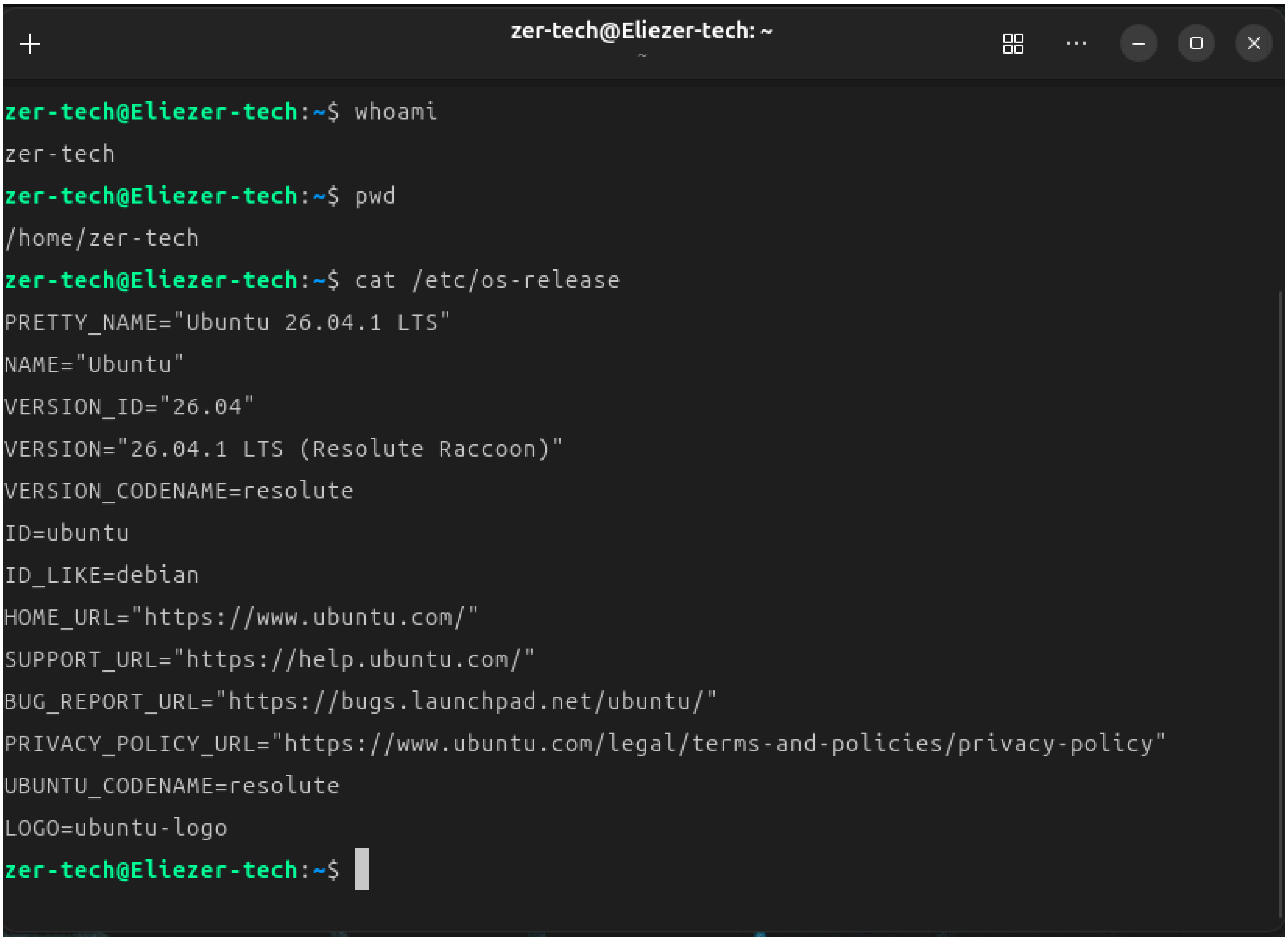
Étant divise en 5 grandes parties, dont la première vise à se repérer et manipuler des fichiers.

EXERCICE 1

1- Afficher :

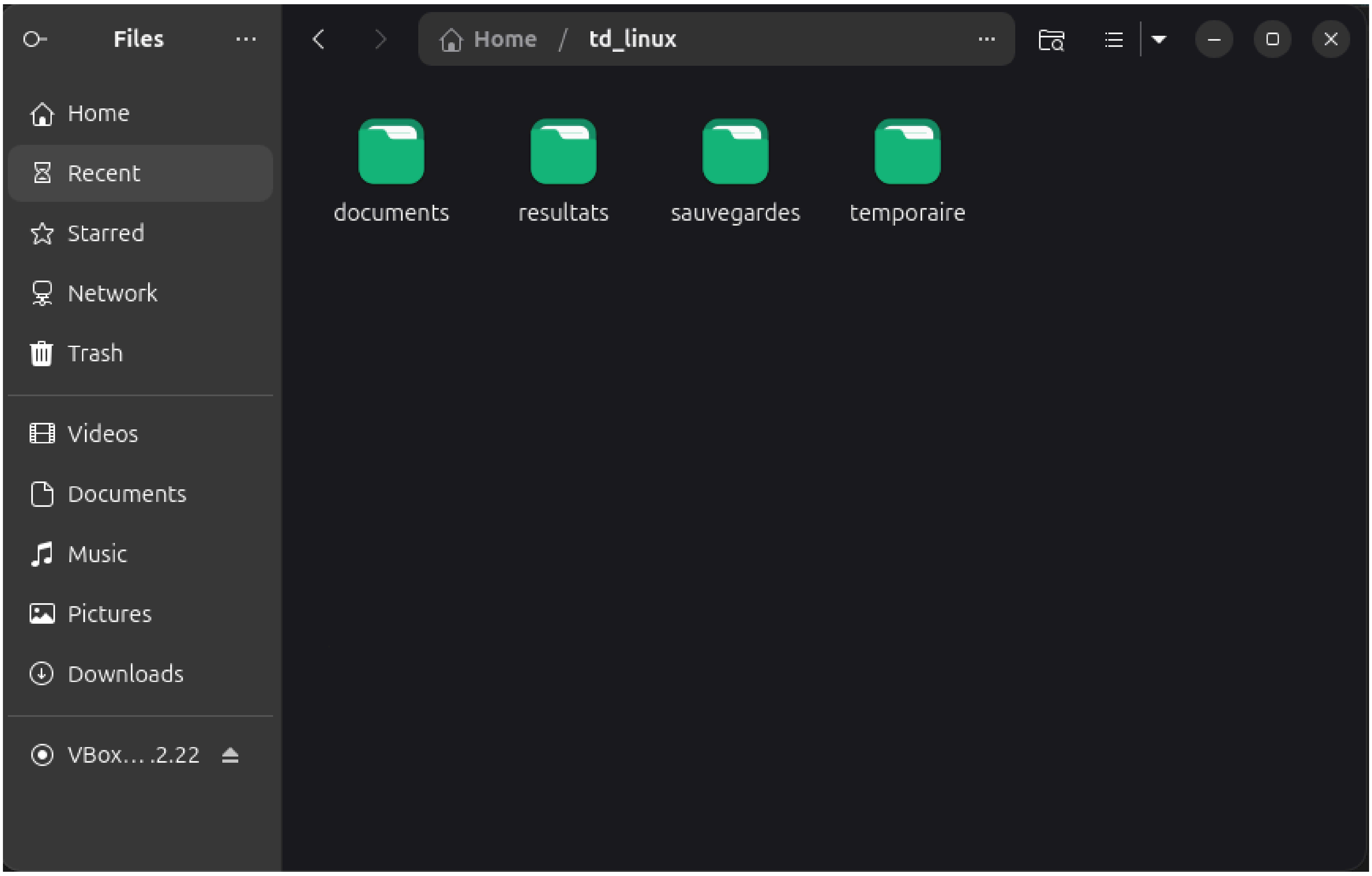
- Le nom de l'utilisateur
- Le chemin du répertoire courant
- Les informations sur la distribution
- La différence entre chemin absolu et chemin relatif.

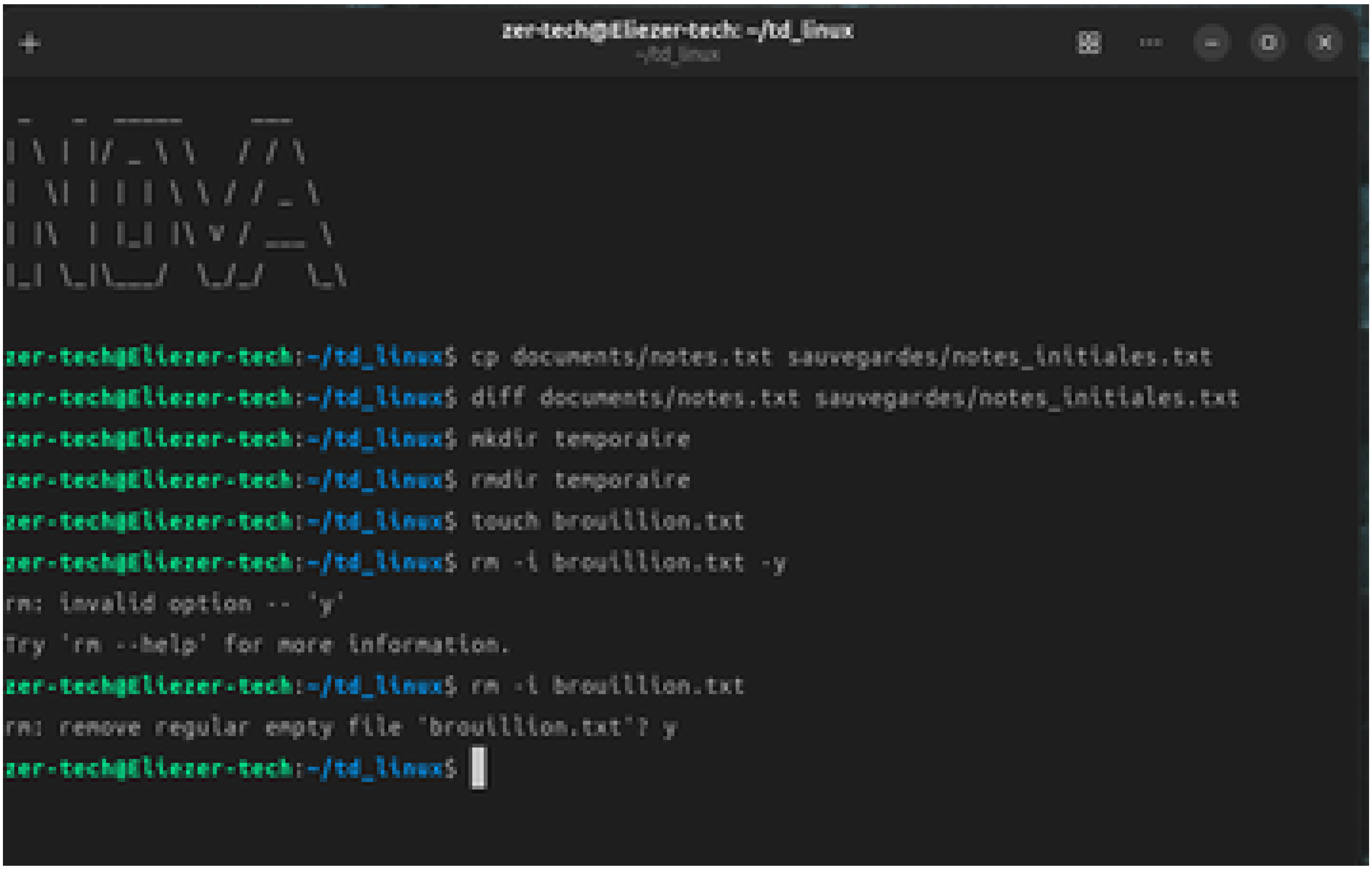
2- créer l'arborescence *~/td_linux(documents/sauvegardes/résultats)* et l'affiché au format détaillé
3-Créer un fichier *notes.txt,* le copier et le renommer dans *sauvegardes*
4- Créer et supprimer un fichier nommé *temporaire* puis faire la différence entre *rmdir et rm*



```
zer-tech@Eliezer-tech:~$ whoami
zer-tech
zer-tech@Eliezer-tech:~$ pwd
/home/zer-tech
zer-tech@Eliezer-tech:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04.1 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04.1 LTS (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
zer-tech@Eliezer-tech:~$ 
```

Les différents dossiers inclus le dossier temporaire sont bel et bien présent


Files
Home
Recent
Starred
Network
Trash
Videos
Documents
Music
Pictures
Downloads
VBox… .2.22
Home / td_linux
documents
resultats
sauvegardes
temporaire


zer-tech@Eliezer-tech: ~/td_linux
~/td_linux

```
 _   _  _____     ___
| \ | |/ _ \ \   / / \
|  \| | | | \ \ / / _ \
| |\  | |_| |\ V / ___ \
|_| \_|\___/  \_/_/   \_\

zer-tech@Eliezer-tech:~/td_linux$ cp documents/notes.txt sauvegardes/notes_initiales.txt
zer-tech@Eliezer-tech:~/td_linux$ diff documents/notes.txt sauvegardes/notes_initiales.txt
zer-tech@Eliezer-tech:~/td_linux$ mkdir temporaire
zer-tech@Eliezer-tech:~/td_linux$ rmdir temporaire
zer-tech@Eliezer-tech:~/td_linux$ touch brouillion.txt
zer-tech@Eliezer-tech:~/td_linux$ rm -i brouillion.txt -y
rm: invalid option -- 'y'
Try 'rm --help' for more information.
zer-tech@Eliezer-tech:~/td_linux$ rm -i brouillion.txt
rm: remove regular empty file 'brouillion.txt'? y
zer-tech@Eliezer-tech:~/td_linux$ 
```

Sous Linux, la différence réside dans le point de départ utilisé pour localiser un fichier ou un dossier.

Un chemin absolu donne l'adresse complète depuis la racine du système, tandis qu'un chemin relatif donne l'adresse par rapport à l'endroit où vous vous trouvez actuellement (le répertoire de travail).

rmdir (Remove Directory) : Elle sert uniquement à supprimer des répertoires vides. Si le dossier contient le moindre fichier ou sous-dossier, la commande échoue par sécurité pour éviter les pertes de données accidentelles.

rm (Remove) : Elle sert principalement à supprimer des fichiers. Cependant, en lui ajoutant des options (comme -r), elle devient capable de supprimer des dossiers et tout leur contenu (fichiers, sous-dossiers), qu'ils soient vides ou non.

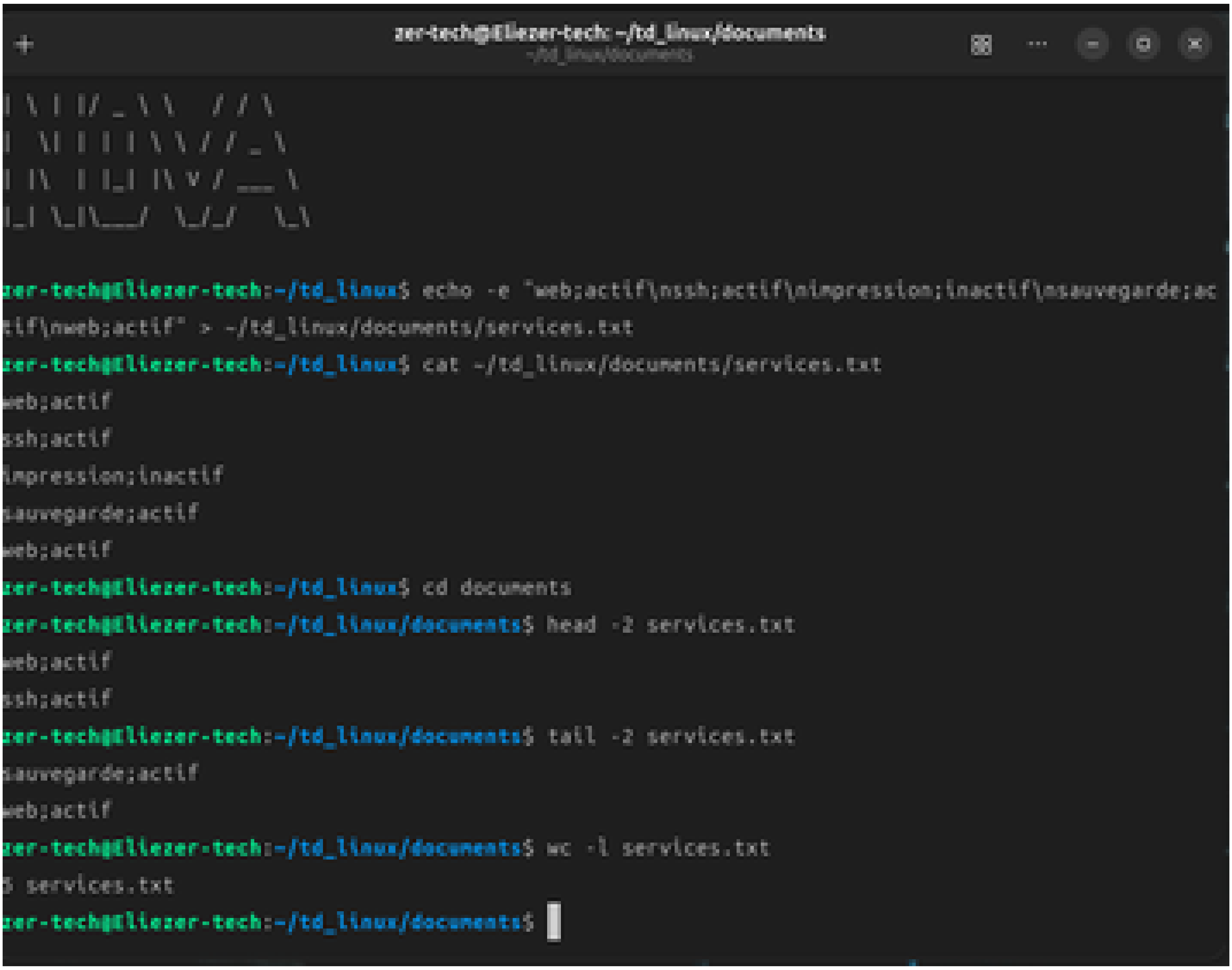
EXERCICE 2

Dans la deuxième partie, vise à exploiter les utilitaires et faire du traitement de texte

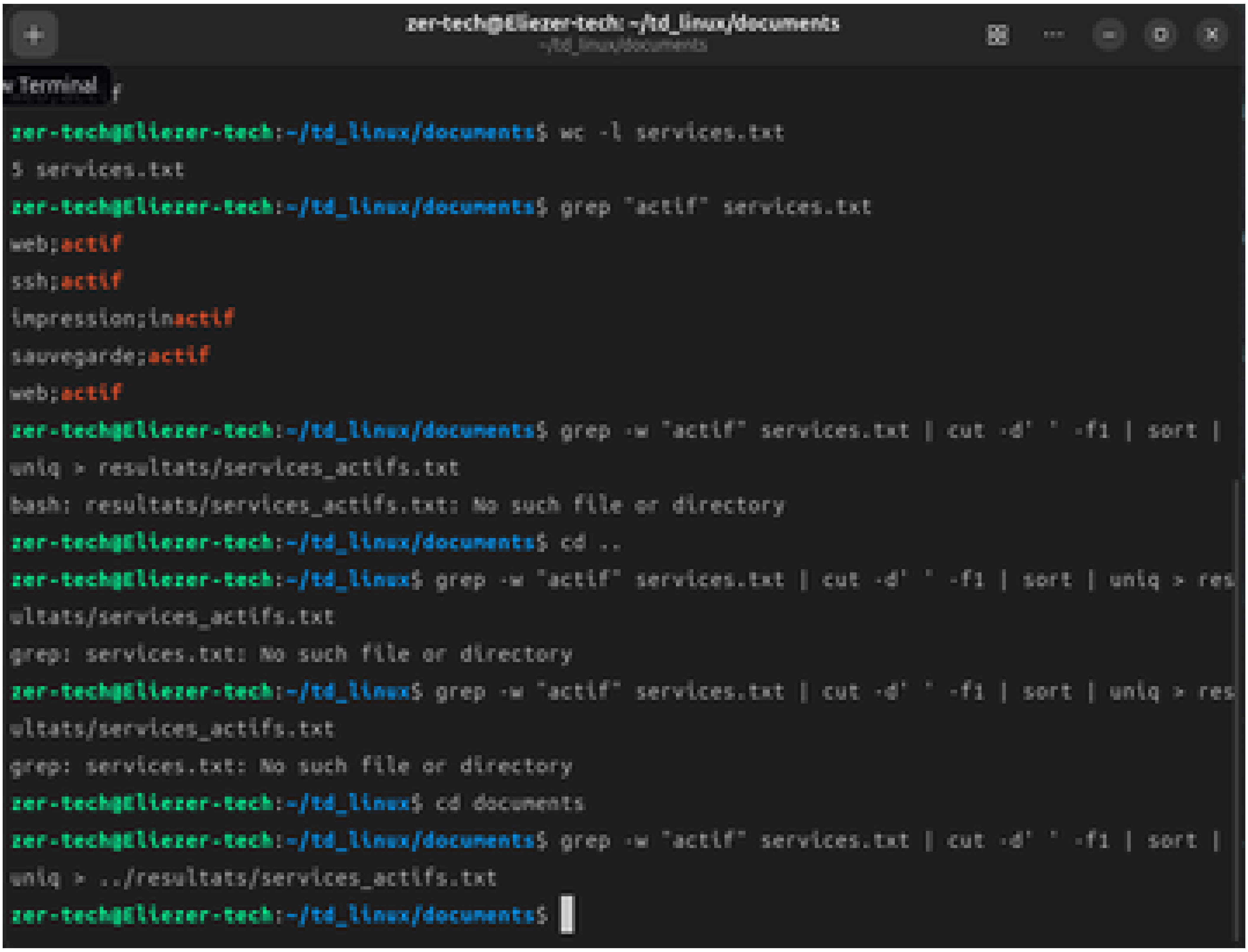
La première tache vise à créer un fichier nommé service.txt et :
1- Afficher

- Le contenu complet du ficher
- Ses deux premières et dernières lignes
- Et son nombre de lignes

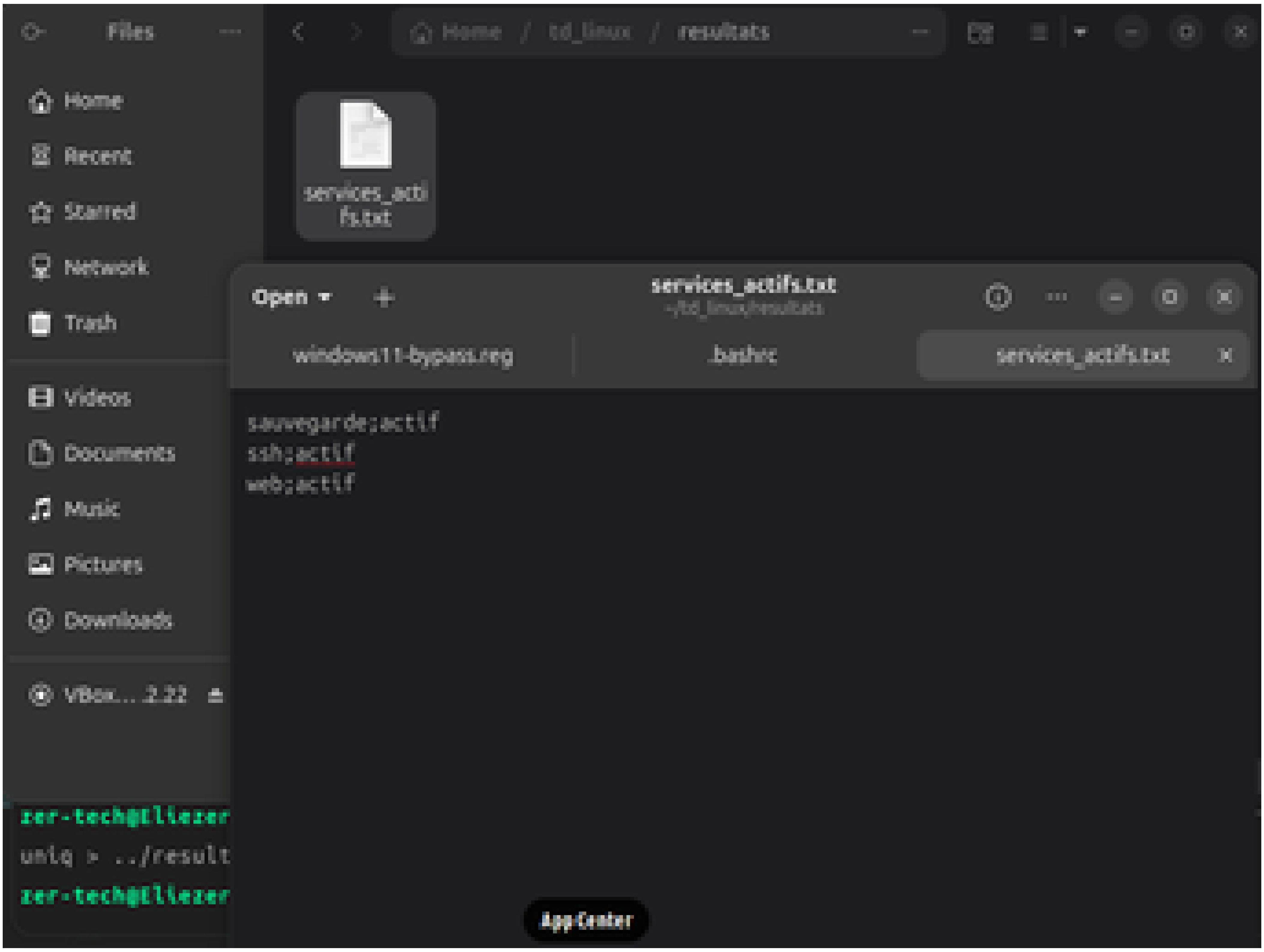
2- Extraire uniquement les lignes dont le deuxième champ vaut actif
3- Avec une seule ligne de commande, sélectionner les services actifs, extraire leur noms, les trier et supprimer les doublons, et enregistrer le résultat dans resultats/services_actifs.txt
4- rechercher les fichiers avec l'extension .txt
5- Créer l'archive sauvegardes/documents.tar.gz, afficher son contenu sans l'extraire, puis l'extraire dans un dossier restauration
6- expliquer la différence entre les operateurs > et >>



```
| \ | |/ _ \ \   / / \
|  \| | | | \ \ / / _ \
| |\  | |_| |\ V / ___ \
|_| \_|\___/  \_/_/   \_\

zer-tech@Eliezer-tech:~/td_linux$ echo -e "web;actif\nssh;actif\nimpression;inactif\nsauvegarde;ac
tif\nweb;actif" > ~/td_linux/documents/services.txt
zer-tech@Eliezer-tech:~/td_linux$ cat ~/td_linux/documents/services.txt
web;actif
ssh;actif
impression;inactif
sauvegarde;actif
web;actif
zer-tech@Eliezer-tech:~/td_linux$ cd documents
zer-tech@Eliezer-tech:~/td_linux/documents$ head -2 services.txt
web;actif
ssh;actif
zer-tech@Eliezer-tech:~/td_linux/documents$ tail -2 services.txt
sauvegarde;actif
web;actif
zer-tech@Eliezer-tech:~/td_linux/documents$ wc -l services.txt
5 services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ 
```

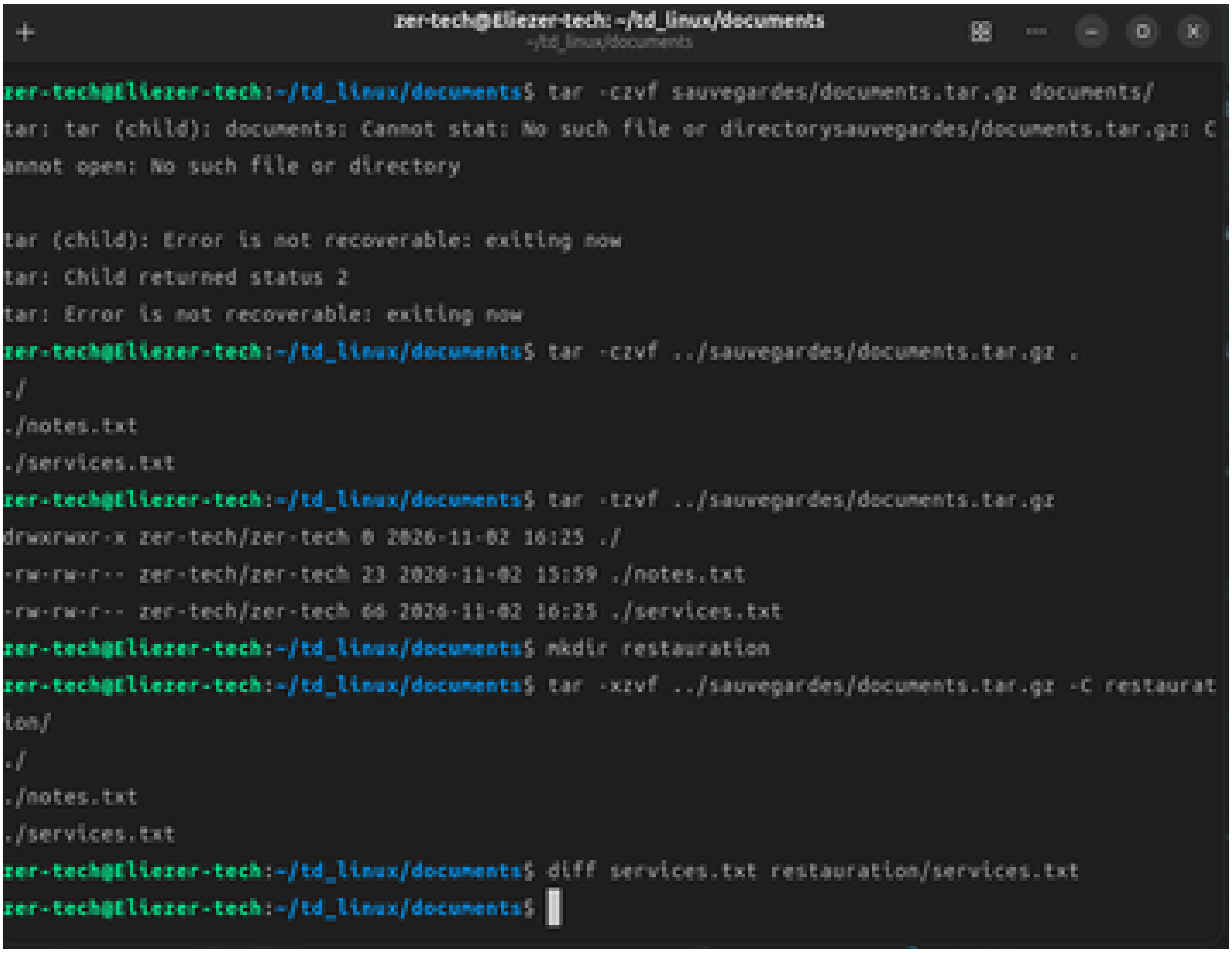


```
zer-tech@Eliezer-tech:~/td_linux/documents$ wc -l services.txt
5 services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ grep "actif" services.txt
web;actif
ssh;actif
impression;inactif
sauvegarde;actif
web;actif
zer-tech@Eliezer-tech:~/td_linux/documents$ grep -w "actif" services.txt | cut -d' ' -f1 | sort |
uniq > resultats/services_actifs.txt
bash: resultats/services_actifs.txt: No such file or directory
zer-tech@Eliezer-tech:~/td_linux/documents$ cd ..
zer-tech@Eliezer-tech:~/td_linux$ grep -w "actif" services.txt | cut -d' ' -f1 | sort | uniq > res
ultats/services_actifs.txt
grep: services.txt: No such file or directory
zer-tech@Eliezer-tech:~/td_linux$ grep -w "actif" services.txt | cut -d' ' -f1 | sort | uniq > res
ultats/services_actifs.txt
grep: services.txt: No such file or directory
zer-tech@Eliezer-tech:~/td_linux$ cd documents
zer-tech@Eliezer-tech:~/td_linux/documents$ grep -w "actif" services.txt | cut -d' ' -f1 | sort |
uniq > ../resultats/services_actifs.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ 
```

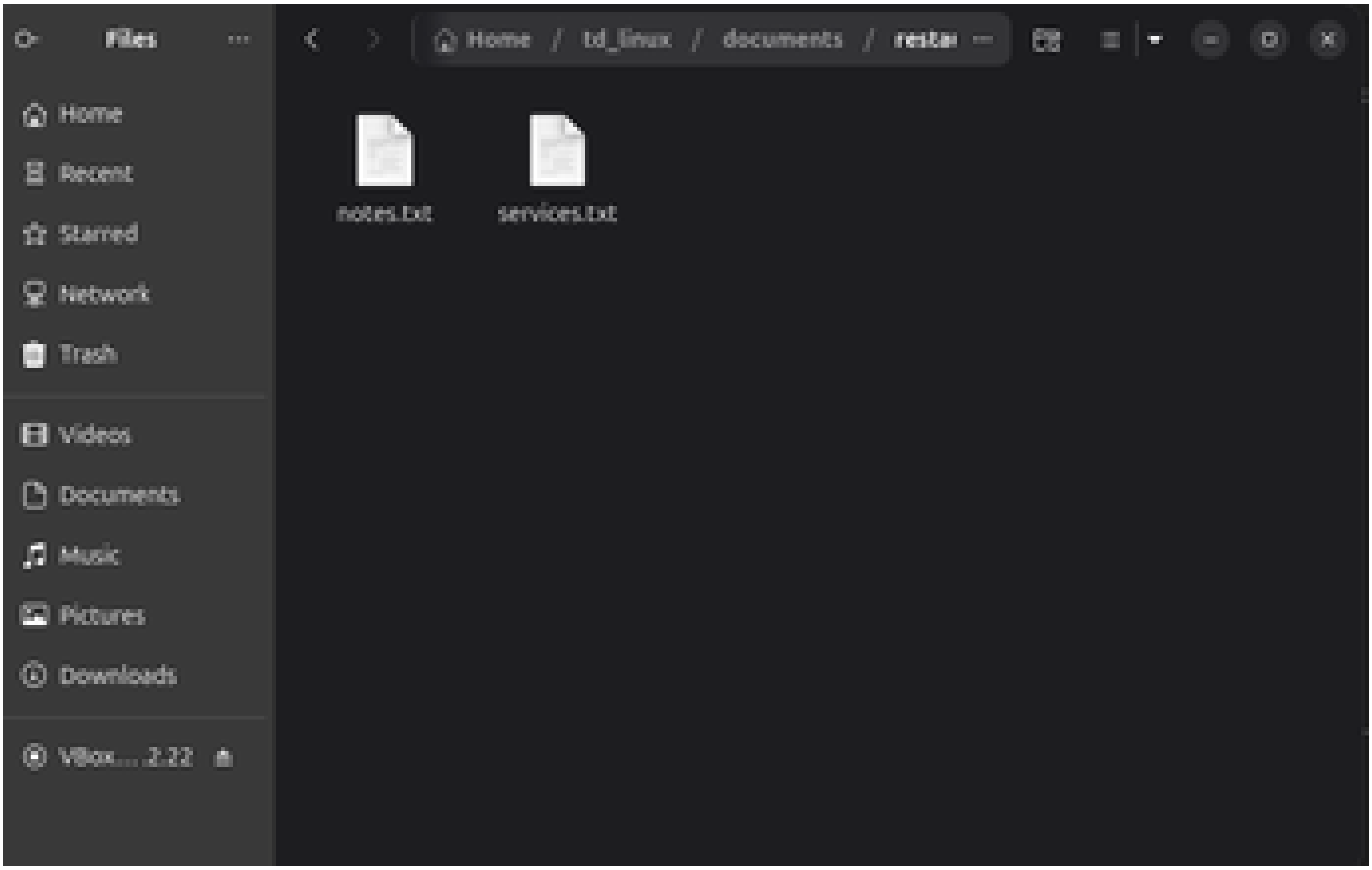

Files
Home / td_linux / resultats
Home
Recent
Starred
Network
Trash
Videos
Documents
Music
Pictures
Downloads
VBox... 2.22
services_actifs.txt
Open
services_actifs.txt
~/td_linux/resultats
windows11-bypass.reg
.bashrc
services_actifs.txt
App Center

```
sauvegarde;actif
ssh;actif
web;actif
```

```
zer-tech@Eliezer
uniq > ../result
zer-tech@Eliezer
```


zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents

```
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -czvf sauvegardes/documents.tar.gz documents/
tar: tar (child): documents: Cannot stat: No such file or directorysauvegardes/documents.tar.gz: C
annot open: No such file or directory

tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -czvf ../sauvegardes/documents.tar.gz .
./
./notes.txt
./services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -tzvf ../sauvegardes/documents.tar.gz
drwxrwxr-x zer-tech/zer-tech 0 2026-11-02 16:25 ./
-rw-rw-r-- zer-tech/zer-tech 23 2026-11-02 15:59 ./notes.txt
-rw-rw-r-- zer-tech/zer-tech 66 2026-11-02 16:25 ./services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ mkdir restauration
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -xzvf ../sauvegardes/documents.tar.gz -C restaurat
ion/
./
./notes.txt
./services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ diff services.txt restauration/services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ 
```


Files
Home / td_linux / documents / restau
Home
Recent
Starred
Network
Trash
Videos
Documents
Music
Pictures
Downloads
VBox_...2.22
notes.txt
services.txt

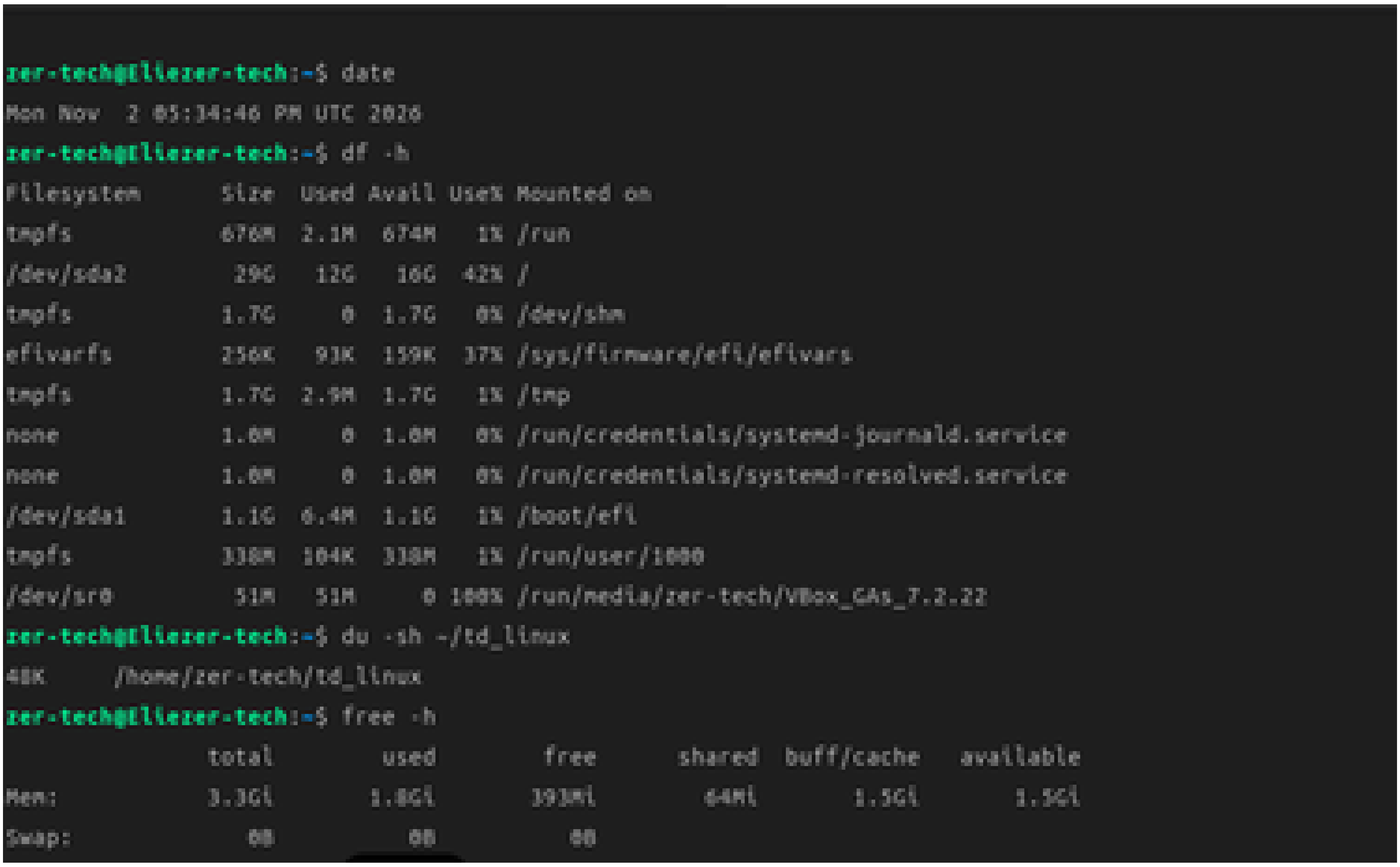
La différence entre l'opérateur > et >> réside dans le traitement du contenu existant lors de la redirection du résultat d'une commande vers un fichier texte. L'opérateur > écrase le contenu actuel du fichier, tandis que l'opérateur >> ajoute le contenu à la fin du fichier sans rien effacer.

EXERCICE 3

La troisième partie quant à elle vise a observer le système et les processus

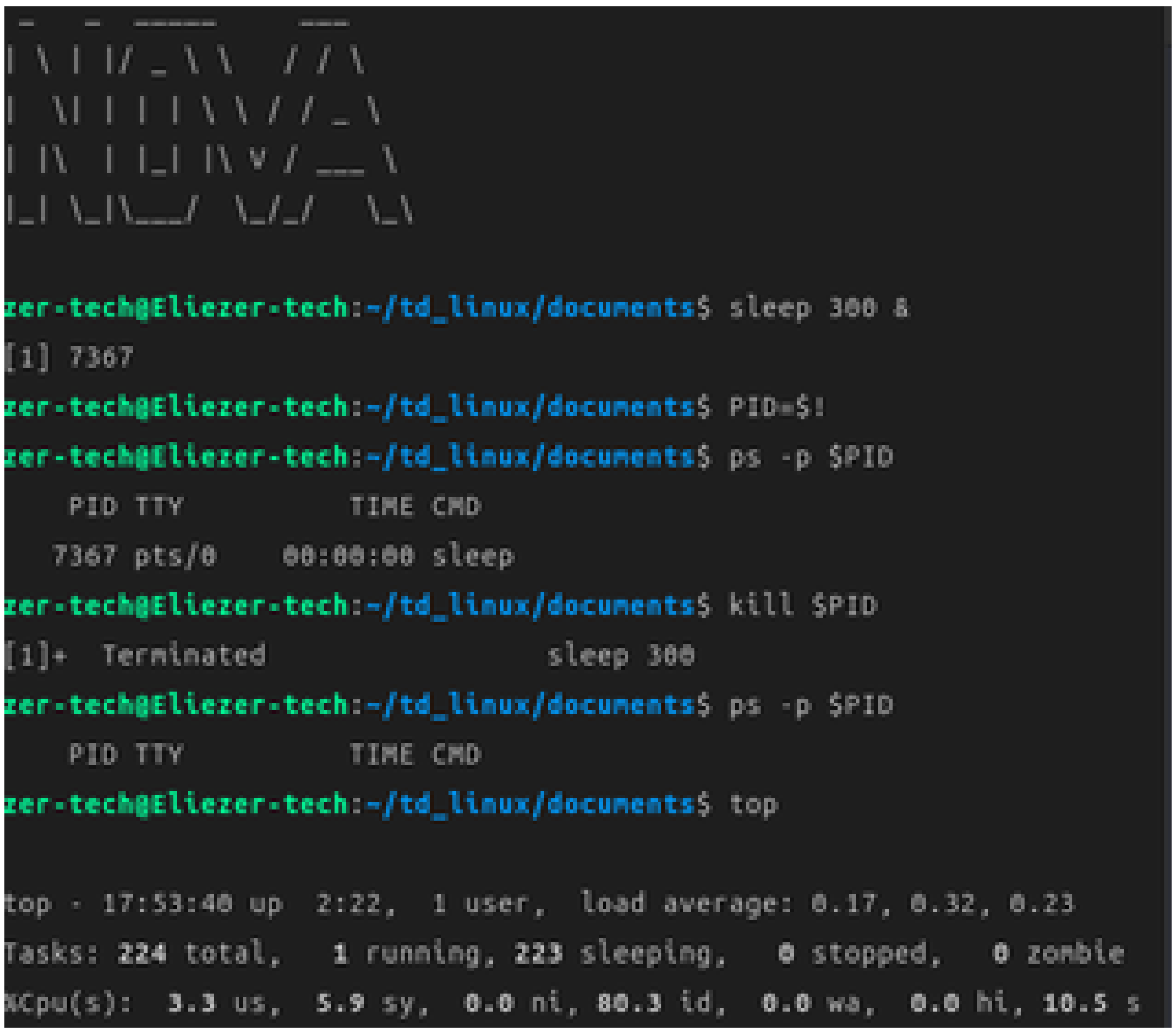
1. Afficher :
    - Expliquer la différence entre *df* et *du* .
    - La date et l’heure.
    - L’espace libre des systèmes de fichiers.
    - L’espace occupé par ~/td_linux .
    - La mémoire disponible.

2. Lancer *sleep* 300 en arrière-plan. Relever immédiatement son PID avec *$!* , puis afficher ce processus avec *ps* .
3. Terminer uniquement ce processus avec *kill* et vérifier son arrêt.
4. Ouvrir *top* , relever un processus et expliquer ses consommations CPU et mémoire. Quitter avec *q* .

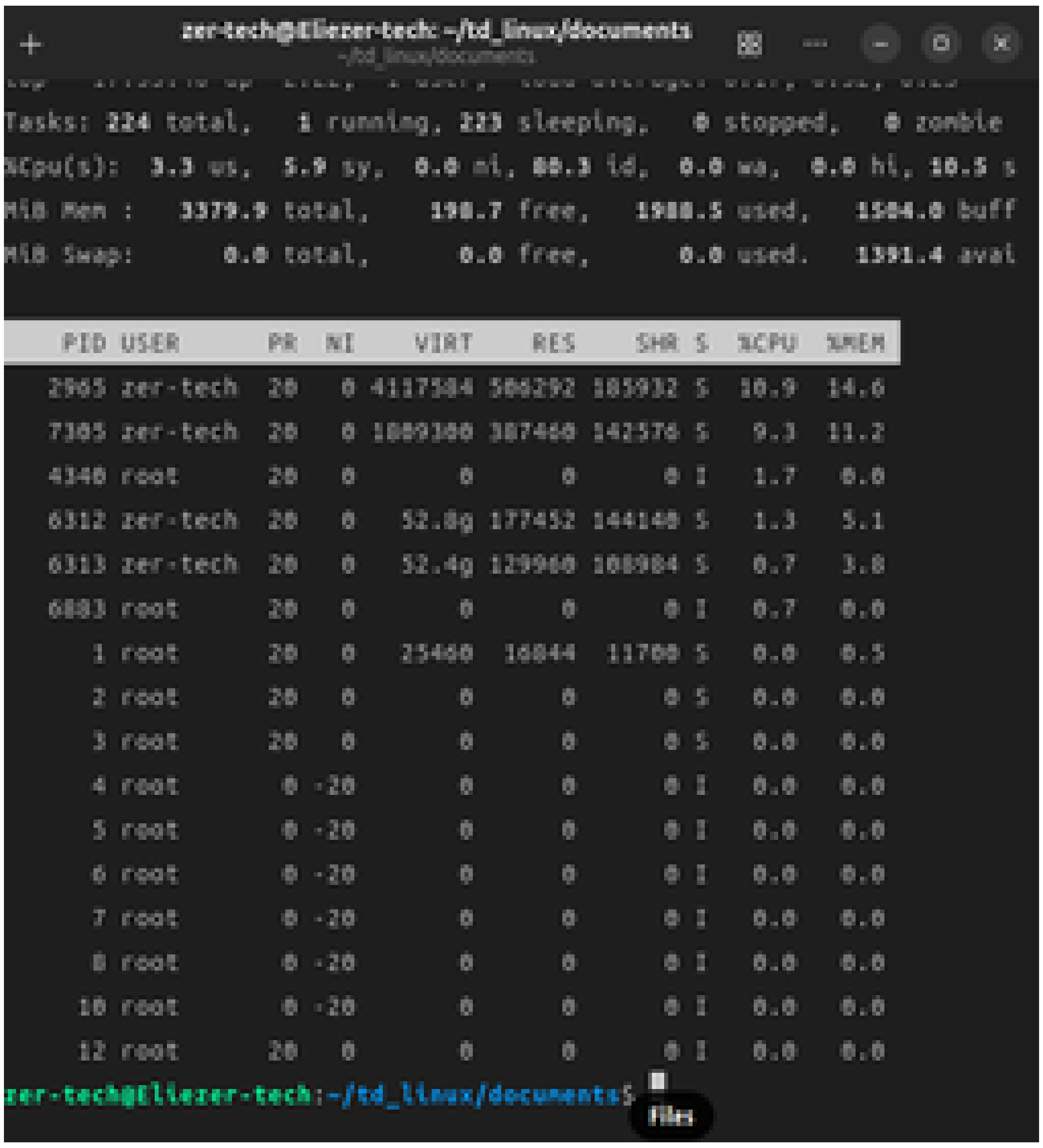


```
zer-tech@Eliezer-tech:~$ date
Mon Nov  2 05:34:46 PM UTC 2026
zer-tech@Eliezer-tech:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           676M  2.1M  674M   1% /run
/dev/sda2        29G   12G   16G  42% /
tmpfs           1.7G     0  1.7G   0% /dev/shm
efivarfs        256K   93K  159K  37% /sys/firmware/efi/efivars
tmpfs           1.7G  2.9M  1.7G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/sda1       1.1G  6.4M  1.1G   1% /boot/efi
tmpfs           338M  104K  338M   1% /run/user/1000
/dev/sr0         51M   51M     0 100% /run/media/zer-tech/VBox_GAs_7.2.22
zer-tech@Eliezer-tech:~$ du -sh ~/td_linux
48K     /home/zer-tech/td_linux
zer-tech@Eliezer-tech:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.3Gi       1.8Gi       393Mi        64Mi       1.5Gi       1.5Gi
Swap:             0B          0B          0B
```

La commande *df* analyse l'ensemble du système de fichiers , tandis que la commande *du* analyse l'espace occupé par des fichiers ou dossiers précis (vision ciblée)



```
 _   _  _____     ___
| \ | |/ _ \ \   / / \
|  \| | | | \ \ / / _ \
| |\  | |_| |\ V / ___ \
|_| \_|\___/  \_/_/   \_\

zer-tech@Eliezer-tech:~/td_linux/documents$ sleep 300 &
[1] 7367
zer-tech@Eliezer-tech:~/td_linux/documents$ PID=$!
zer-tech@Eliezer-tech:~/td_linux/documents$ ps -p $PID
    PID TTY          TIME CMD
   7367 pts/0    00:00:00 sleep
zer-tech@Eliezer-tech:~/td_linux/documents$ kill $PID
[1]+  Terminated                 sleep 300
zer-tech@Eliezer-tech:~/td_linux/documents$ ps -p $PID
    PID TTY          TIME CMD
zer-tech@Eliezer-tech:~/td_linux/documents$ top

top - 17:53:40 up  2:22,  1 user,  load average: 0.17, 0.32, 0.23
Tasks: 224 total,   1 running, 223 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.3 us,  5.9 sy,  0.0 ni, 80.3 id,  0.0 wa,  0.0 hi, 10.5 s
```

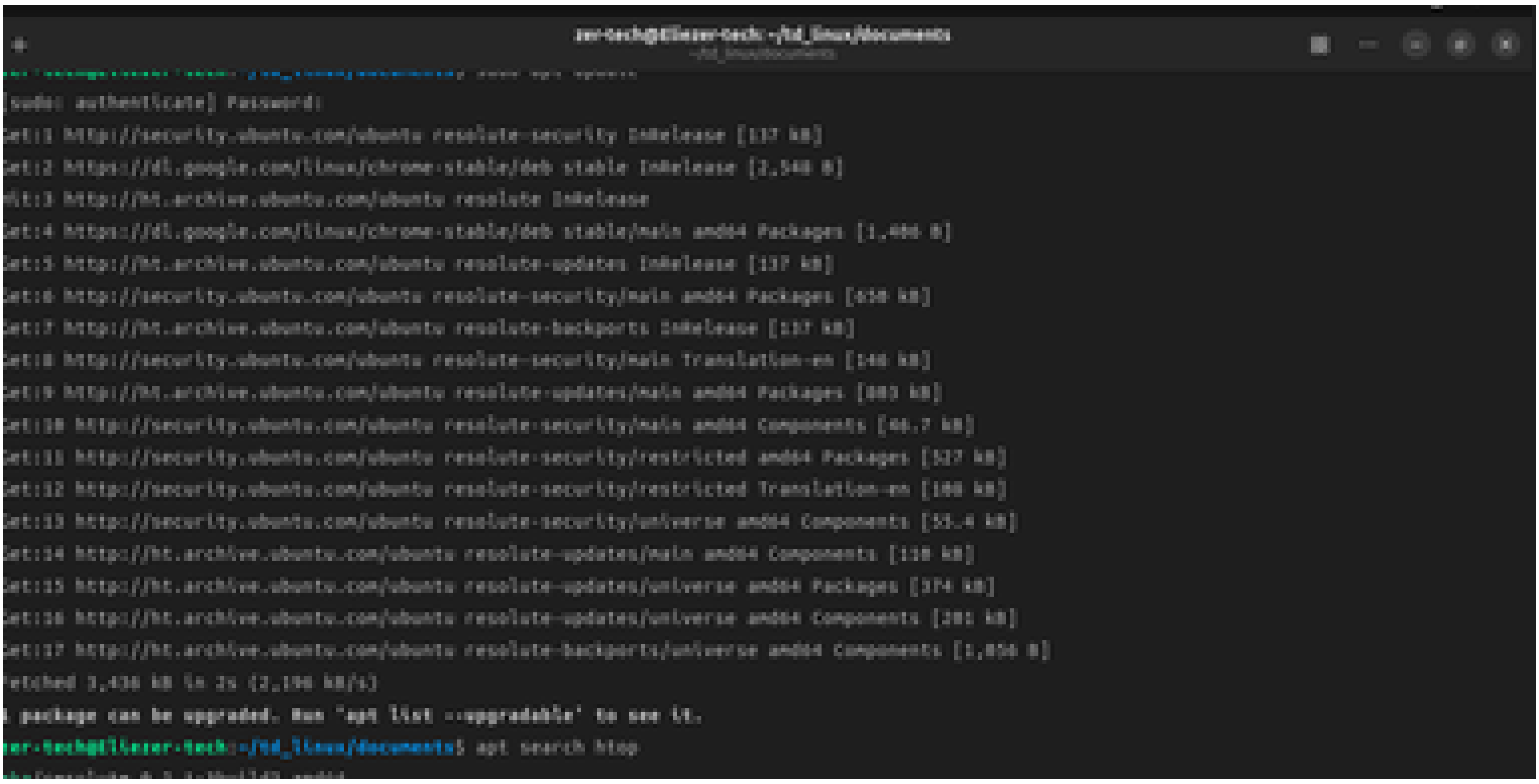


```
Tasks: 224 total,   1 running, 223 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.3 us,  5.9 sy,  0.0 ni, 80.3 id,  0.0 wa,  0.0 hi, 10.5 s
MiB Mem :   3379.9 total,    190.7 free,   1988.5 used,   1504.0 buff
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   1391.4 avai

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
   2965 zer-tech  20   0 4117584 506292 183932 S  10.9  14.6
   7305 zer-tech  20   0 1089300 387460 142576 S   9.3  11.2
   4340 root      20   0       0      0      0 I   1.7   0.0
   6312 zer-tech  20   0   52.8g 177452 144140 S   1.3   5.1
   6313 zer-tech  20   0   52.4g 129960 108984 S   0.7   3.8
   6883 root      20   0       0      0      0 I   0.7   0.0
      1 root      20   0   25460  16044  11700 S   0.0   0.5
      2 root      20   0       0      0      0 S   0.0   0.0
      3 root      20   0       0      0      0 S   0.0   0.0
      4 root       0 -20       0      0      0 I   0.0   0.0
      5 root       0 -20       0      0      0 I   0.0   0.0
      6 root       0 -20       0      0      0 I   0.0   0.0
      7 root       0 -20       0      0      0 I   0.0   0.0
      8 root       0 -20       0      0      0 I   0.0   0.0
     10 root       0 -20       0      0      0 I   0.0   0.0
     12 root      20   0       0      0      0 I   0.0   0.0
zer-tech@Eliezer-tech:~/td_linux/documents$
```

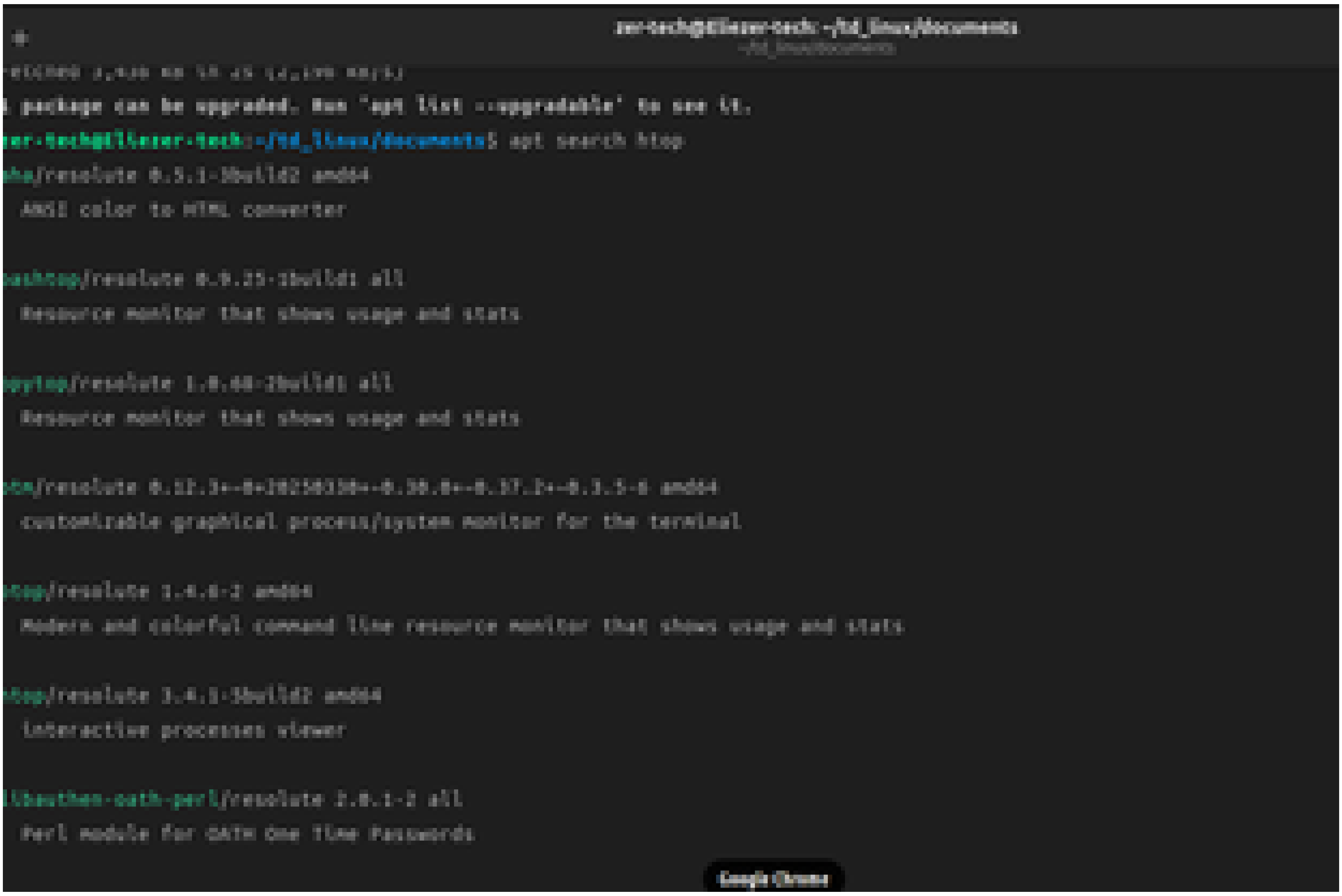
EXERCICE 4

La partie 4 vise à faire l'installation et la gestion des packets

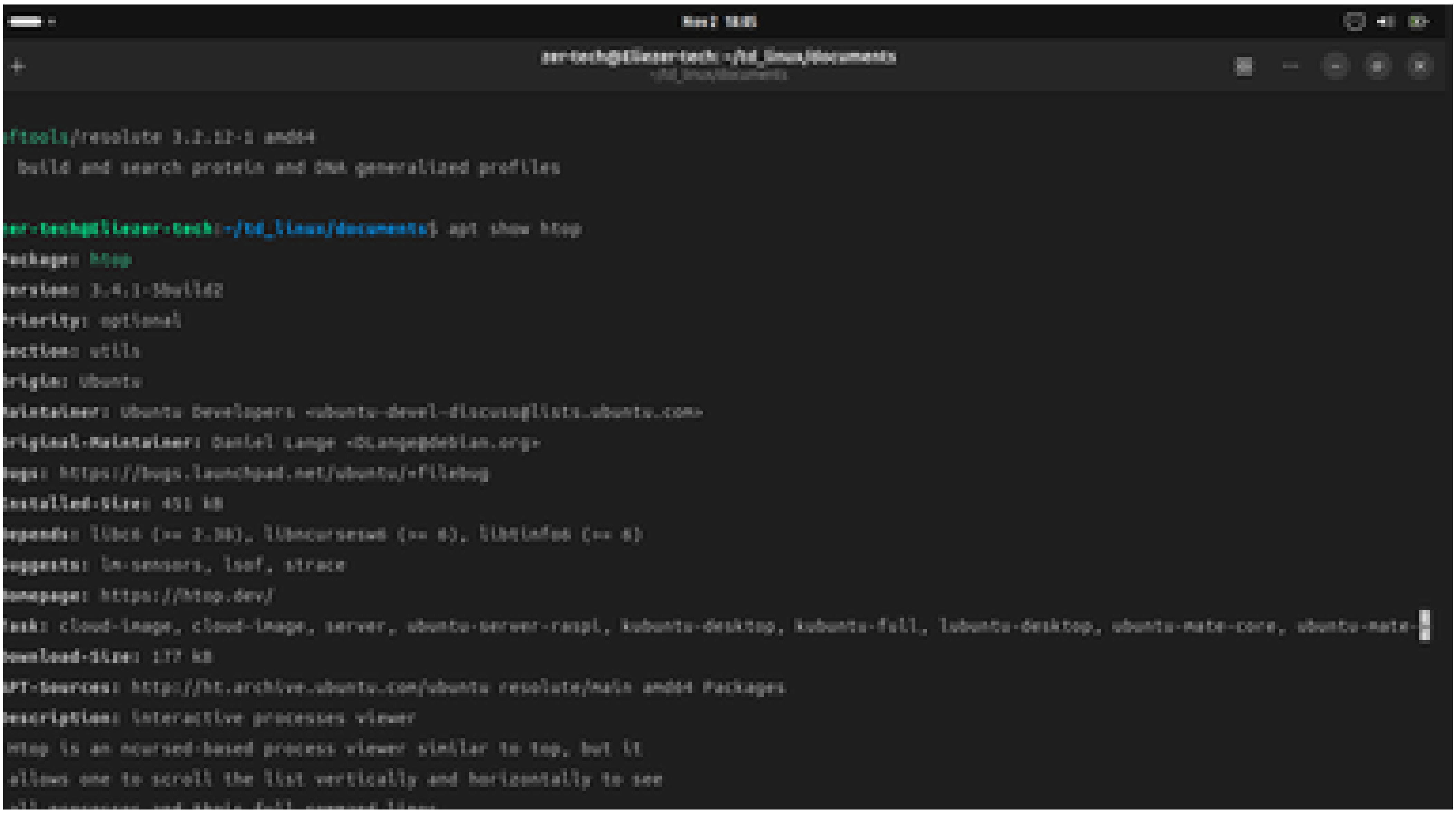
Sous Linux la différence réside dans l'action effectuée. En effet, tandis que l'une met à jour la liste des logiciels disponibles, l'autre installe concrètement les nouvelles versions. Donc *sudo apt update actualise l'index* des paquets, tandis que *sudo apt upgrade* télécharge et installe les mises à jour logicielles sur la machine.



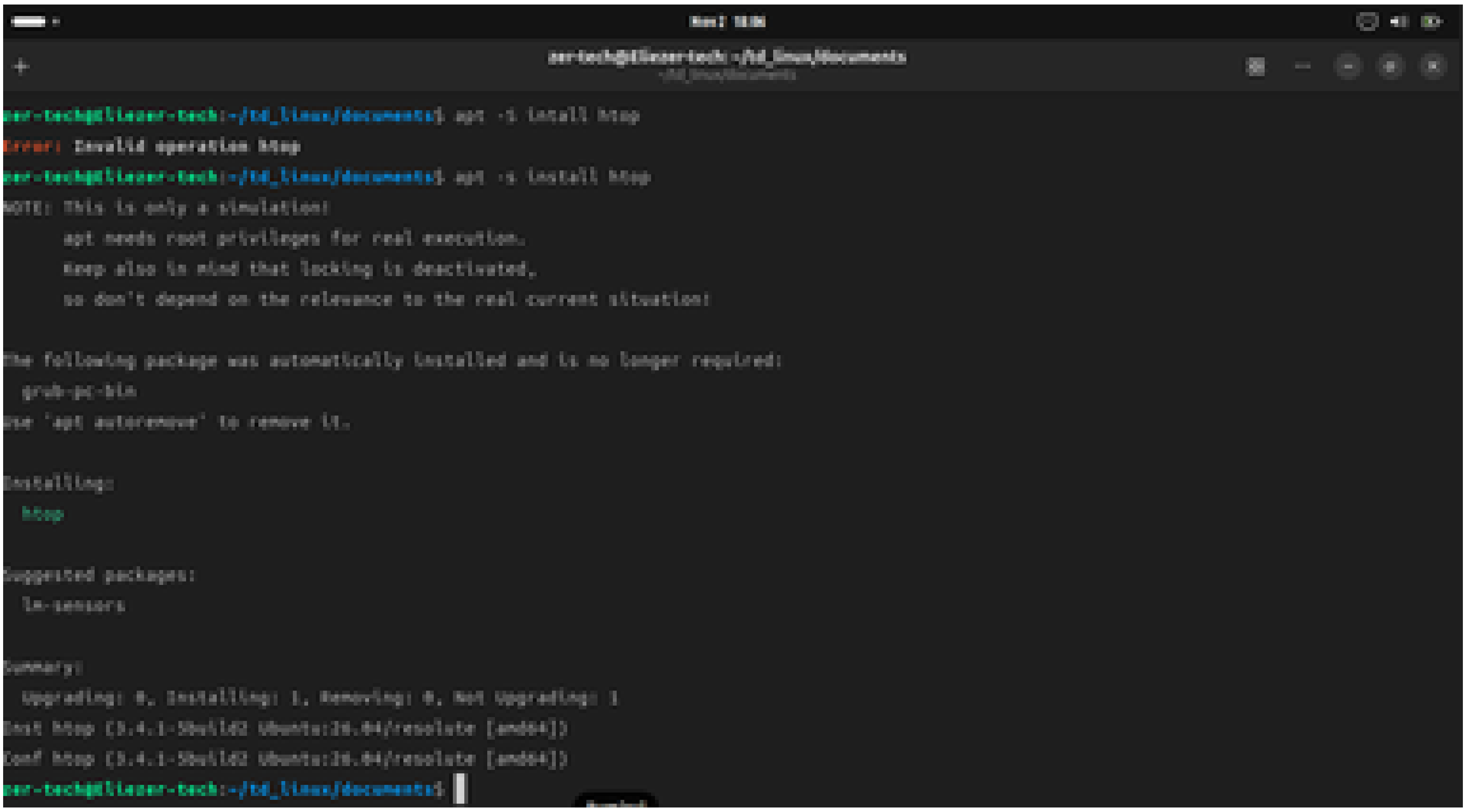
```
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 https://dl.google.com/linux/chrome-stable/deb stable InRelease [2,548 B]
Hit:3 http://ht.archive.ubuntu.com/ubuntu resolute InRelease
Get:4 https://dl.google.com/linux/chrome-stable/deb stable/main amd64 Packages [1,486 B]
Get:5 http://ht.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [638 kB]
Get:7 http://ht.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [146 kB]
Get:9 http://ht.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [883 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [527 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [108 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.4 kB]
Get:14 http://ht.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [118 kB]
Get:15 http://ht.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [374 kB]
Get:16 http://ht.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:17 http://ht.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1,056 B]
Fetched 3,436 kB in 2s (2,196 kB/s)
1 package can be upgraded. Run 'apt list --upgradable' to see it.
er-tech@Eliezer-tech:~/td_linux/documents$ apt search htop
```



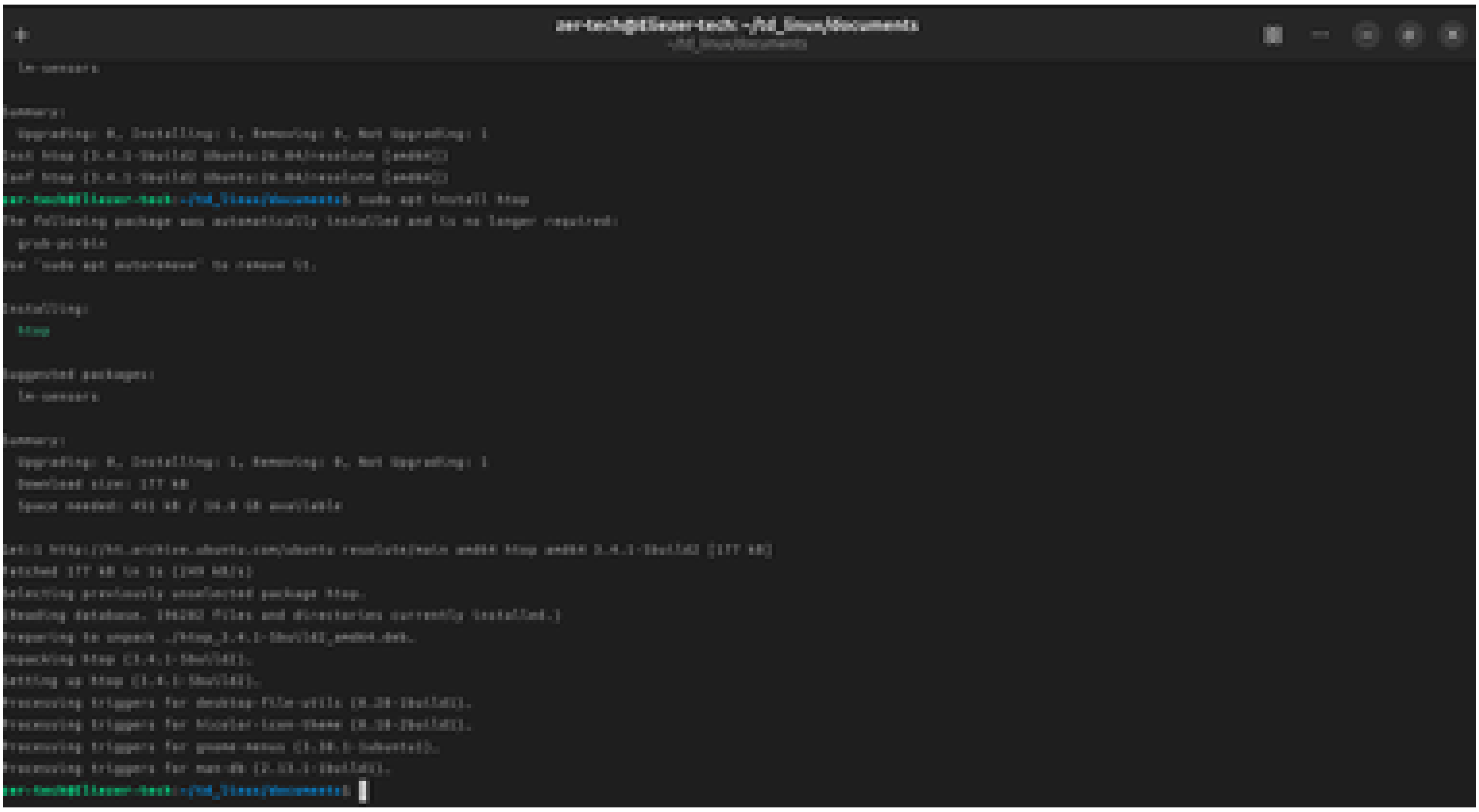
```
1 package can be upgraded. Run 'apt list --upgradable' to see it.
er-tech@Eliezer-tech:~/td_linux/documents$ apt search htop
aha/resolute 0.5.1-3build2 amd64
  ANSI color to HTML converter

ashtop/resolute 0.9.25-1build1 all
  Resource monitor that shows usage and stats

pytop/resolute 1.0.68-2build1 all
  Resource monitor that shows usage and stats

tm/resolute 0.12.3+~0+20250330+-0.30.0+-0.37.2+-0.3.5-6 amd64
  customizable graphical process/system monitor for the terminal

top/resolute 1.4.6-2 amd64
  Modern and colorful command line resource monitor that shows usage and stats

top/resolute 3.4.1-3build2 amd64
  interactive processes viewer

libauthen-oath-perl/resolute 2.0.1-2 all
  Perl module for OATH One Time Passwords
```



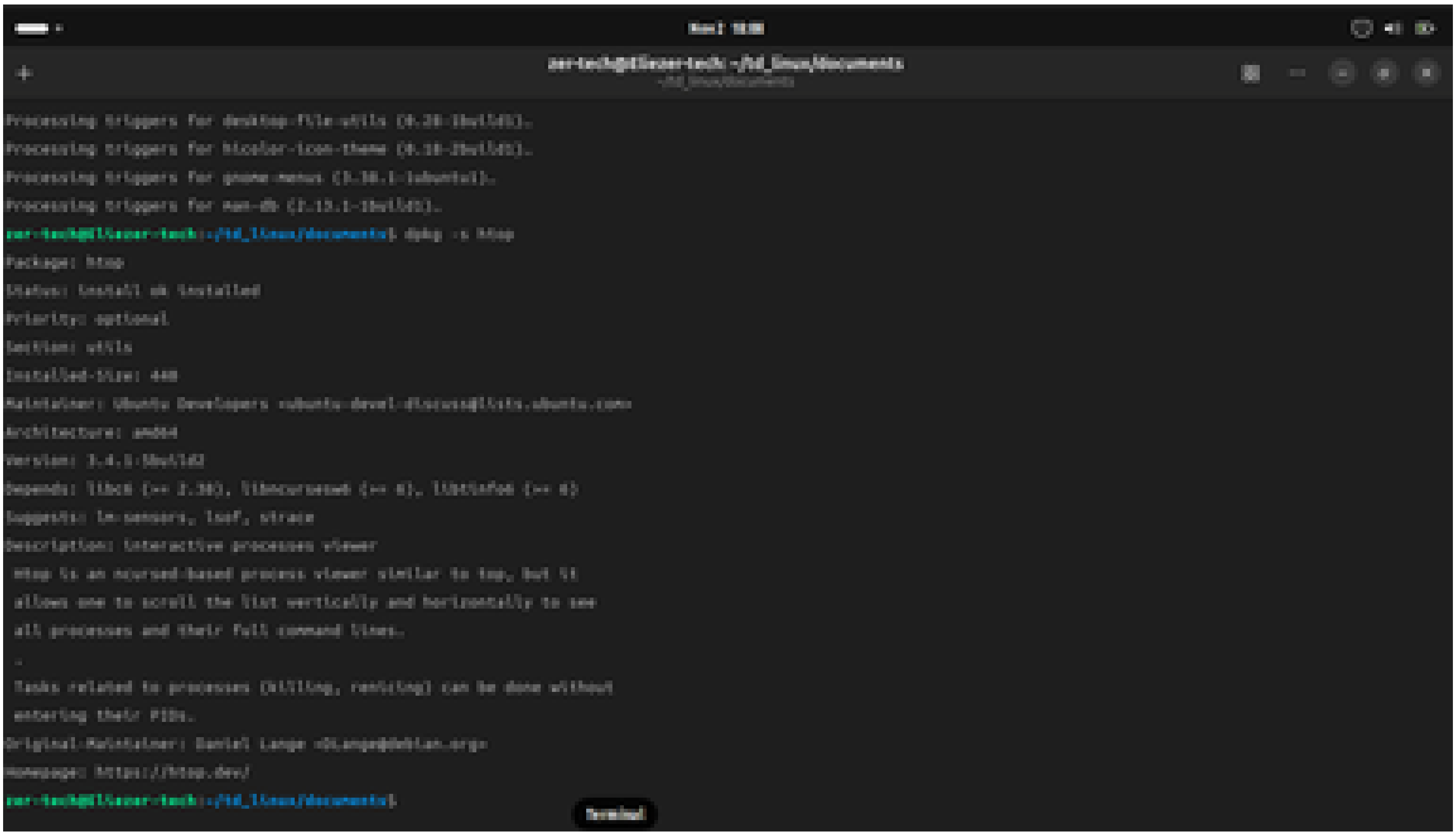
```
ftools/resolute 3.2.12-1 amd64
  build and search protein and DNA generalized profiles

zer-tech@Eliezer-tech:~/td_linux/documents$ apt show htop
Package: htop
Version: 3.4.1-5build2
Priority: optional
Section: utils
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Daniel Lange <DLange@debian.org>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 451 kB
Depends: libc6 (>= 2.38), libncursesw6 (>= 6), libtinfo6 (>= 6)
Suggests: lm-sensors, lsof, strace
Homepage: https://htop.dev/
Task: cloud-image, cloud-image, server, ubuntu-server-raspi, kubuntu-desktop, kubuntu-full, lubuntu-desktop, ubuntu-mate-core, ubuntu-mate-
Download-Size: 177 kB
APT-Sources: http://ht.archive.ubuntu.com/ubuntu resolute/main amd64 Packages
Description: interactive processes viewer
 Htop is an ncursed-based process viewer similar to top, but it
 allows one to scroll the list vertically and horizontally to see
 all processes and their full command lines.
```



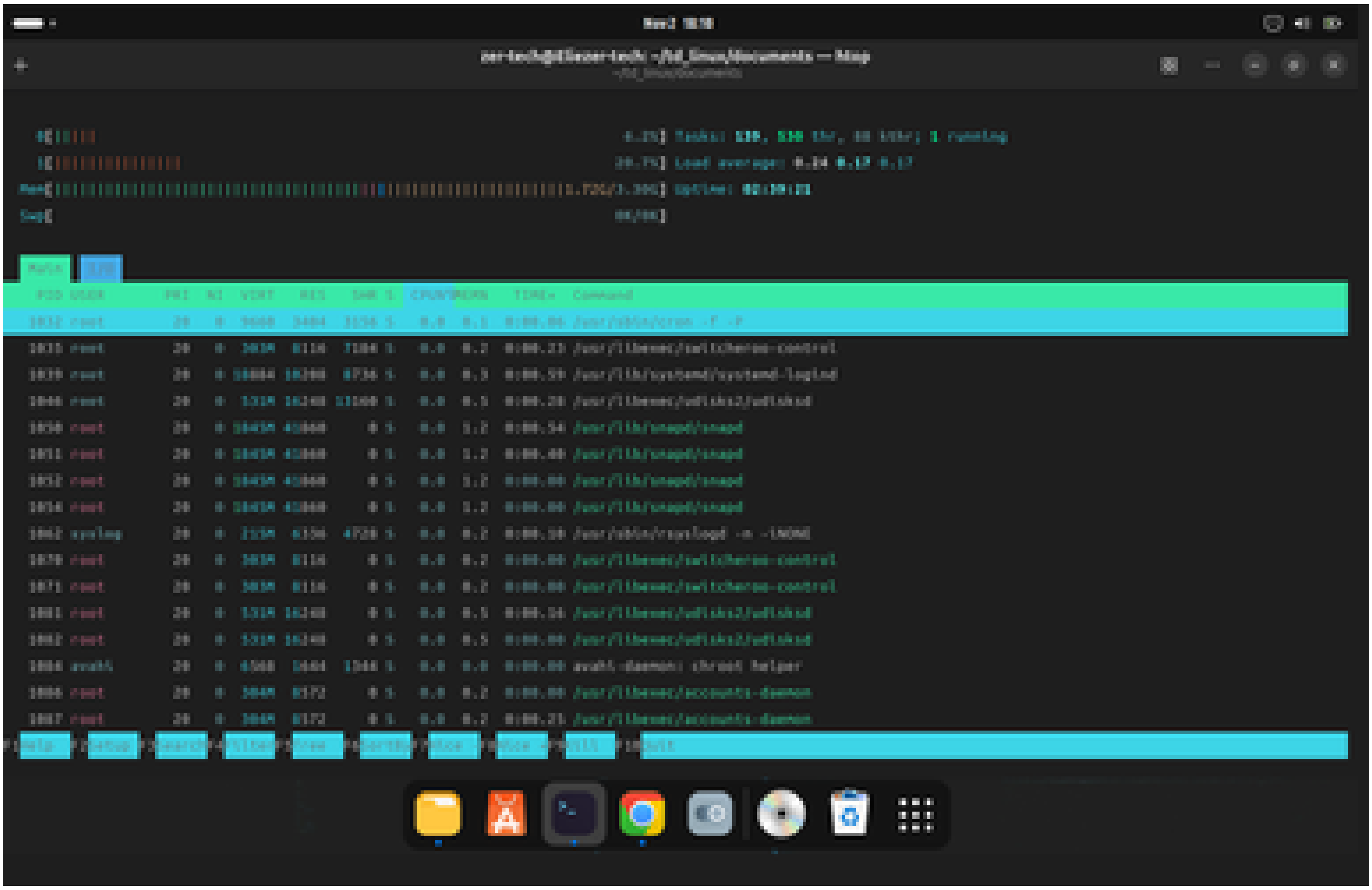
```
zer-tech@Eliezer-tech:~/td_linux/documents$ apt -i intall htop
Error: Invalid operation htop
zer-tech@Eliezer-tech:~/td_linux/documents$ apt -s install htop
NOTE: This is only a simulation!
      apt needs root privileges for real execution.
      Keep also in mind that locking is deactivated,
      so don't depend on the relevance to the real current situation!

The following package was automatically installed and is no longer required:
  grub-pc-bin
Use 'apt autoremove' to remove it.

Installing:
  htop

Suggested packages:
  lm-sensors

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Inst htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
Conf htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
zer-tech@Eliezer-tech:~/td_linux/documents$ 
```



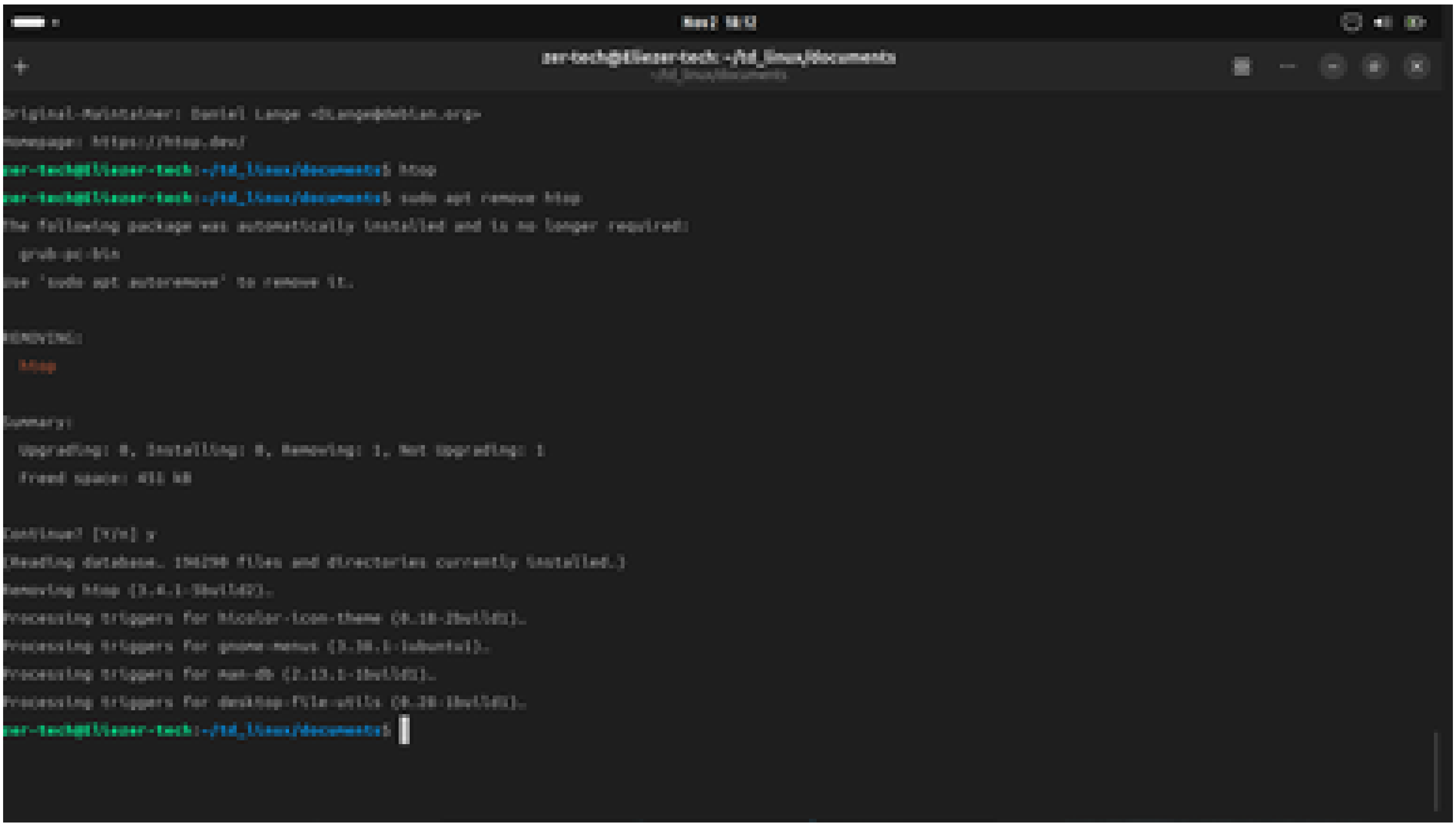
```
  lm-sensors

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Inst htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
Conf htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
zer-tech@Eliezer-tech:~/td_linux/documents$ sudo apt install htop
The following package was automatically installed and is no longer required:
  grub-pc-bin
Use 'sudo apt autoremove' to remove it.

Installing:
  htop

Suggested packages:
  lm-sensors

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
  Download size: 177 kB
  Space needed: 451 kB / 16.8 GB available

Get:1 http://fr.archive.ubuntu.com/ubuntu resolute/main amd64 htop amd64 3.4.1-5build2 [177 kB]
Fetched 177 kB in 1s (249 kB/s)
Selecting previously unselected package htop.
(Reading database ... 196282 files and directories currently installed.)
Preparing to unpack .../htop_3.4.1-5build2_amd64.deb ...
Unpacking htop (3.4.1-5build2) ...
Setting up htop (3.4.1-5build2) ...
Processing triggers for desktop-file-utils (0.28-1build1) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
Processing triggers for gnome-menus (3.38.1-1ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
zer-tech@Eliezer-tech:~/td_linux/documents$ 
```



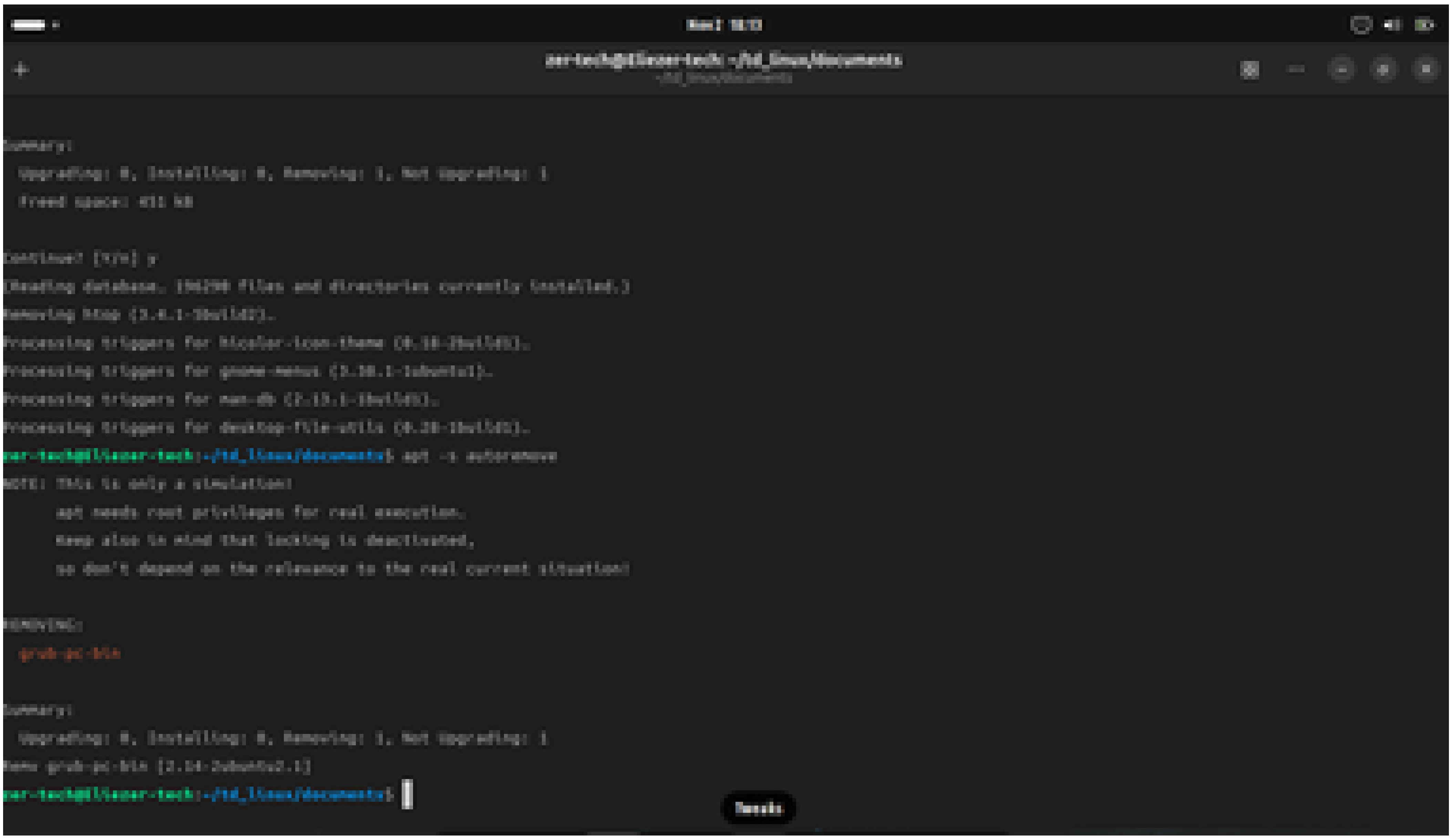
```
Processing triggers for desktop-file-utils (0.28-1build1) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
Processing triggers for gnome-menus (3.38.1-1ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
zer-tech@Eliezer-tech:~/td_linux/documents$ dpkg -s htop
Package: htop
Status: install ok installed
Priority: optional
Section: utils
Installed-Size: 448
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Architecture: amd64
Version: 3.4.1-5build2
Depends: libc6 (>= 2.38), libncursesw6 (>= 6), libtinfo6 (>= 6)
Suggests: lm-sensors, lsof, strace
Description: interactive processes viewer
 Htop is an ncursed-based process viewer similar to top, but it
 allows one to scroll the list vertically and horizontally to see
 all processes and their full command lines.
 .
 Tasks related to processes (killing, renicing) can be done without
 entering their PIDs.
Original-Maintainer: Daniel Lange <DLange@debian.org>
Homepage: https://htop.dev/
zer-tech@Eliezer-tech:~/td_linux/documents$
```



```
  0[|||||                                             4.2%] Tasks: 139, 530 thr, 88 kthr; 1 running
  1[||||||||||||||                                   20.7%] Load average: 0.24 0.17 0.17
Mem[||||||||||||||||||||||||||||||||||||||||||||||||1.72G/3.38G] Uptime: 02:39:21
Swp[                                                  0K/0K]

 Main  I/O
   PID USER       PRI  NI  VIRT   RES   SHR S  CPU% MEM%   TIME+  Command
  1832 root        20   0  9660  3404  3156 S   0.0  0.1  0:00.06 /usr/sbin/cron -f -P
  1835 root        20   0  303M  8116  7184 S   0.0  0.2  0:00.23 /usr/libexec/switcheroo-control
  1839 root        20   0 18084 10208  8736 S   0.0  0.3  0:00.59 /usr/lib/systemd/systemd-logind
  1846 root        20   0  531M 16248 11108 S   0.0  0.5  0:00.28 /usr/libexec/udisks2/udisksd
  1850 root        20   0 1845M 41860     0 S   0.0  1.2  0:00.54 /usr/lib/snapd/snapd
  1851 root        20   0 1845M 41860     0 S   0.0  1.2  0:00.40 /usr/lib/snapd/snapd
  1852 root        20   0 1845M 41860     0 S   0.0  1.2  0:00.00 /usr/lib/snapd/snapd
  1854 root        20   0 1845M 41860     0 S   0.0  1.2  0:00.00 /usr/lib/snapd/snapd
  1862 syslog      20   0  215M  6336  4720 S   0.0  0.2  0:00.10 /usr/sbin/rsyslogd -n -iNONE
  1870 root        20   0  303M  8116     0 S   0.0  0.2  0:00.00 /usr/libexec/switcheroo-control
  1871 root        20   0  303M  8116     0 S   0.0  0.2  0:00.00 /usr/libexec/switcheroo-control
  1881 root        20   0  531M 16248     0 S   0.0  0.5  0:00.16 /usr/libexec/udisks2/udisksd
  1882 root        20   0  531M 16248     0 S   0.0  0.5  0:00.00 /usr/libexec/udisks2/udisksd
  1884 avahi       20   0  6508  1604  1344 S   0.0  0.0  0:00.00 avahi-daemon: chroot helper
  1886 root        20   0  304M  8572     0 S   0.0  0.2  0:00.00 /usr/libexec/accounts-daemon
  1887 root        20   0  304M  8572     0 S   0.0  0.2  0:00.21 /usr/libexec/accounts-daemon
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```



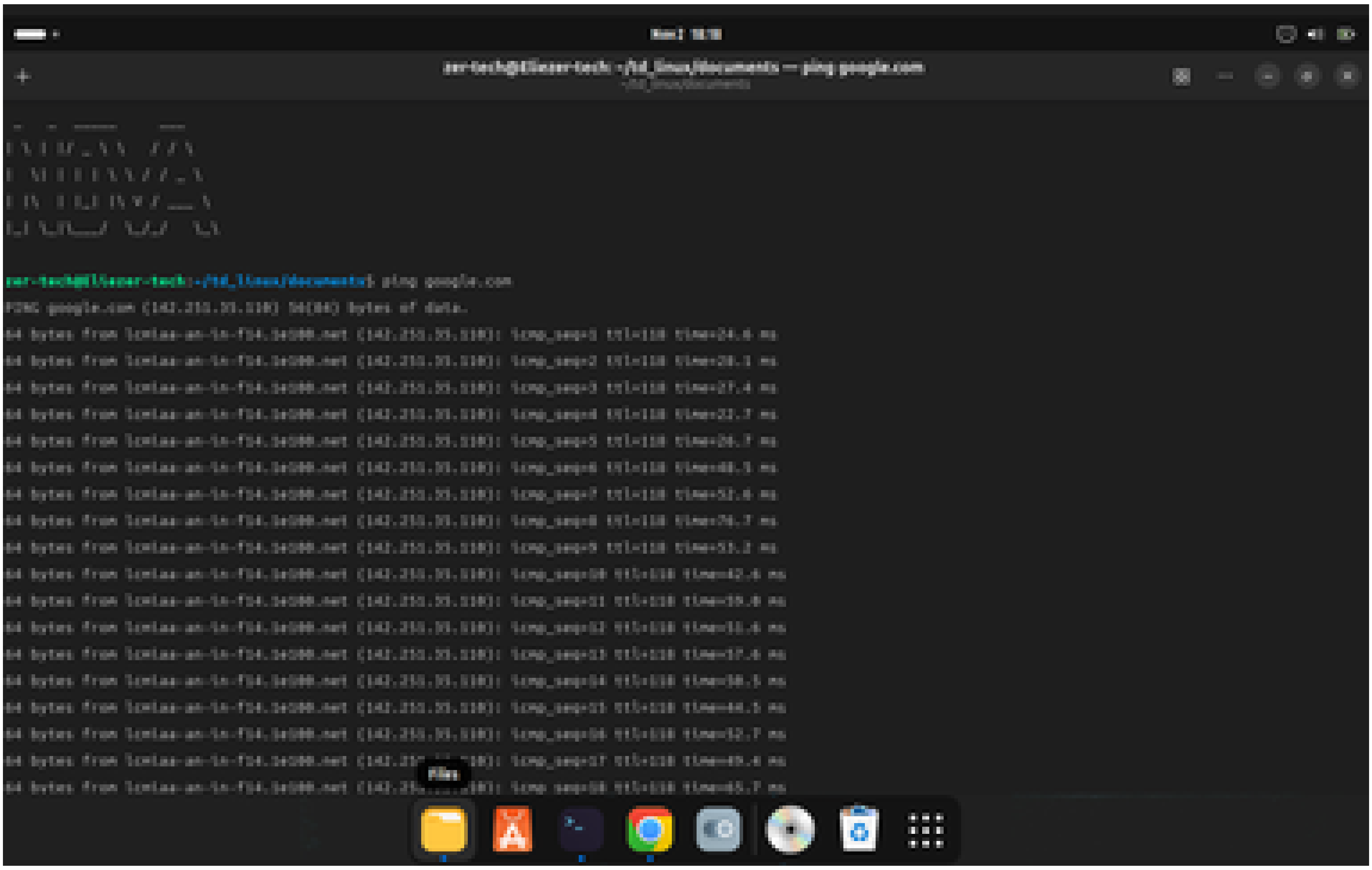
```
Original-Maintainer: Daniel Lange <DLange@debian.org>
Homepage: https://htop.dev/
zer-tech@Eliezer-tech:~/td_linux/documents$ htop
zer-tech@Eliezer-tech:~/td_linux/documents$ sudo apt remove htop
The following package was automatically installed and is no longer required:
  grub-pc-bin
Use 'sudo apt autoremove' to remove it.

REMOVING:
  htop

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1
  Freed space: 411 kB

Continue? [Y/n] y
(Reading database ... 196298 files and directories currently installed.)
Removing htop (3.4.1-5build2) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
Processing triggers for gnome-menus (3.38.1-1ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for desktop-file-utils (0.28-1build1) ...
zer-tech@Eliezer-tech:~/td_linux/documents$
```

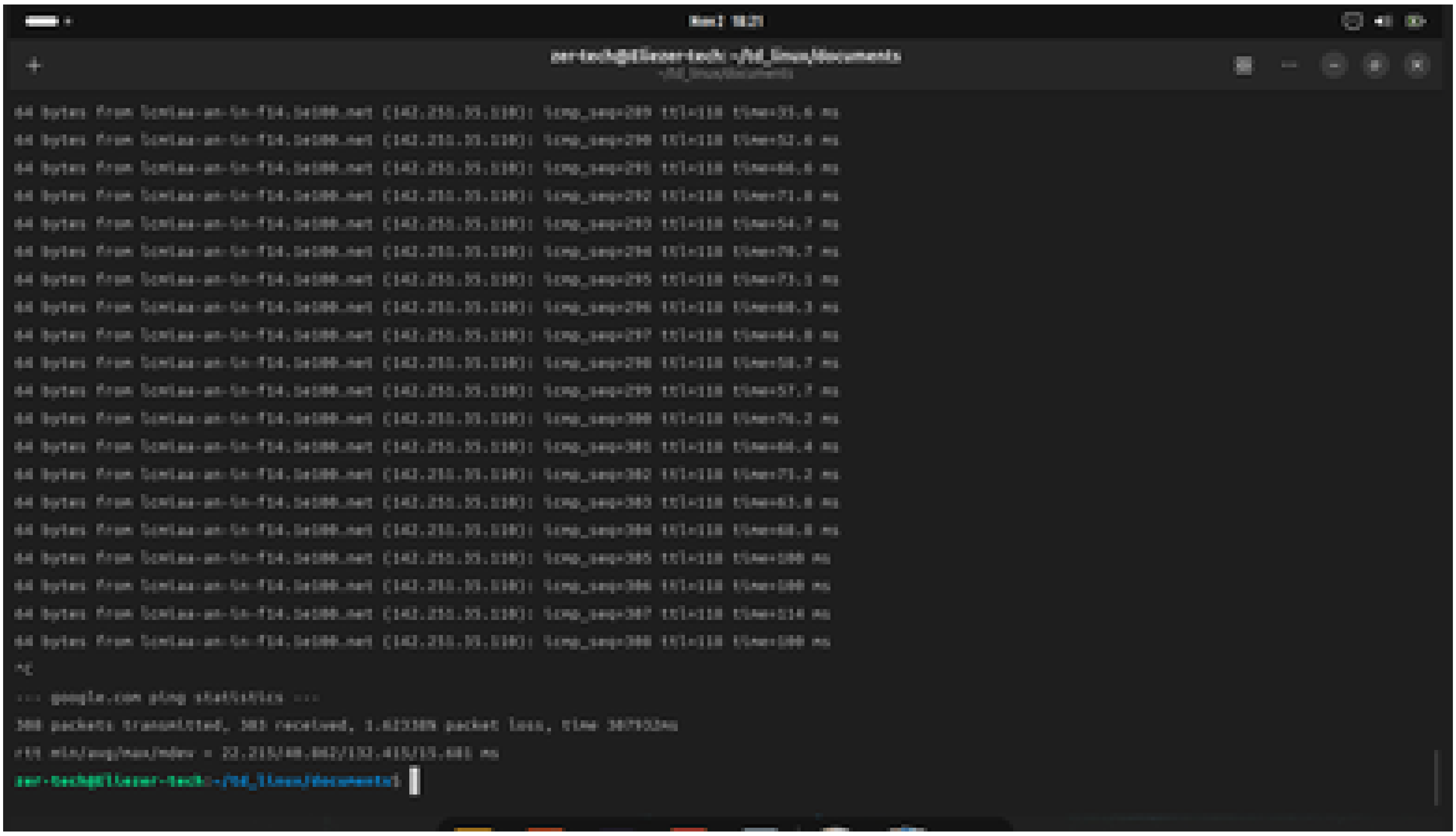


```
Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1
  Freed space: 431 kB

Continue? [Y/n] y
(Reading database ... 196290 files and directories currently installed.)
Removing htop (3.4.1-5build2) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
Processing triggers for gnome-menus (3.38.1-1ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for desktop-file-utils (0.28-1build1) ...
zer-tech@Eliezer-tech:~/td_linux/documents$ apt -s autoremove
NOTE: This is only a simulation!
      apt needs root privileges for real execution.
      Keep also in mind that locking is deactivated,
      so don't depend on the relevance to the real current situation!

REMOVING:
  grub-pc-bin

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1
Remv grub-pc-bin [2.14-2ubuntu2.1]
zer-tech@Eliezer-tech:~/td_linux/documents$ 
```

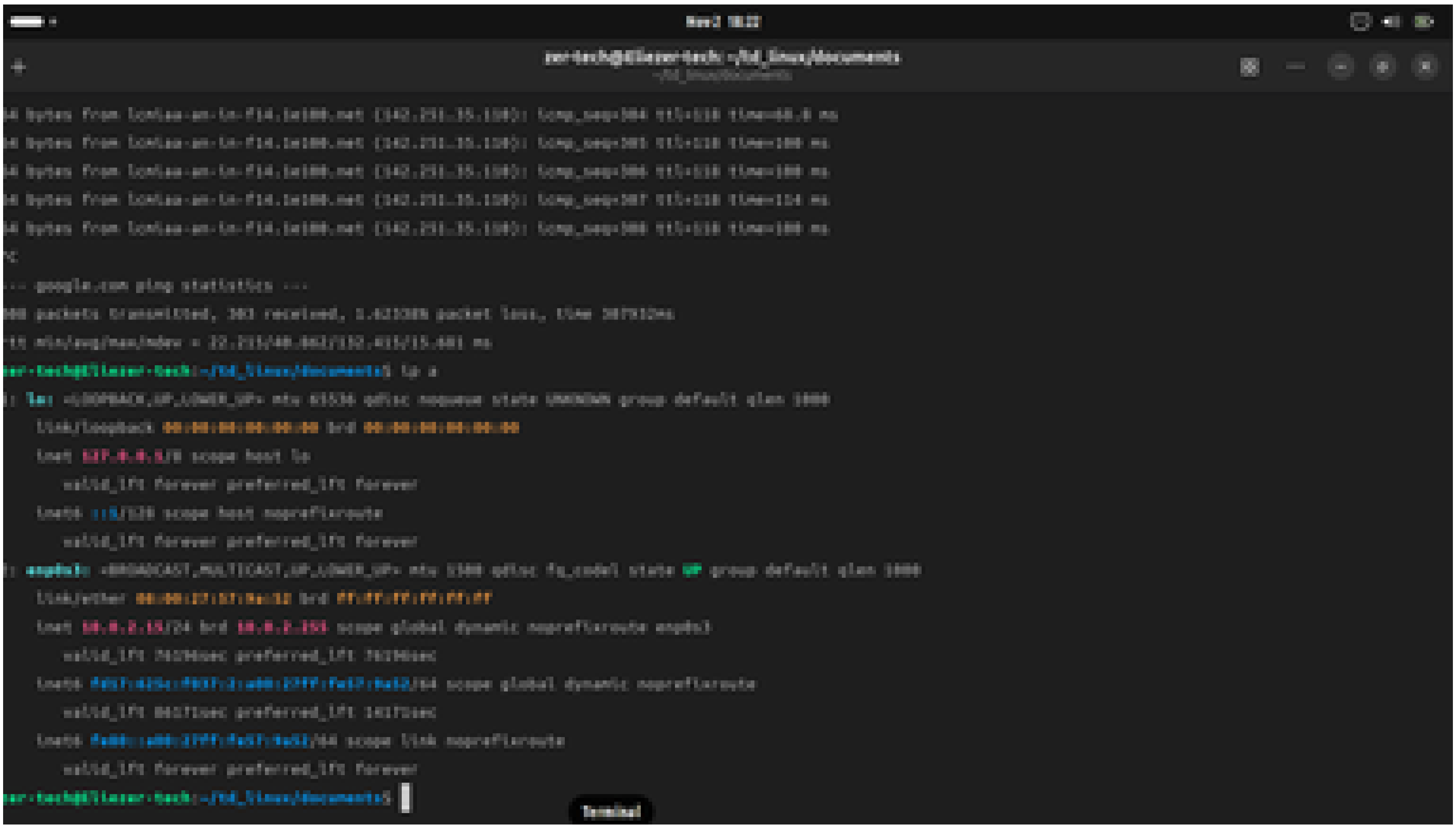
Exécutons ces commandes

```
ping google.com # Tester la connectivité réseau

ip a # Affiche les interfaces réseau et IP

history # Affiche l’historique des commandes

cowsay "Bienvenue dans le cours Linux" # Le terminal te parle avec une vache

cmatrix # Effet “Matrix” dans le terminal

neofetch # Affiche les infos système avec logo Linux

journalctl -xe # Consulter les logs système récents

sudo dmesg

sudo dmesg | grep -i memory

date

cal

figlet "UBUNTU"# Écrire un grand titre

sl # Écrire un grand titre

seq 1 10 # Écrire un grand titre

seq 2 2 20 # Affiche les nombres pairs de 2 à 20.

shuf -i 1-100 -n 1 # Affiche un nombre aléatoire entre 1 et 100.
```

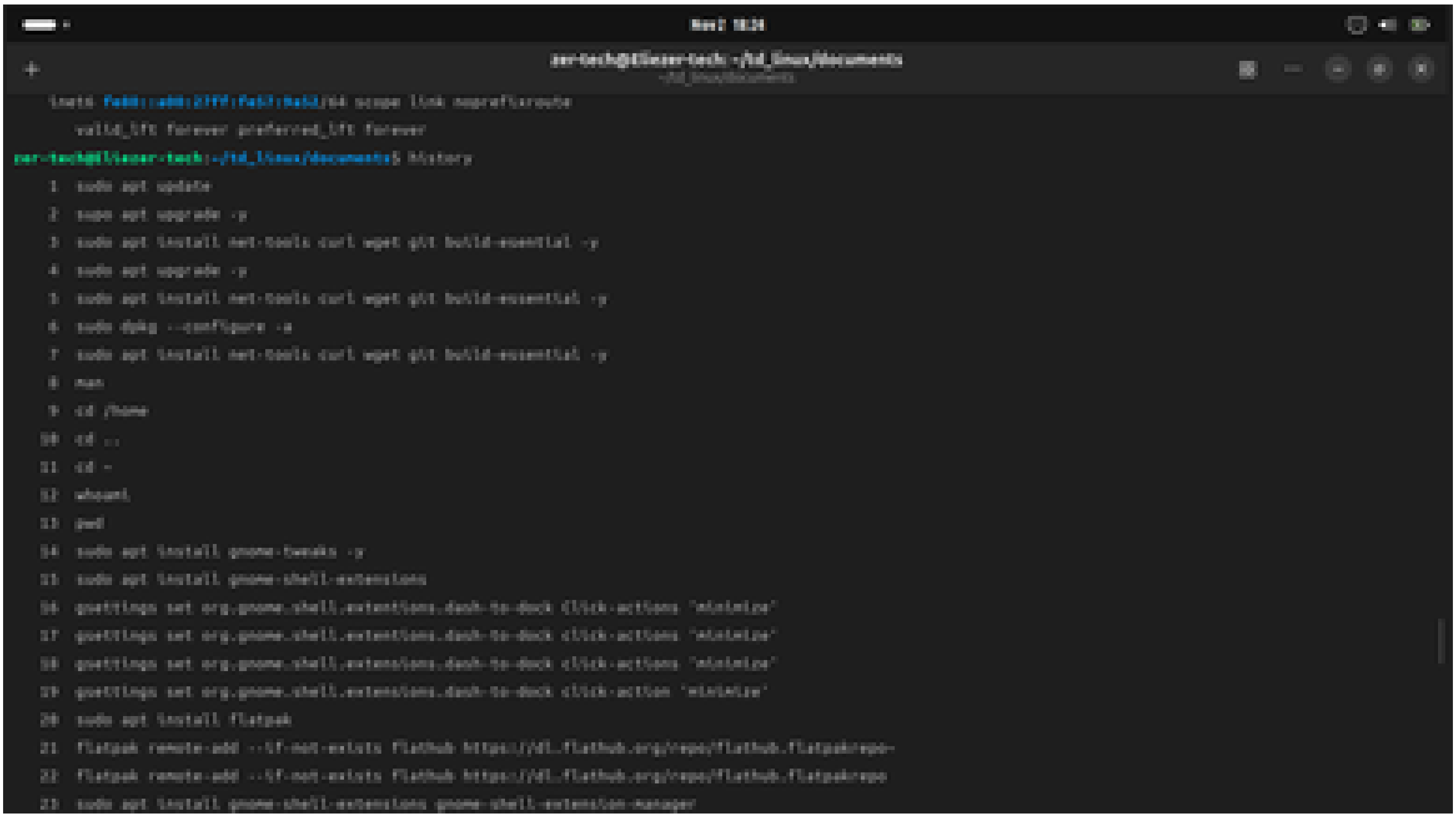




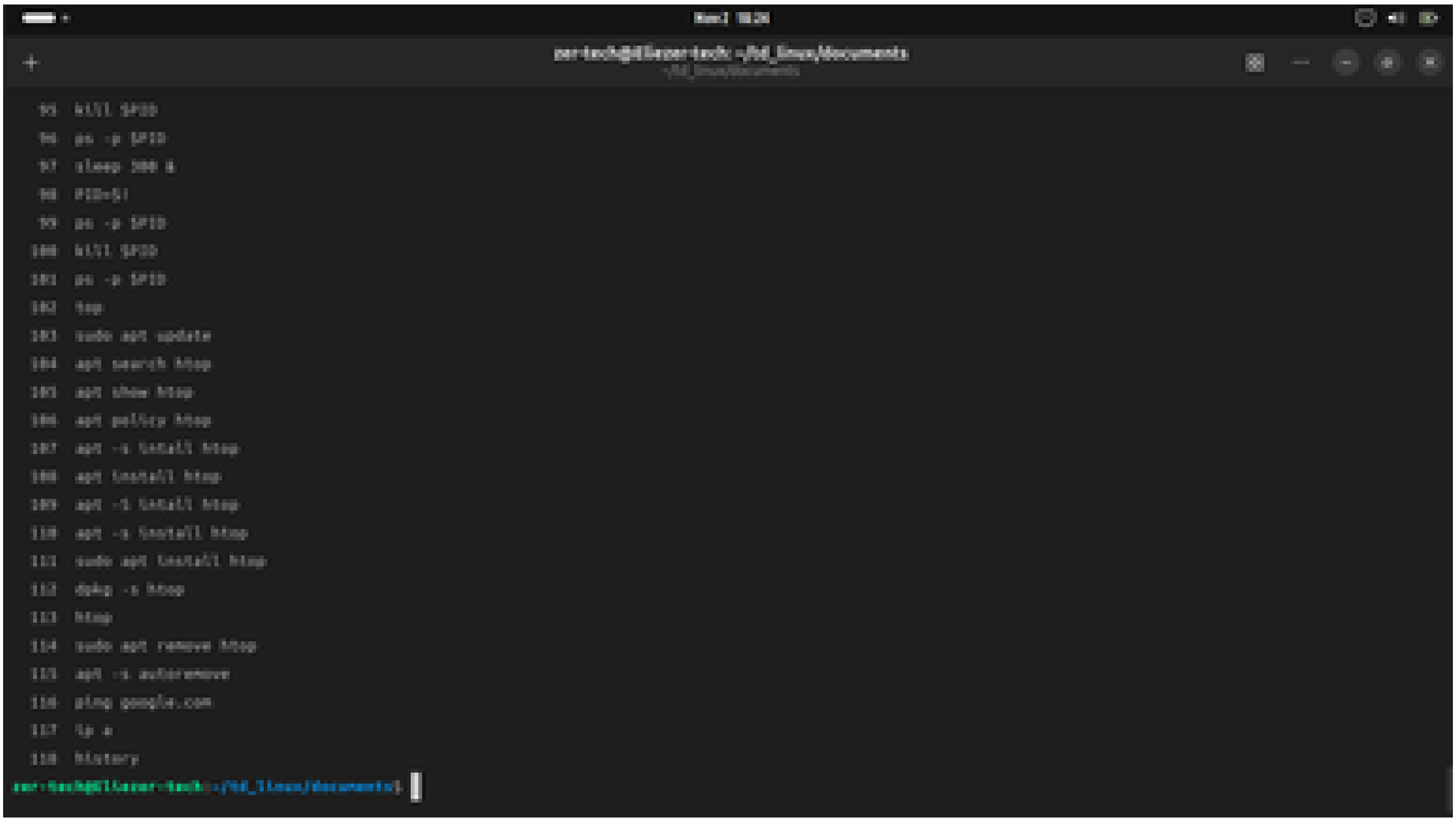
```
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=289 ttl=118 time=53.6 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=290 ttl=118 time=52.6 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=291 ttl=118 time=66.6 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=292 ttl=118 time=71.8 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=293 ttl=118 time=54.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=294 ttl=118 time=70.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=295 ttl=118 time=73.1 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=296 ttl=118 time=68.3 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=297 ttl=118 time=64.8 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=298 ttl=118 time=58.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=299 ttl=118 time=57.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=300 ttl=118 time=76.2 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=301 ttl=118 time=66.4 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=302 ttl=118 time=71.2 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=303 ttl=118 time=63.8 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=304 ttl=118 time=68.8 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=305 ttl=118 time=100 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=306 ttl=118 time=100 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=307 ttl=118 time=114 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=308 ttl=118 time=100 ms
^C
--- google.com ping statistics ---
308 packets transmitted, 303 received, 1.62338% packet loss, time 307932ms
rtt min/avg/max/mdev = 22.215/68.862/132.415/15.681 ms
zer-tech@Eliezer-tech:~/td_linux/documents$ 
```



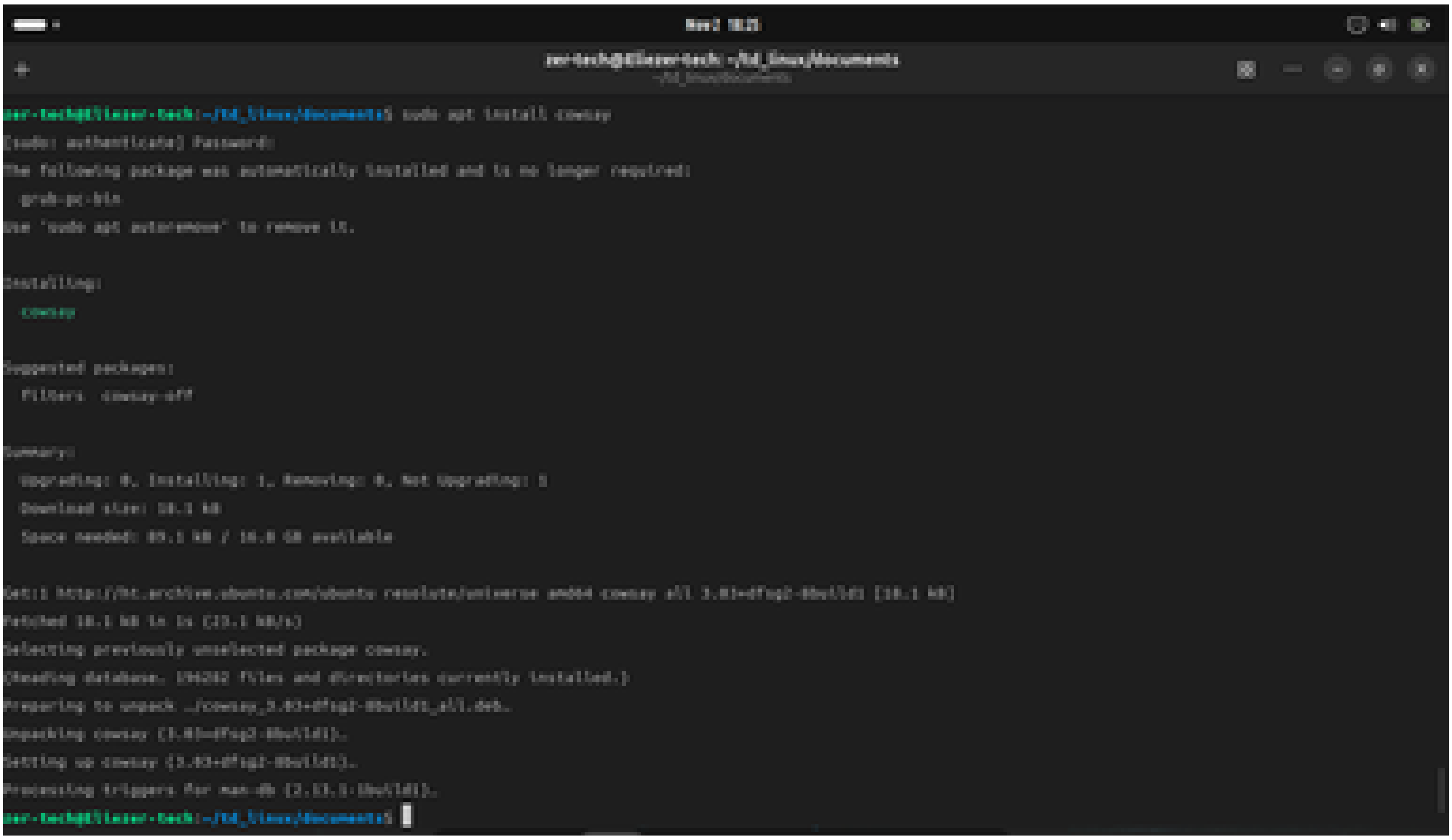
```
4 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=304 ttl=118 time=68.8 ms
4 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=305 ttl=118 time=100 ms
4 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=306 ttl=118 time=100 ms
4 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=307 ttl=118 time=114 ms
4 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=308 ttl=118 time=100 ms
C
-- google.com ping statistics ---
08 packets transmitted, 303 received, 1.62338% packet loss, time 307932ms
tt min/avg/max/mdev = 22.215/68.862/132.415/15.681 ms
er-tech@Eliezer-tech:~/td_linux/documents$ ip a
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
      valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
      valid_lft forever preferred_lft forever
: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:57:9a:12 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
      valid_lft 76190sec preferred_lft 76190sec
   inet6 fd17:625c:f037:2:a00:27ff:fe57:9a12/64 scope global dynamic noprefixroute
      valid_lft 86171sec preferred_lft 14171sec
   inet6 fe80::a00:27ff:fe57:9a12/64 scope link noprefixroute
      valid_lft forever preferred_lft forever
er-tech@Eliezer-tech:~/td_linux/documents$ 
```



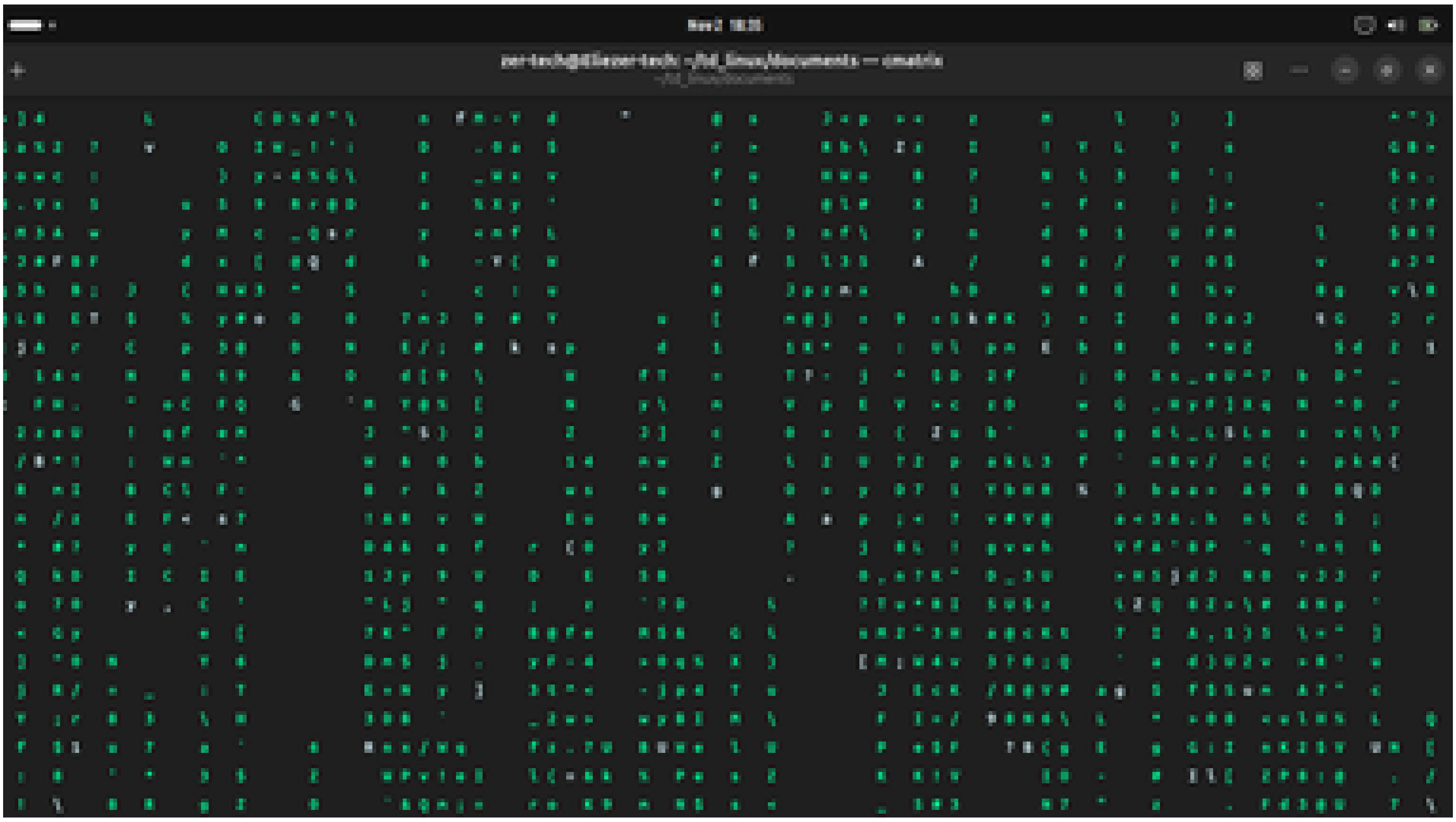
```
    inet6 fe80::a00:27ff:fe57:9a52/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
zer-tech@Eliezer-tech:~/td_linux/documents$ history
    1  sudo apt update
    2  supo apt upgrade -y
    3  sudo apt install net-tools curl wget git build-essential -y
    4  sudo apt upgrade -y
    5  sudo apt install net-tools curl wget git build-essential -y
    6  sudo dpkg --configure -a
    7  sudo apt install net-tools curl wget git build-essential -y
    8  man
    9  cd /home
   10  cd ..
   11  cd ~
   12  whoami
   13  pwd
   14  sudo apt install gnome-tweaks -y
   15  sudo apt install gnome-shell-extensions
   16  gsettings set org.gnome.shell.extentions.dash-to-dock click-actions 'minimize'
   17  gsettings set org.gnome.shell.extentions.dash-to-dock click-actions 'minimize'
   18  gsettings set org.gnome.shell.extensions.dash-to-dock click-actions 'minimize'
   19  gsettings set org.gnome.shell.extensions.dash-to-dock click-action 'minimize'
   20  sudo apt install flatpak
   21  flatpak remote-add --if-not-exists flathub https://dl.flathub.org/repo/flathub.flatpakrepo-
   22  flatpak remote-add --if-not-exists flathub https://dl.flathub.org/repo/flathub.flatpakrepo
   23  sudo apt install gnome-shell-extensions gnome-shell-extension-manager
```



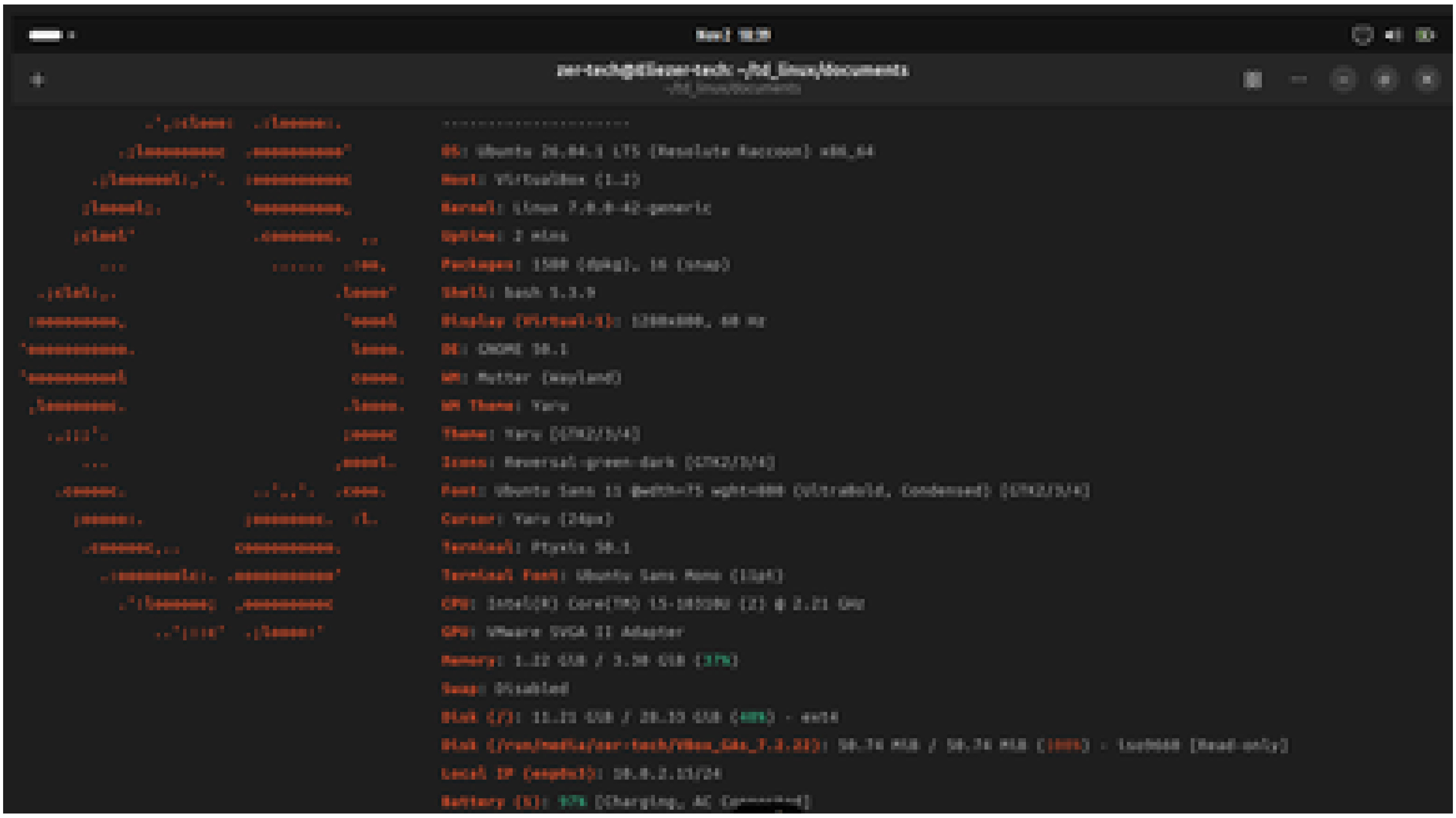
```
   95  kill $PID
   96  ps -p $PID
   97  sleep 300 &
   98  PID=$!
   99  ps -p $PID
  100  kill $PID
  101  ps -p $PID
  102  top
  103  sudo apt update
  104  apt search htop
  105  apt show htop
  106  apt policy htop
  107  apt -s install htop
  108  apt install htop
  109  apt -i install htop
  110  apt -s install htop
  111  sudo apt install htop
  112  dpkg -s htop
  113  htop
  114  sudo apt remove htop
  115  apt -s autoremove
  116  ping google.com
  117  ip a
  118  history
zer-tech@Eliezer-tech:~/td_linux/documents$ 
```


Nov 2 18:25
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents

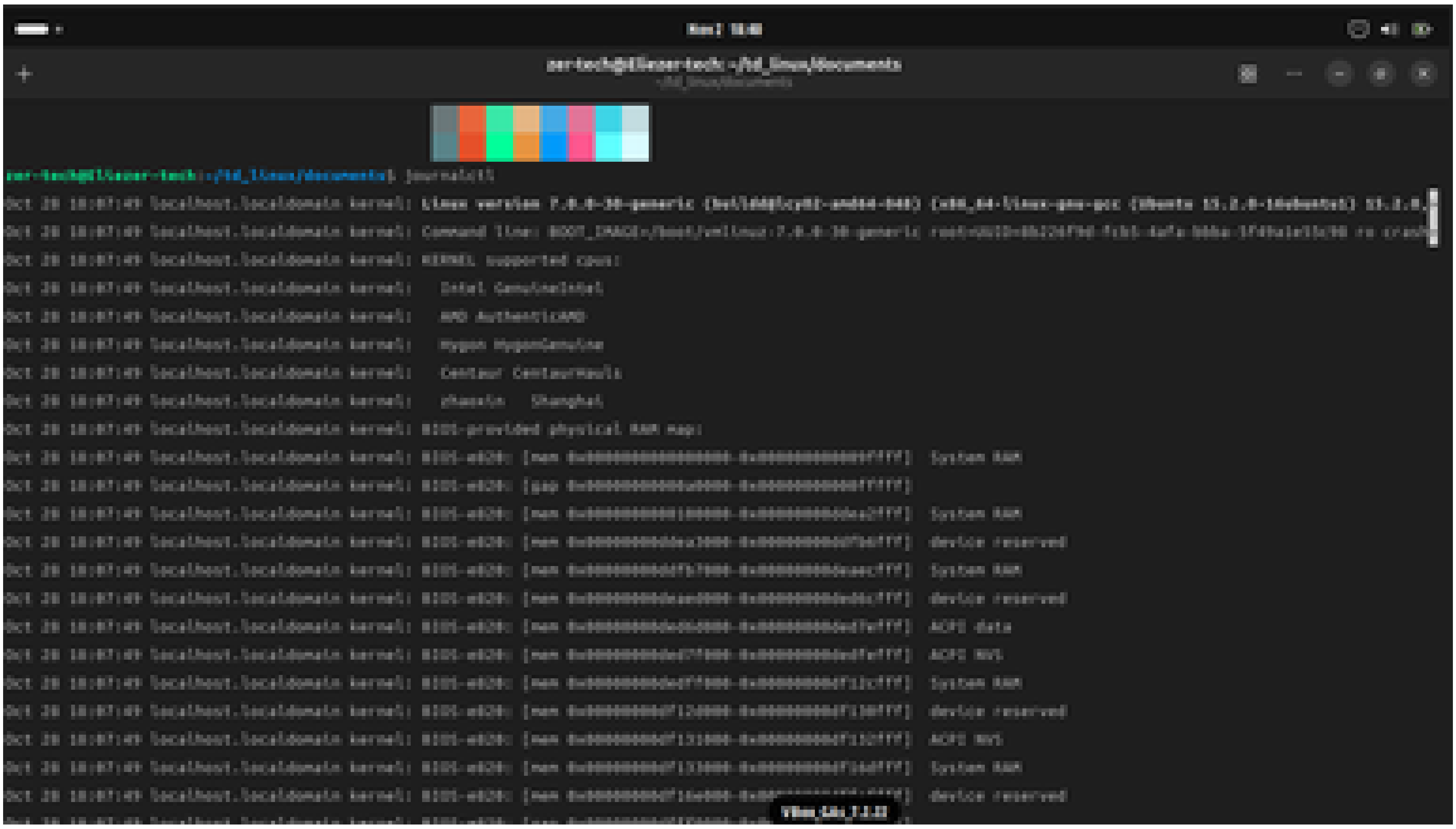
```
zer-tech@Eliezer-tech:~/td_linux/documents$ sudo apt install cowsay
[sudo: authenticate] Password:
The following package was automatically installed and is no longer required:
  grub-pc-bin
Use 'sudo apt autoremove' to remove it.

Installing:
  cowsay

Suggested packages:
  filters  cowsay-off

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
  Download size: 18.1 kB
  Space needed: 89.1 kB / 16.8 GB available

Get:1 http://ht.archive.ubuntu.com/ubuntu resolute/universe amd64 cowsay all 3.03+dfsg2-8build1 [18.1 kB]
Fetched 18.1 kB in 1s (23.1 kB/s)
Selecting previously unselected package cowsay.
(Reading database ... 196282 files and directories currently installed.)
Preparing to unpack .../cowsay_3.03+dfsg2-8build1_all.deb ...
Unpacking cowsay (3.03+dfsg2-8build1) ...
Setting up cowsay (3.03+dfsg2-8build1) ...
Processing triggers for man-db (2.13.1-1build1) ...
zer-tech@Eliezer-tech:~/td_linux/documents$ 
```


Nov 2 18:26
zer-tech@Eliezer-tech: ~/td_linux/documents — cmatrix
~/td_linux/documents

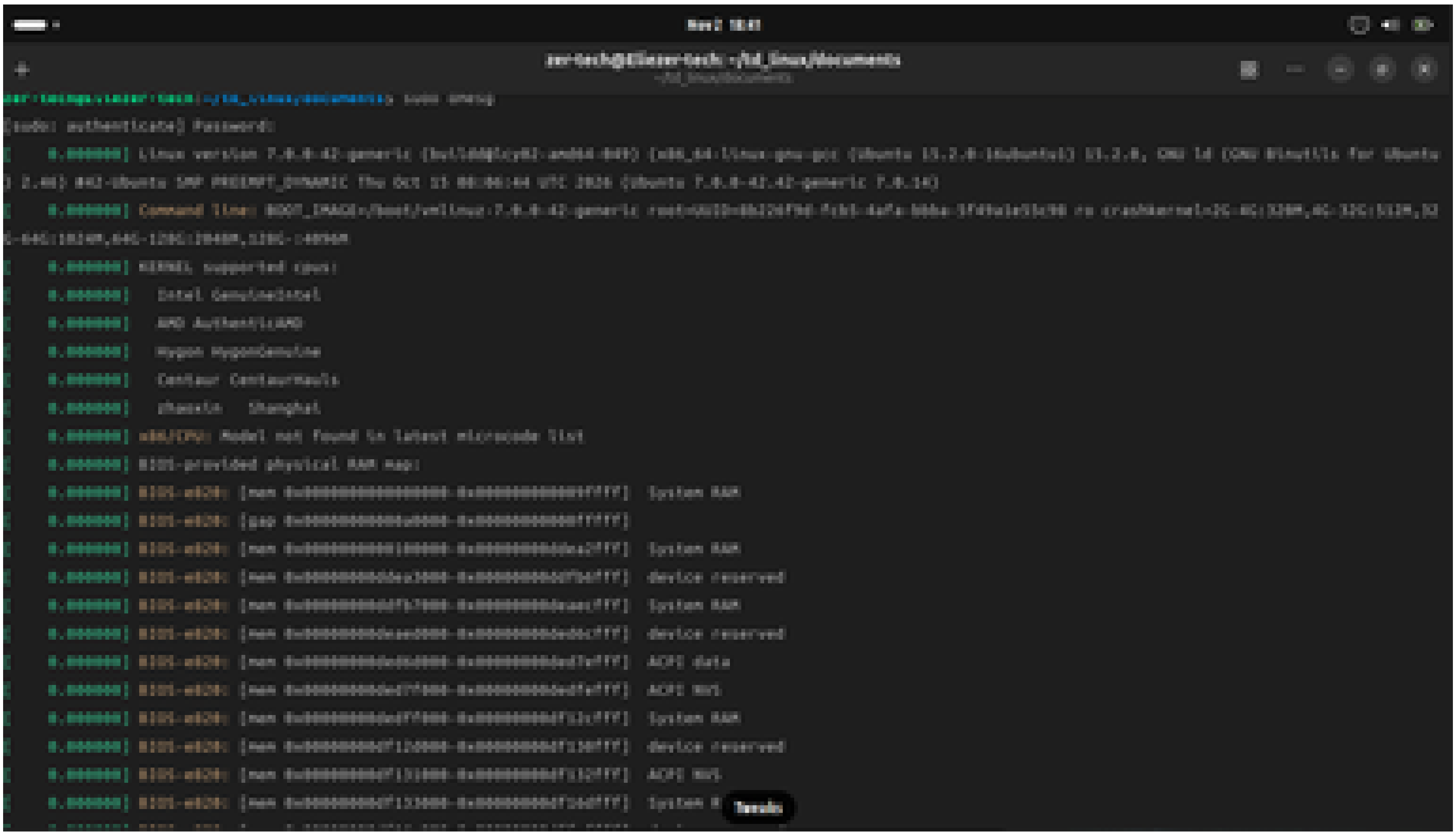
nous avons utilisé Fastfetch au lieu de Neofetch car le packet était absent

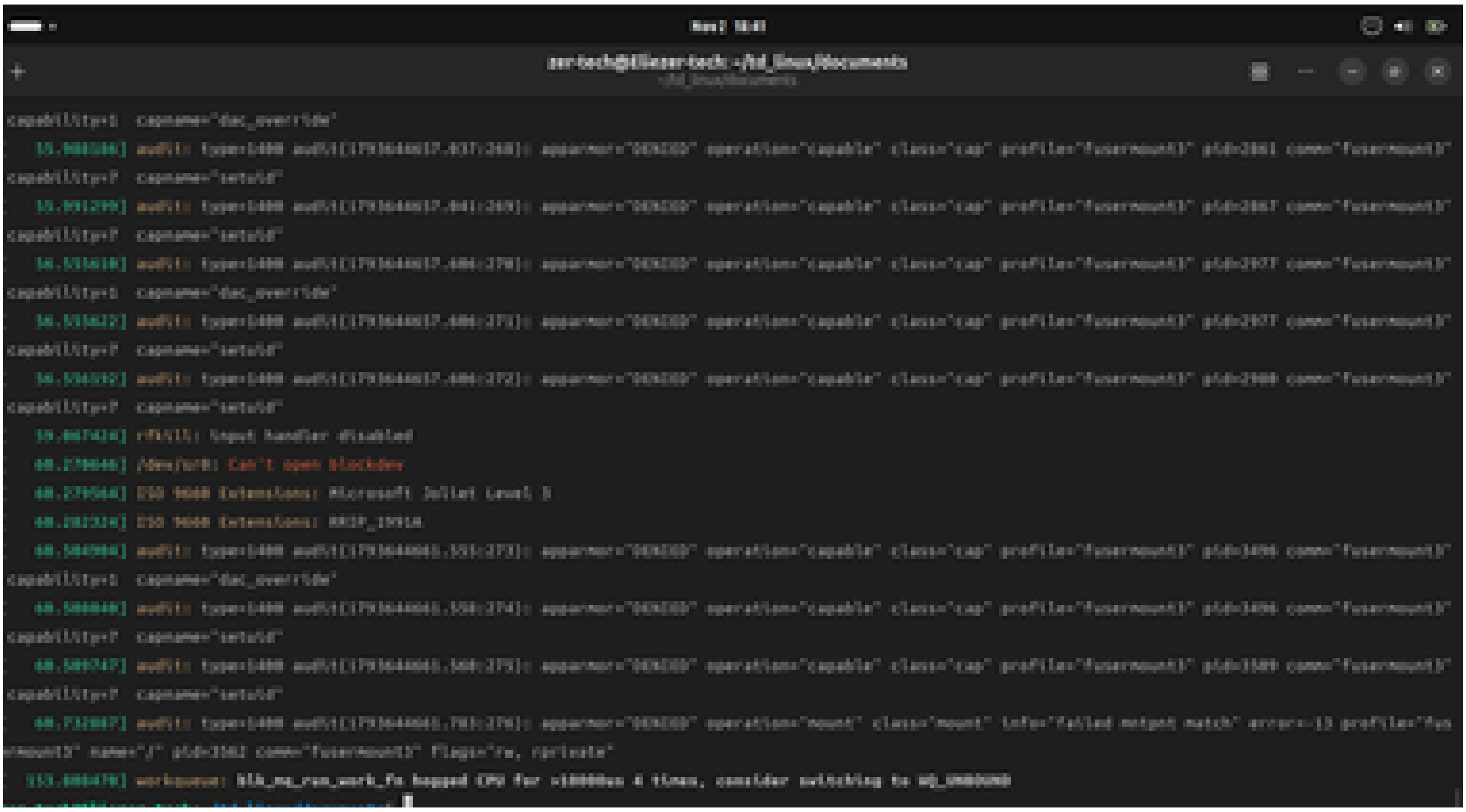


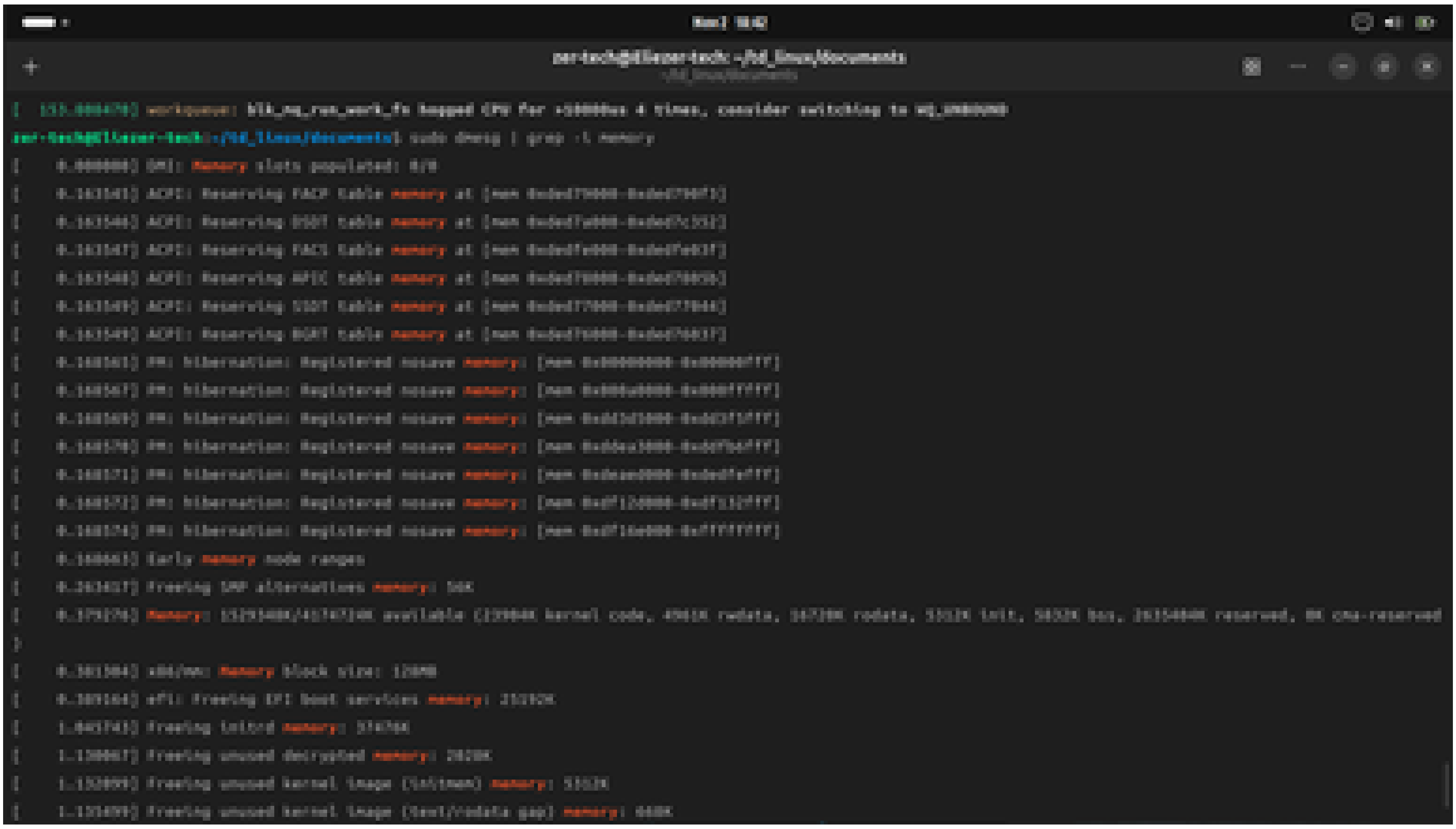
```
            .',;clooo:   .:looooo:.          ----------------------
         .;looooooooc   .oooooooooo'         OS: Ubuntu 26.04.1 LTS (Resolute Raccoon) x86_64
      .;looooool:,''.  :ooooooooooc          Host: VirtualBox (1.2)
     ;looool;.          'oooooooooo,         Kernel: Linux 7.0.0-42-generic
    ;clool'              .cooooooc.  ,,      Uptime: 2 mins
       ...                ......  .:oo,      Packages: 1500 (dpkg), 16 (snap)
  .;clol:,.                      .loooo'     Shell: bash 5.3.9
 :ooooooooo,                      'ooool     Display (Virtual-1): 1280x800, 60 Hz
'ooooooooooo.                      loooo.    DE: GNOME 50.1
'ooooooooool                       coooo.    WM: Mutter (Wayland)
 ,loooooooc.                      .loooo.    WM Theme: Yaru
   .,;;;'.                        ;ooooc     Theme: Yaru [GTK2/3/4]
       ...                       ,ooool.     Icons: Reversal-green-dark [GTK2/3/4]
    .cooooc.              ..',,'.  .cooo.    Font: Ubuntu Sans 11 @wdth=75 wght=800 (UltraBold, Condensed) [GTK2/3/4]
      ;ooooo:.           ;oooooooc.  :l.     Cursor: Yaru (24px)
       .coooooc,..      coooooooooo.         Terminal: Ptyxis 50.1
         .:oooooolc:. .ooooooooooo'          Terminal Font: Ubuntu Sans Mono (11pt)
           .':loooooo;  ,oooooooooc          CPU: Intel(R) Core(TM) i5-10310U (2) @ 2.21 GHz
               ..';::c'  .;loooo:'           GPU: VMware SVGA II Adapter
                                             Memory: 1.22 GiB / 3.30 GiB (37%)
                                             Swap: Disabled
                                             Disk (/): 11.21 GiB / 28.13 GiB (40%) - ext4
                                             Disk (/run/media/zer-tech/VBox_GAs_7.2.22): 58.74 MiB / 58.74 MiB (100%) - iso9660 [Read-only]
                                             Local IP (enp0s3): 10.0.2.15/24
                                             Battery (1): 97% [Charging, AC Connected]
```



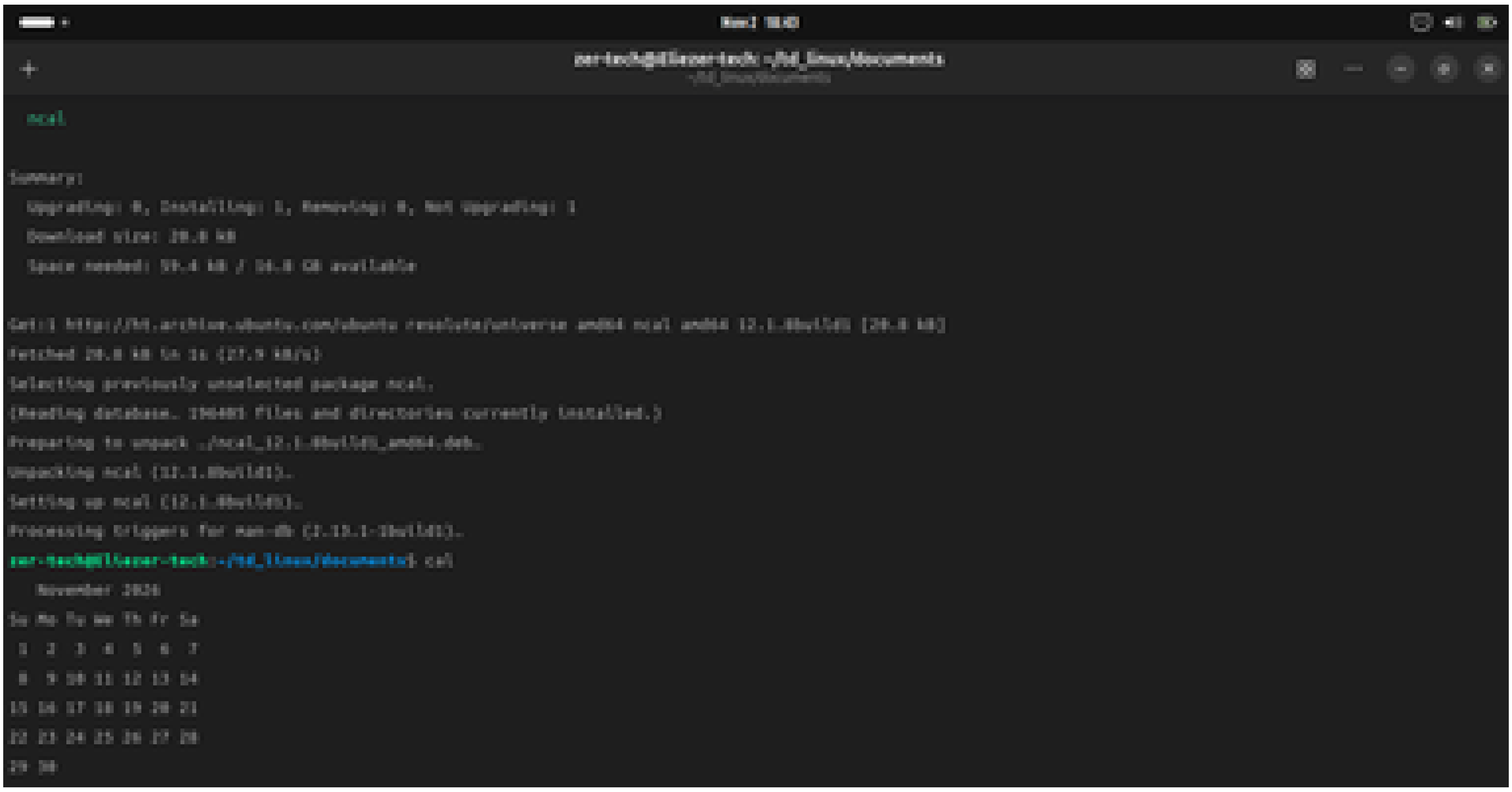
```
zer-tech@Eliezer-tech:~/td_linux/documents$ journalctl
Oct 28 18:07:49 localhost.localdomain kernel: Linux version 7.0.0-38-generic (buildd@lcy02-amd64-048) (x86_64-linux-gnu-gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0,
Oct 28 18:07:49 localhost.localdomain kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-7.0.0-38-generic root=UUID=8b226f90-fcb5-4afa-bbba-1f49a1e15c98 ro crash
Oct 28 18:07:49 localhost.localdomain kernel: KERNEL supported cpus:
Oct 28 18:07:49 localhost.localdomain kernel:   Intel GenuineIntel
Oct 28 18:07:49 localhost.localdomain kernel:   AMD AuthenticAMD
Oct 28 18:07:49 localhost.localdomain kernel:   Hygon HygonGenuine
Oct 28 18:07:49 localhost.localdomain kernel:   Centaur CentaurHauls
Oct 28 18:07:49 localhost.localdomain kernel:   zhaoxin   Shanghai
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-provided physical RAM map:
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff]  System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [gap 0x00000000000a0000-0x00000000000fffff]
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000ddea2fff]  System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000ddea3000-0x00000000ddf6bfff]  device reserved
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000ddf6c000-0x00000000deaecfff]  System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000deaed000-0x00000000deddcfff]  device reserved
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000dedd0000-0x00000000ded7efff]  ACPI data
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000ded7f000-0x00000000dedfefff]  ACPI NVS
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000dedff000-0x00000000df12cfff]  System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000df12d000-0x00000000df130fff]  device reserved
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000df131000-0x00000000df132fff]  ACPI NVS
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000df133000-0x00000000df16dfff]  System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000df16e000-0x00000000df[illegible]fff]  device reserved
```



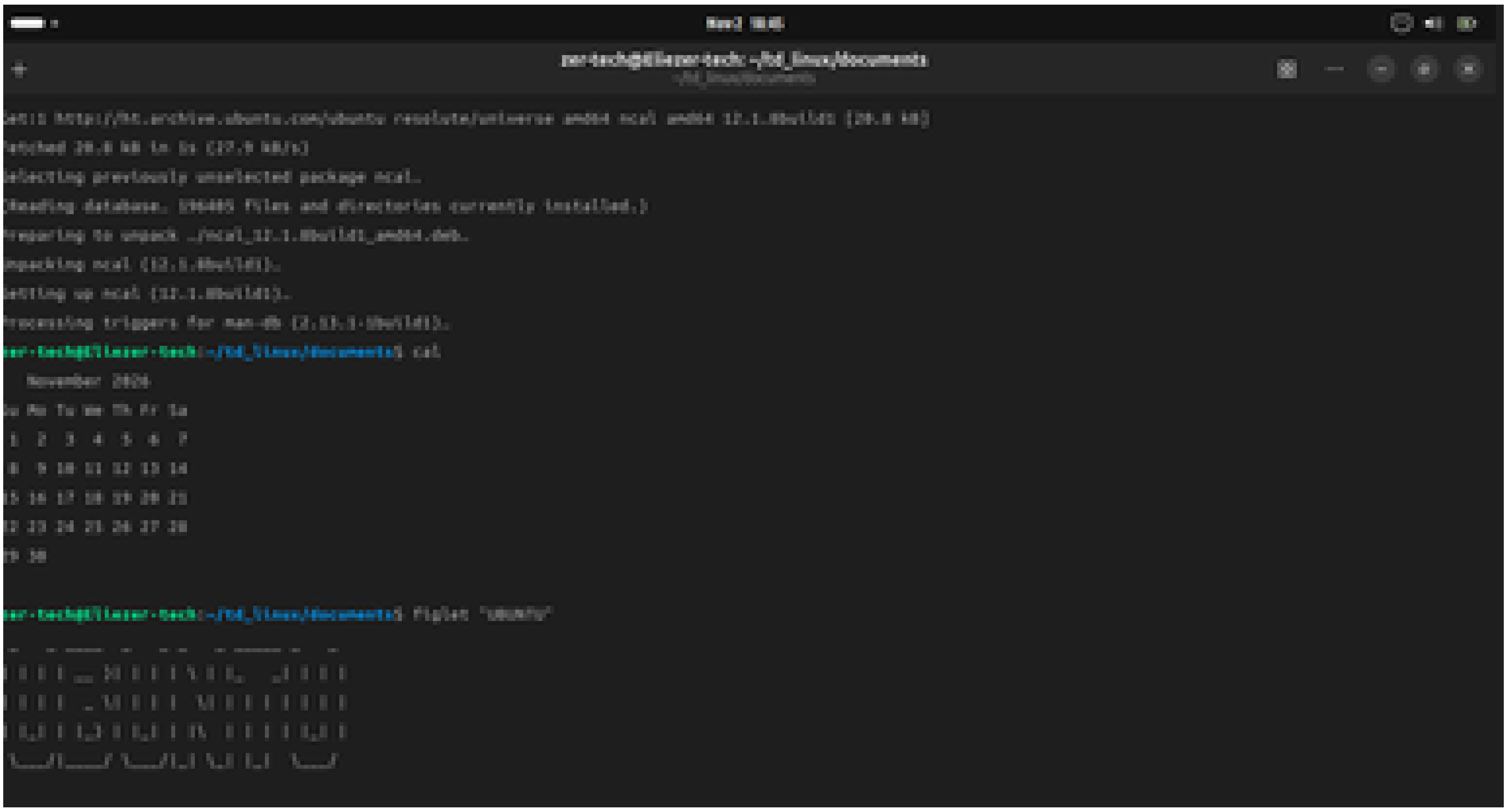




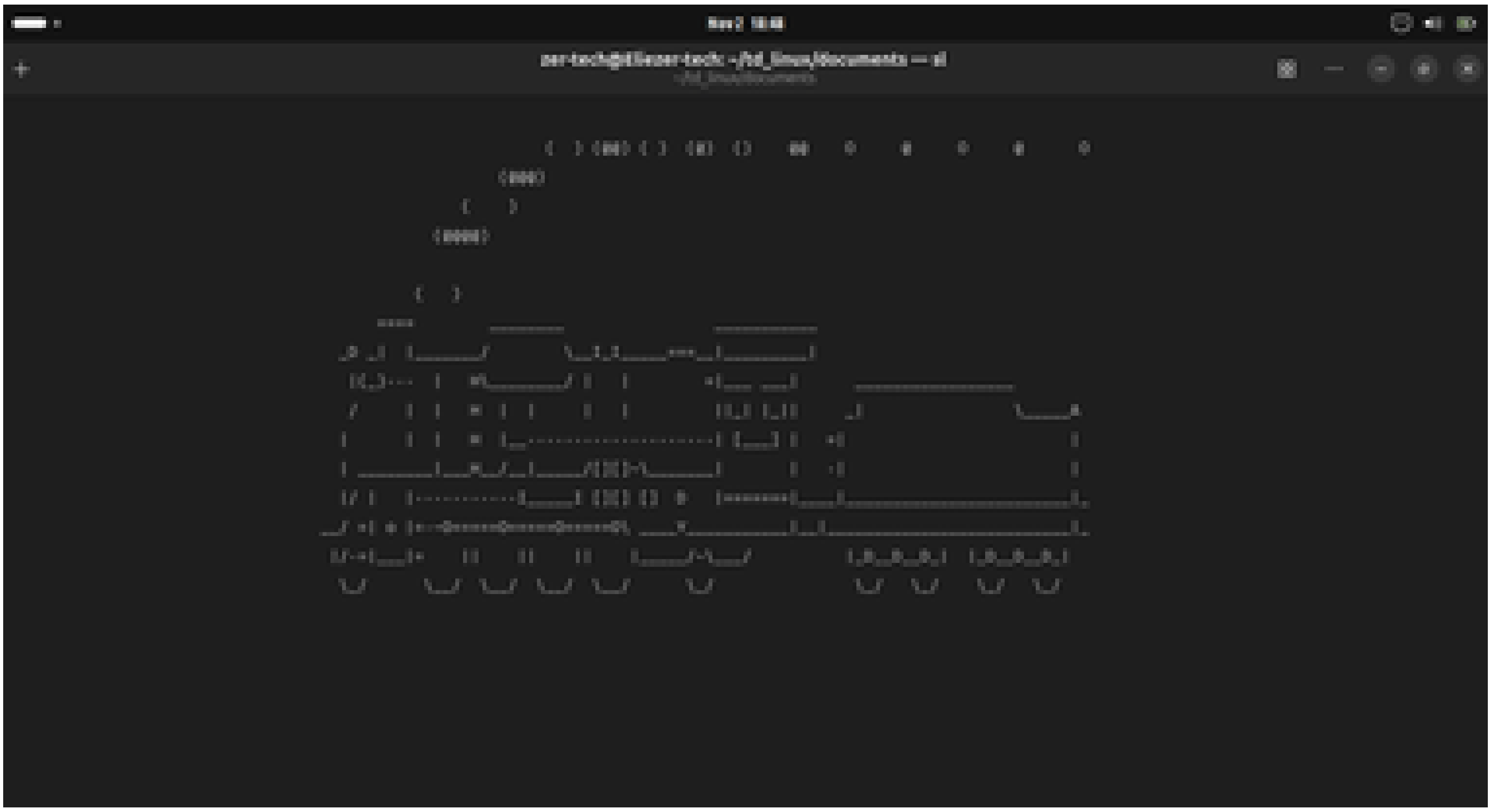
```
[  133.088479] workqueue: blk_mq_run_work_fn hogged CPU for >10000us 4 times, consider switching to WQ_UNBOUND
zer-tech@Elazer-tech:~/td_linux/documents$ sudo dmesg | grep -i memory
[    0.000000] DMI: Memory slots populated: 4/8
[    0.163541] ACPI: Reserving FACP table memory at [mem 0xded79000-0xded790f3]
[    0.163546] ACPI: Reserving DSDT table memory at [mem 0xded7a000-0xded7c352]
[    0.163547] ACPI: Reserving FACS table memory at [mem 0xdedfe000-0xdedfe03f]
[    0.163548] ACPI: Reserving APIC table memory at [mem 0xded78000-0xded7805b]
[    0.163549] ACPI: Reserving SSDT table memory at [mem 0xded77000-0xded77044]
[    0.163549] ACPI: Reserving BGRT table memory at [mem 0xded76000-0xded7603f]
[    0.168565] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
[    0.168567] PM: hibernation: Registered nosave memory: [mem 0x000a0000-0x000fffff]
[    0.168569] PM: hibernation: Registered nosave memory: [mem 0xdd3d1000-0xdd3f1fff]
[    0.168570] PM: hibernation: Registered nosave memory: [mem 0xddea3000-0xddf5afff]
[    0.168571] PM: hibernation: Registered nosave memory: [mem 0xdeae0000-0xdedfefff]
[    0.168572] PM: hibernation: Registered nosave memory: [mem 0xdf12d000-0xdf132fff]
[    0.168574] PM: hibernation: Registered nosave memory: [mem 0xdf16e000-0xffffffff]
[    0.168663] Early memory node ranges
[    0.263417] Freeing SMP alternatives memory: 56K
[    0.379276] Memory: 15293480K/41747240K available (23984K kernel code, 4961K rwdata, 16728K rodata, 5312K init, 5832K bss, 26354840K reserved, 0K cma-reserved
)
[    0.381304] x86/mm: Memory block size: 128MB
[    0.389164] efi: Freeing EFI boot services memory: 23192K
[    1.045743] Freeing initrd memory: 37476K
[    1.130067] Freeing unused decrypted memory: 2028K
[    1.132099] Freeing unused kernel image (initmem) memory: 5312K
[    1.133499] Freeing unused kernel image (text/rodata gap) memory: 668K
```

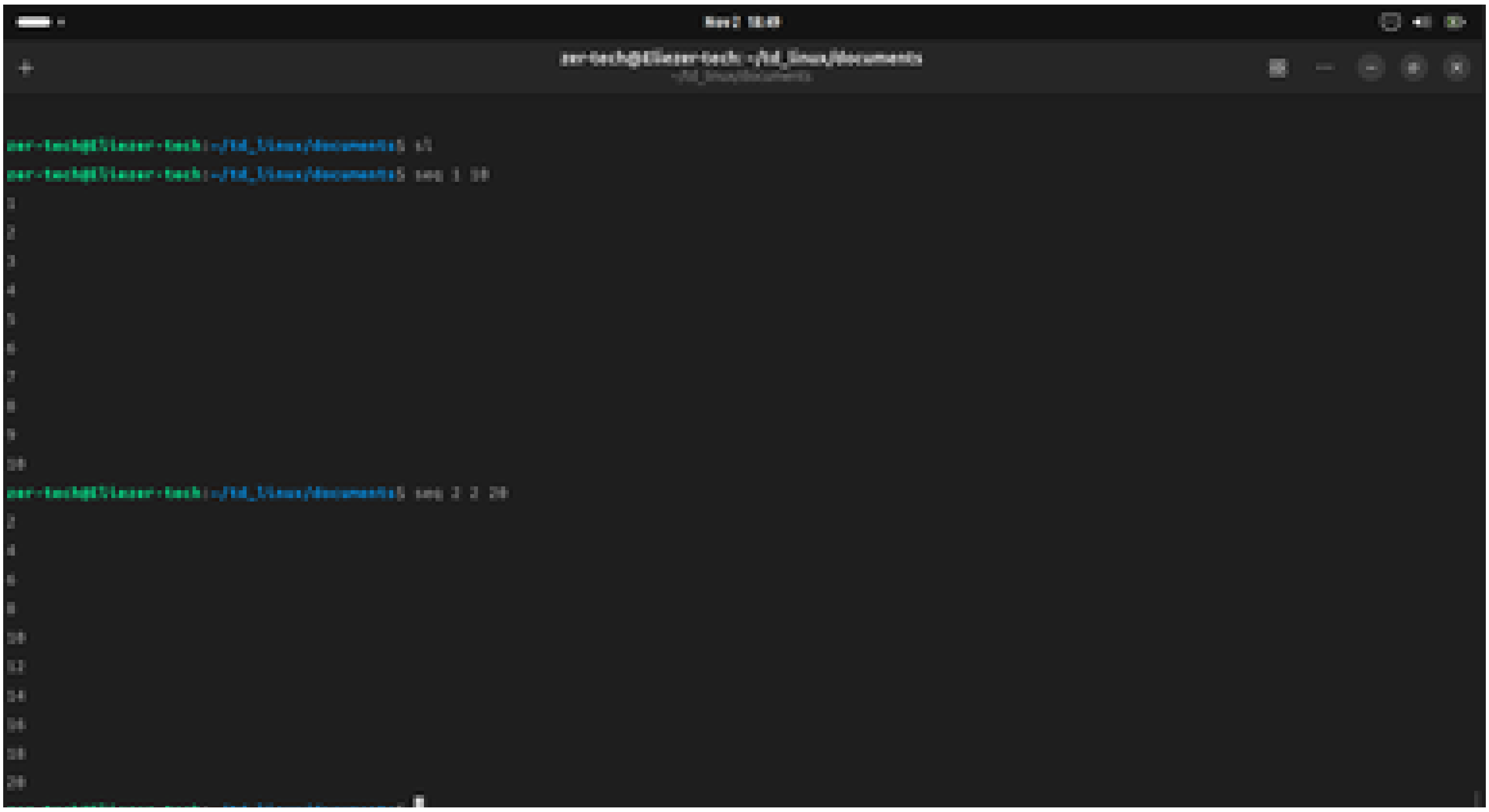


```
  ncal

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
  Download size: 20.8 kB
  Space needed: 59.4 kB / 16.8 GB available

Get:1 http://th.archive.ubuntu.com/ubuntu resolute/universe amd64 ncal amd64 12.1.8build1 [20.8 kB]
Fetched 20.8 kB in 1s (27.9 kB/s)
Selecting previously unselected package ncal.
(Reading database ... 196485 files and directories currently installed.)
Preparing to unpack .../ncal_12.1.8build1_amd64.deb ...
Unpacking ncal (12.1.8build1) ...
Setting up ncal (12.1.8build1) ...
Processing triggers for man-db (2.13.1-1build1) ...
zer-tech@Elazer-tech:~/td_linux/documents$ cal
   November 2026
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30
```

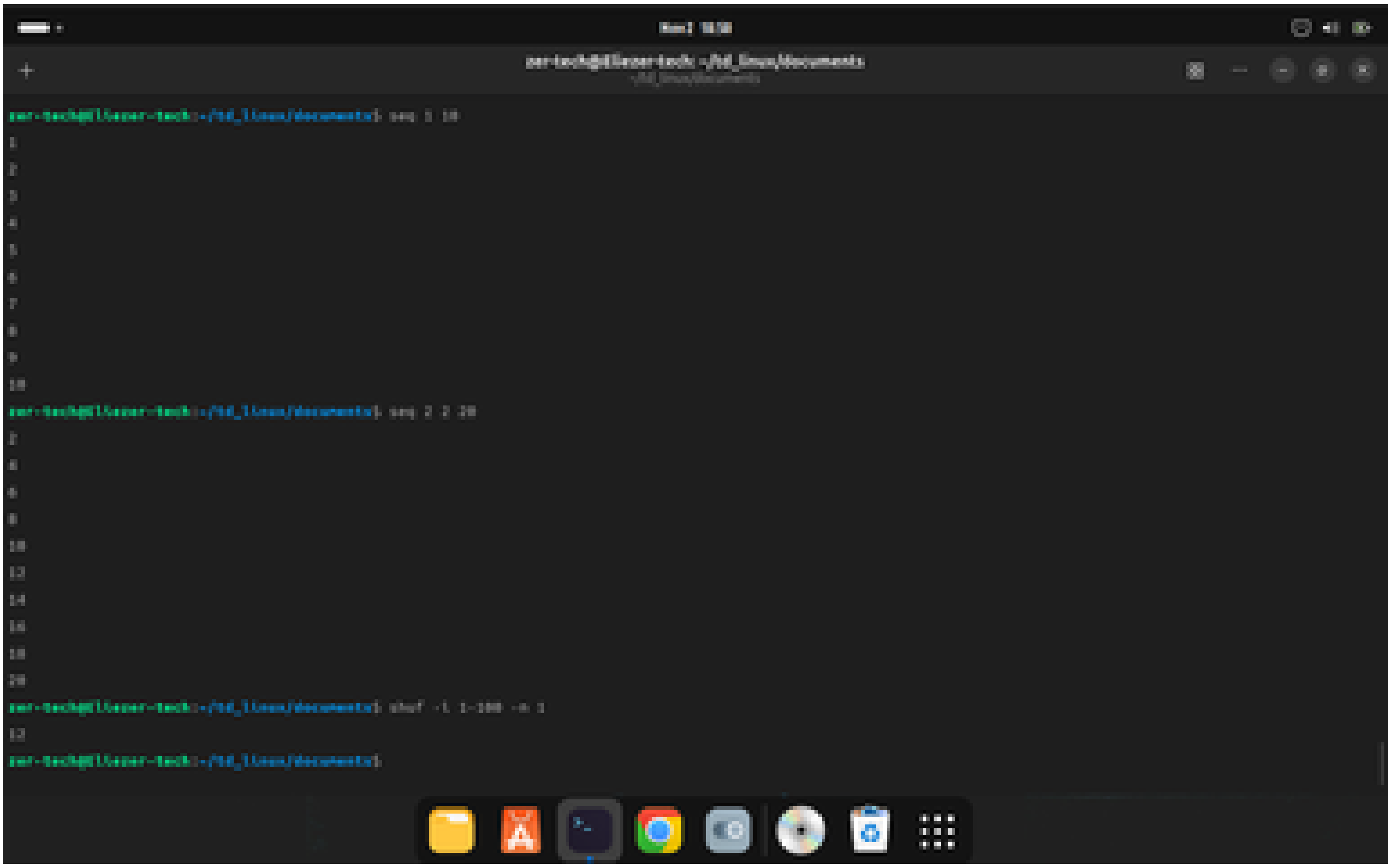


```
et:1 http://ht.archive.ubuntu.com/ubuntu resolute/universe amd64 ncal amd64 12.1.8build1 [20.8 kB]
etched 20.8 kB in 1s (27.9 kB/s)
electing previously unselected package ncal.
(Reading database ... 196485 files and directories currently installed.)
reparing to unpack .../ncal_12.1.8build1_amd64.deb ...
npacking ncal (12.1.8build1) ...
etting up ncal (12.1.8build1) ...
rocessing triggers for man-db (2.13.1-1build1) ...
er-tech@Eliezer-tech:~/td_linux/documents$ cal
   November 2026
u Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

er-tech@Eliezer-tech:~/td_linux/documents$ figlet "UBUNTU"
 _   _ ____  _   _ _   _ _____ _   _
| | | | __ )| | | | \ | |_   _| | | |
| | | |  _ \| | | |  \| | | | | | | |
| |_| | |_) | |_| | |\  | | | | |_| |
 \___/|____/ \___/|_| \_| |_|  \___/
```




zer-tech@Eliezer-tech: ~/td_linux/documents

```
zer-tech@Eliezer-tech:~/td_linux/documents$ cl
zer-tech@Eliezer-tech:~/td_linux/documents$ seq 1 10
1
2
3
4
5
6
7
8
9
10
zer-tech@Eliezer-tech:~/td_linux/documents$ seq 2 2 20
2
4
6
8
10
12
14
16
18
20
```


zer-tech@Eliezer-tech: ~/td_linux/documents

```
zer-tech@Eliezer-tech:~/td_linux/documents$ seq 1 10
1
2
3
4
5
6
7
8
9
10
zer-tech@Eliezer-tech:~/td_linux/documents$ seq 2 2 20
2
4
6
8
10
12
14
16
18
20
zer-tech@Eliezer-tech:~/td_linux/documents$ shuf -i 1-100 -n 1
12
zer-tech@Eliezer-tech:~/td_linux/documents$
```

EXERCICE 5

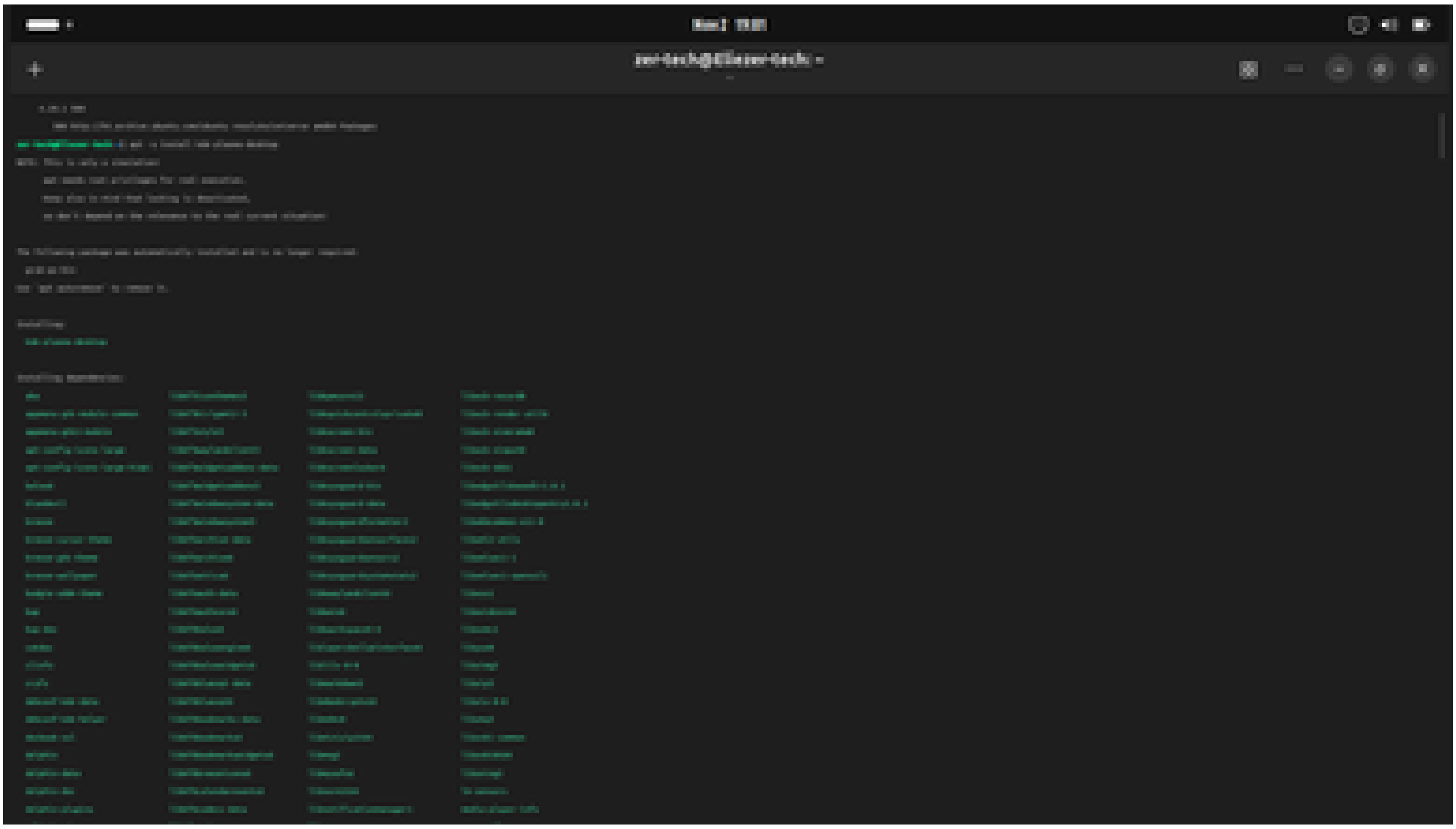
Cette expérience vise à installer, configurer et comparer deux environnements de bureau distincts (KDE Plasma et XFCE) afin d'analyser leur comportement et leur impact sur le système.

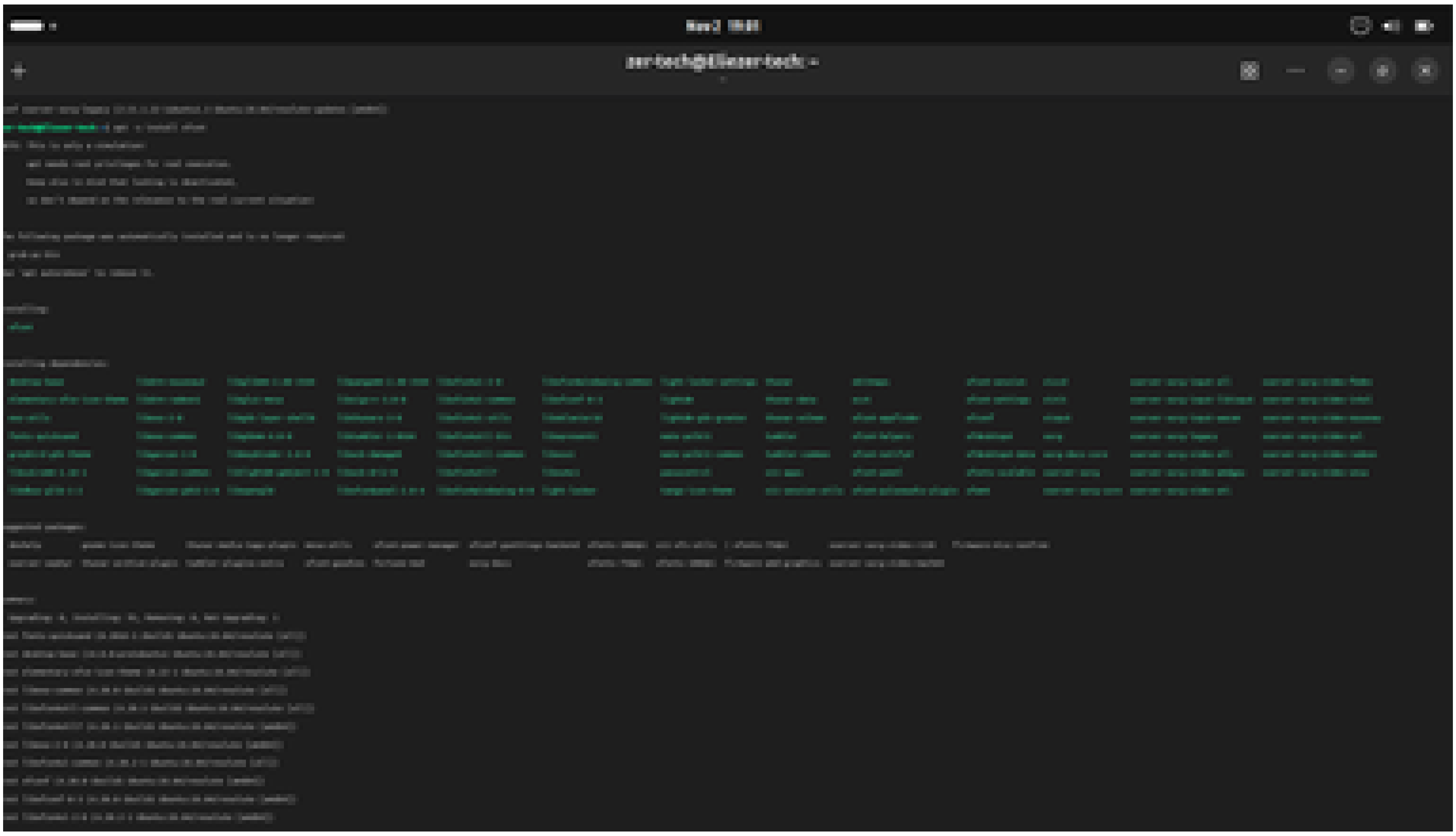
1. Identification de l'environnement de bureau actif et du type de session graphique (serveur d'affichage) via l'analyse des variables d'environnement du système.
2. Vérification de la disponibilité et du statut des paquets logiciels requis dans le gestionnaire de paquets de la distribution.
3. Simulation des installations pour évaluer en amont l'espace disque nécessaire et examiner l'arbre des dépendances logicielles de chaque interface.
4. Installation effective des deux environnements avec les privilèges d'administration, en prenant soin de conserver le gestionnaire de connexion d'origine en cas d'invite système.
5. Déconnexion de la session courante après sauvegarde des travaux, puis tests successifs d'ouverture de session sous Plasma, puis sous XFCE.

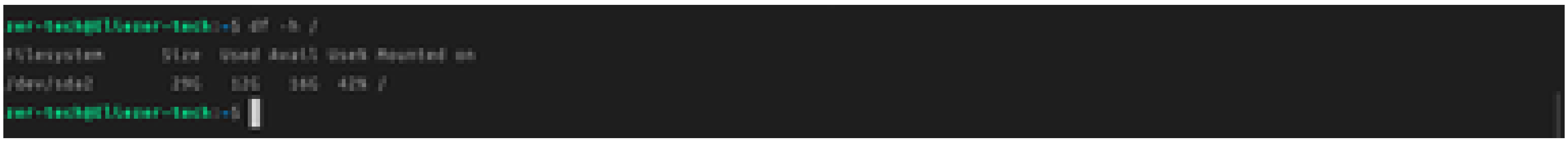
Évaluation et comparaison

Pour chaque interface, un banc d'essai standardisé est appliqué afin de tester l'ergonomie et la configuration :

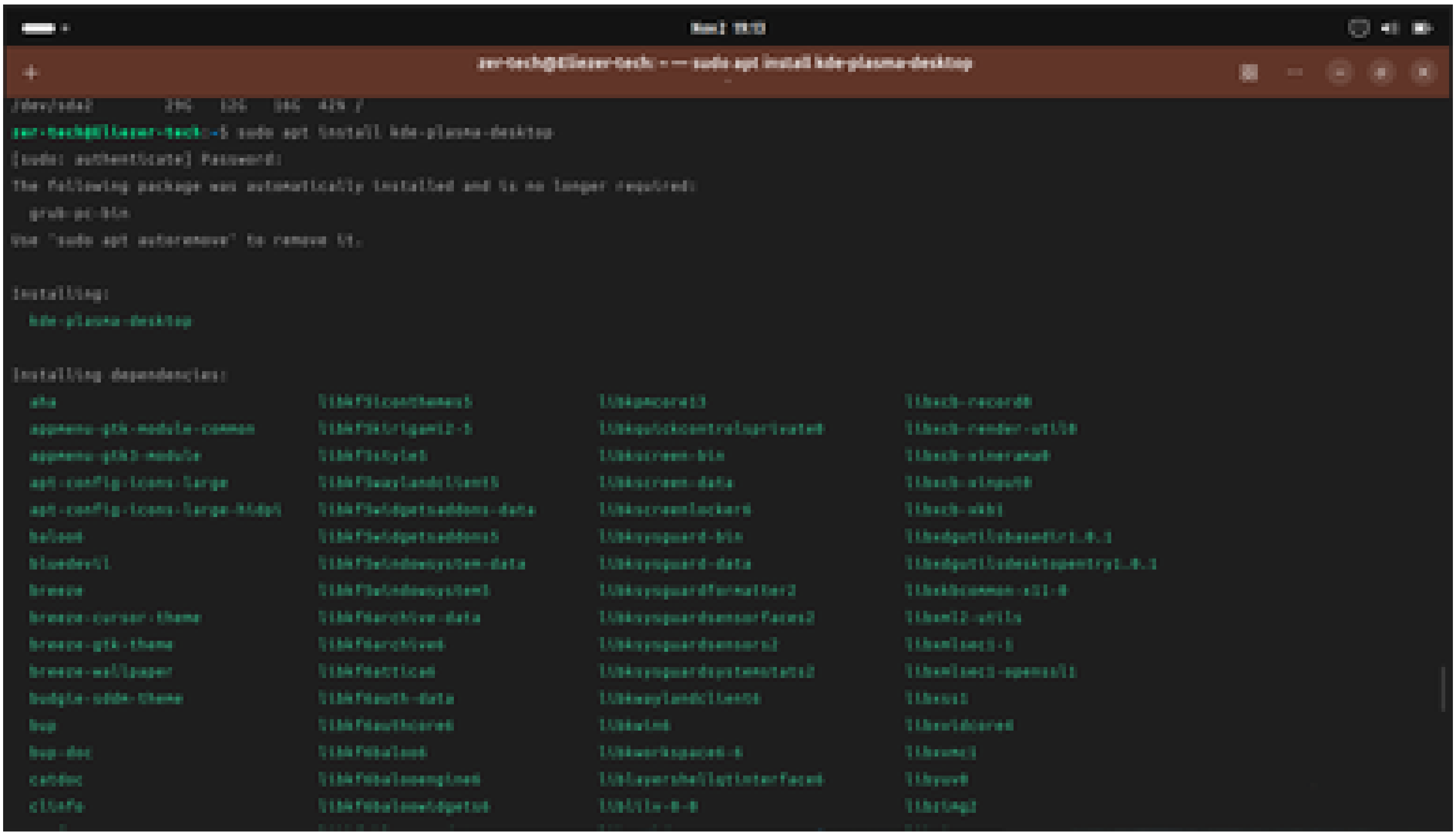
- L'exécution des outils de base (terminal, gestionnaire de fichiers).
- L'accès aux documents de travail.
- La personnalisation de l'interface (fond d'écran, création de raccourcis).
- L'exploration du panneau d'administration des paramètres système.



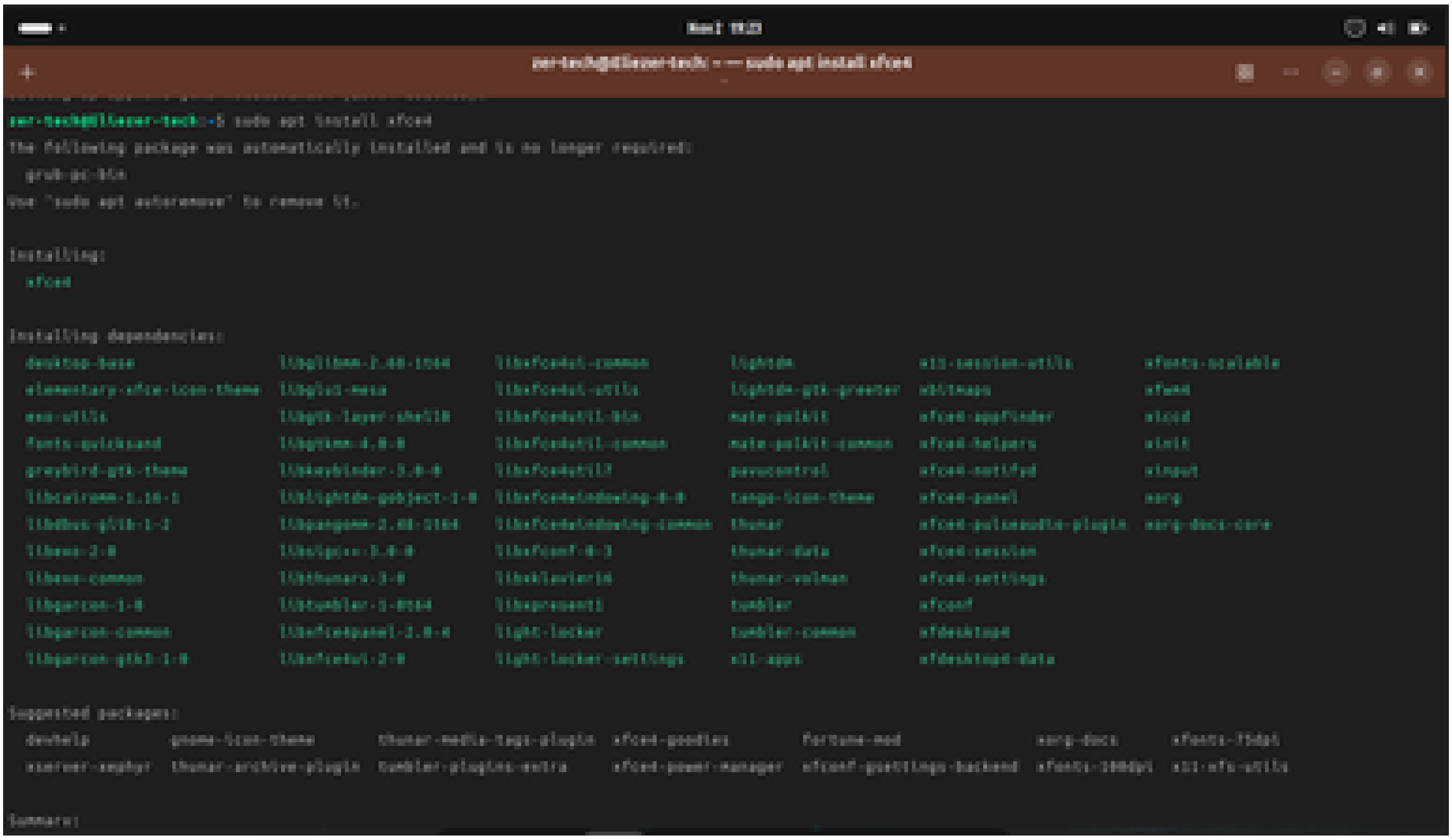




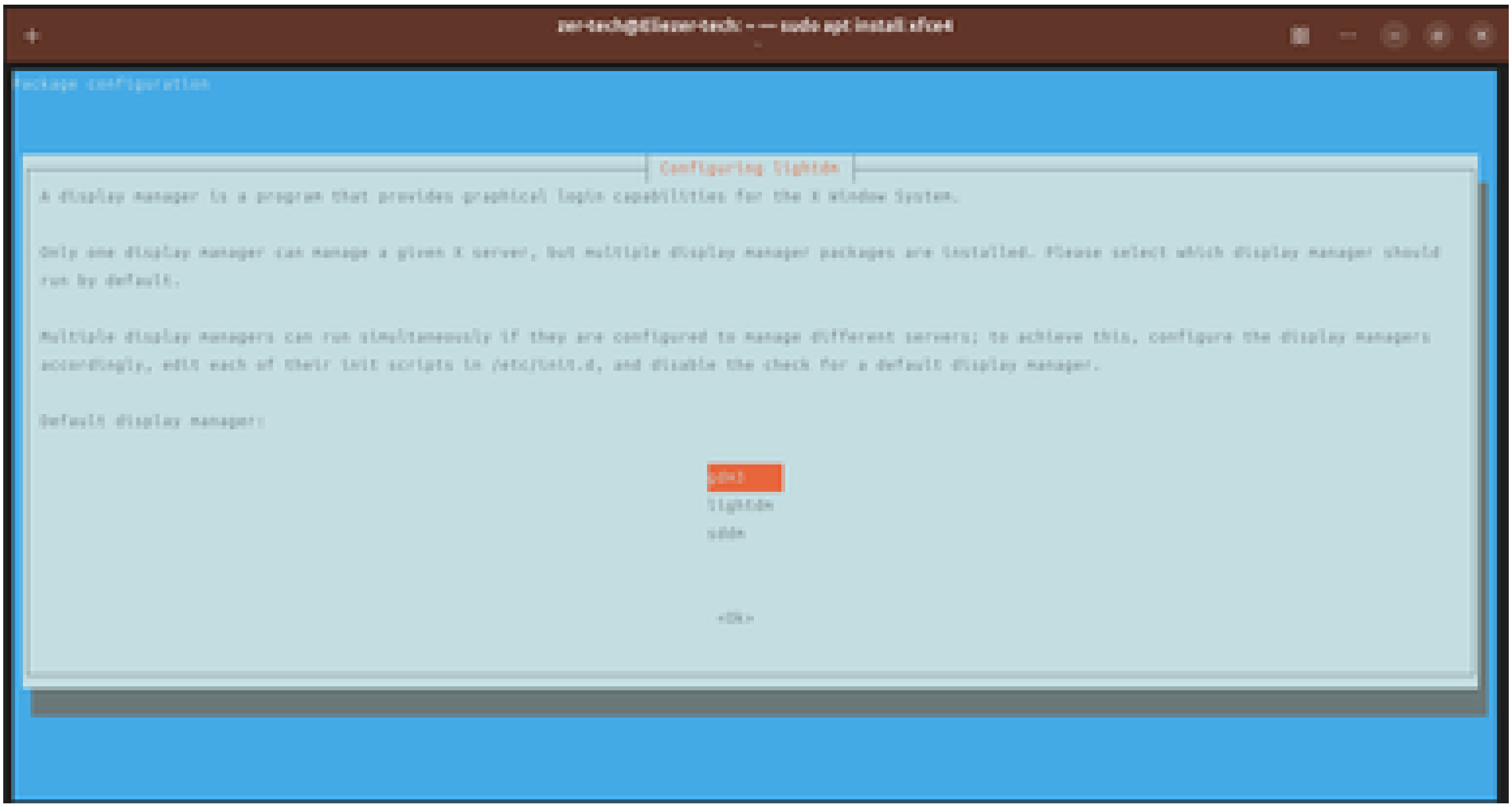
```
zer-tech@Eliezer-tech:~$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2        29G   12G   16G  42% /
zer-tech@Eliezer-tech:~$ 
```


Nov 2 19:13
zer-tech@Eliezer-tech: ~ — sudo apt install kde-plasma-desktop

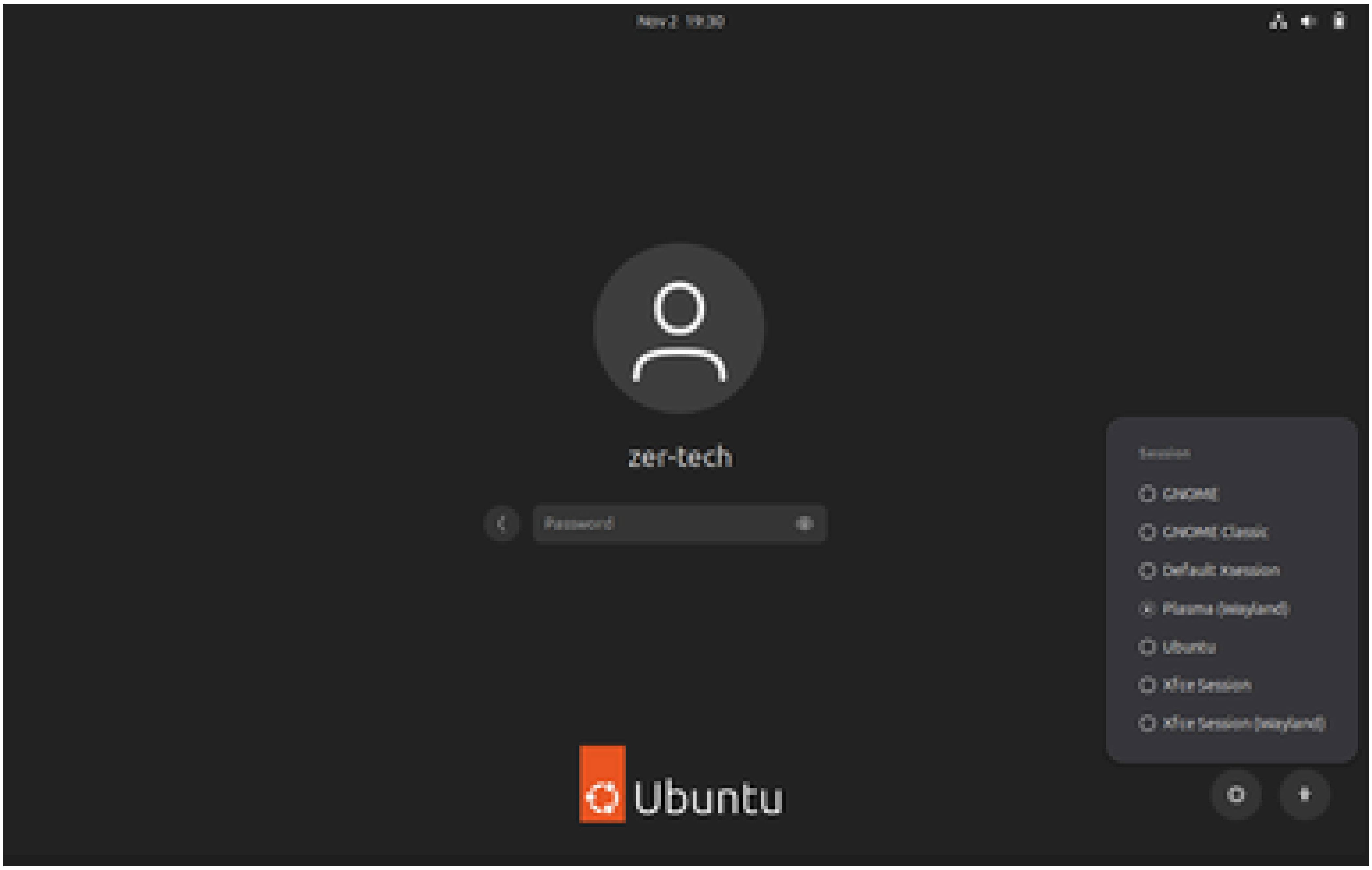
```
/dev/sda2        29G   12G   16G  42% /
zer-tech@Eliezer-tech:~$ sudo apt install kde-plasma-desktop
[sudo: authenticate] Password:
The following package was automatically installed and is no longer required:
  grub-pc-bin
Use 'sudo apt autoremove' to remove it.

Installing:
  kde-plasma-desktop

Installing dependencies:
  aha                           libkf5iconthemes5            libkpmcore13                  libxcb-record0
  appmenu-gtk-module-common     libkf5kirigami2-5            libkquickcontrolsprivate0     libxcb-render-util0
  appmenu-gtk3-module           libkf5style5                 libkscreen-bin                libxcb-xinerama0
  apt-config-icons-large        libkf5waylandclient5         libkscreen-data               libxcb-xinput0
  apt-config-icons-large-hidpi  libkf5widgetsaddons-data     libkscreenlocker6             libxcb-xkb1
  baloo6                        libkf5widgetsaddons5         libksysguard-bin              libxdgutilsbasedir1.0.1
  bluedevil                     libkf5windowsystem-data      libksysguard-data             libxdgutilsdesktopentry1.0.1
  breeze                        libkf5windowsystem5          libksysguardformatter2        libxkbcommon-x11-0
  breeze-cursor-theme           libkf6archive-data           libksysguardsensorfaces2      libxml2-utils
  breeze-gtk-theme              libkf6archive6               libksysguardsensors2          libxmlsec1-1
  breeze-wallpaper              libkf6attica6                libksysguardsystemstats2      libxmlsec1-openssl1
  budgie-sddm-theme             libkf6auth-data              libkwaylandclient6            libxss1
  bup                           libkf6authcore6              libkwin6                      libxvidcore4
  bup-doc                       libkf6baloo6                 libkworkspace6-6              libxvmc1
  catdoc                        libkf6balooengine6           liblayershellqtinterface6     libyuv0
  clifo                         libkf6baloowidgets6          liblilv-0-0                   libzimg2
```



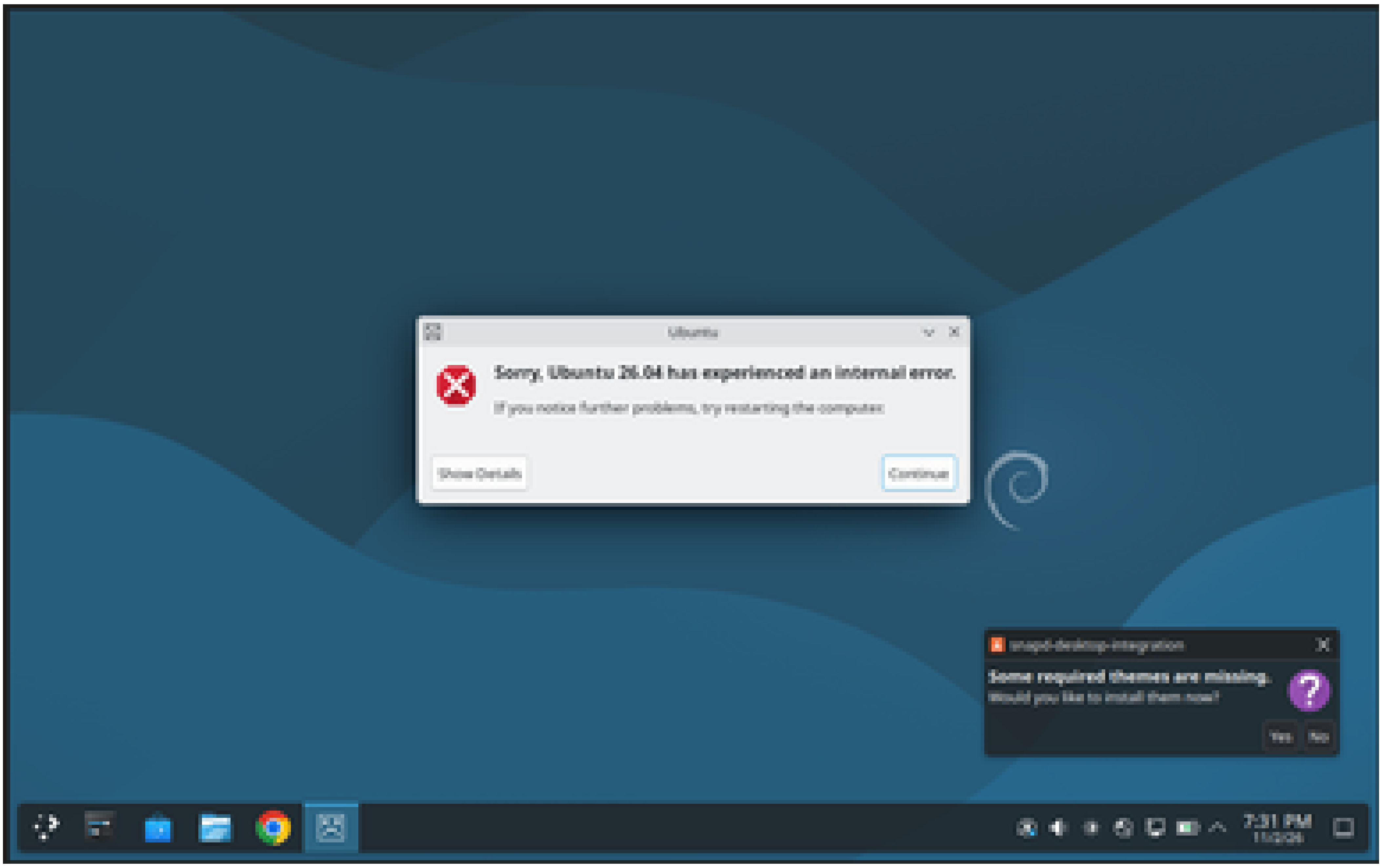
```
zer-tech@Eliezer-tech:~$ sudo apt install xfce4
The following package was automatically installed and is no longer required:
  grub-pc-bin
Use 'sudo apt autoremove' to remove it.

Installing:
  xfce4

Installing dependencies:
  desktop-base               libglibmm-2.68-1t64    libxfce4ui-common        lightdm             x11-session-utils       xfonts-scalable
  elementary-xfce-icon-theme libglu1-mesa           libxfce4ui-utils         lightdm-gtk-greeter xbitmaps                xfwm4
  exo-utils                  libgtk-layer-shell0    libxfce4util-bin         mate-polkit         xfce4-appfinder         xiccd
  fonts-quicksand            libgtkmm-4.0-0         libxfce4util-common      mate-polkit-common  xfce4-helpers           xinit
  greybird-gtk-theme         libkeybinder-3.0-0     libxfce4util7            pavucontrol         xfce4-notifyd           xinput
  libcairomm-1.16-1          liblightdm-gobject-1-0 libxfce4windowing-0-0    tango-icon-theme    xfce4-panel             xorg
  libdbus-glib-1-2           libpangomm-2.48-1t64   libxfce4windowing-common thunar              xfce4-pulseaudio-plugin xorg-docs-core
  libexo-2-0                 libsigc++-3.0-0        libxfconf-0-3            thunar-data         xfce4-session
  libexo-common              libthunarx-3-0         libxklavier16            thunar-volman       xfce4-settings
  libgarcon-1-0              libtumbler-1-0t64      libxpresent1             tumbler             xfconf
  libgarcon-common           libxfce4panel-2.0-4    light-locker             tumbler-common      xfdesktop4
  libgarcon-gtk3-1-0         libxfce4ui-2-0         light-locker-settings    x11-apps            xfdesktop4-data

Suggested packages:
  devhelp        gnome-icon-theme      thunar-media-tags-plugin xfce4-goodies       fortune-mod              xorg-docs     xfonts-75dpi
  xserver-xephyr thunar-archive-plugin tumbler-plugins-extra    xfce4-power-manager xfconf-gsettings-backend xfonts-100dpi x11-xfs-utils

Summary:
```



```
Package configuration

                              Configuring lightdm
 A display manager is a program that provides graphical login capabilities for the X Window System.

 Only one display manager can manage a given X server, but multiple display manager packages are installed. Please select which display manager should
 run by default.

 Multiple display managers can run simultaneously if they are configured to manage different servers; to achieve this, configure the display managers
 accordingly, edit each of their init scripts in /etc/init.d, and disable the check for a default display manager.

 Default display manager:

                                        gdm3
                                        lightdm
                                        sddm


                                         <Ok>
```


Nov 2 19:30
zer-tech
Password
Ubuntu

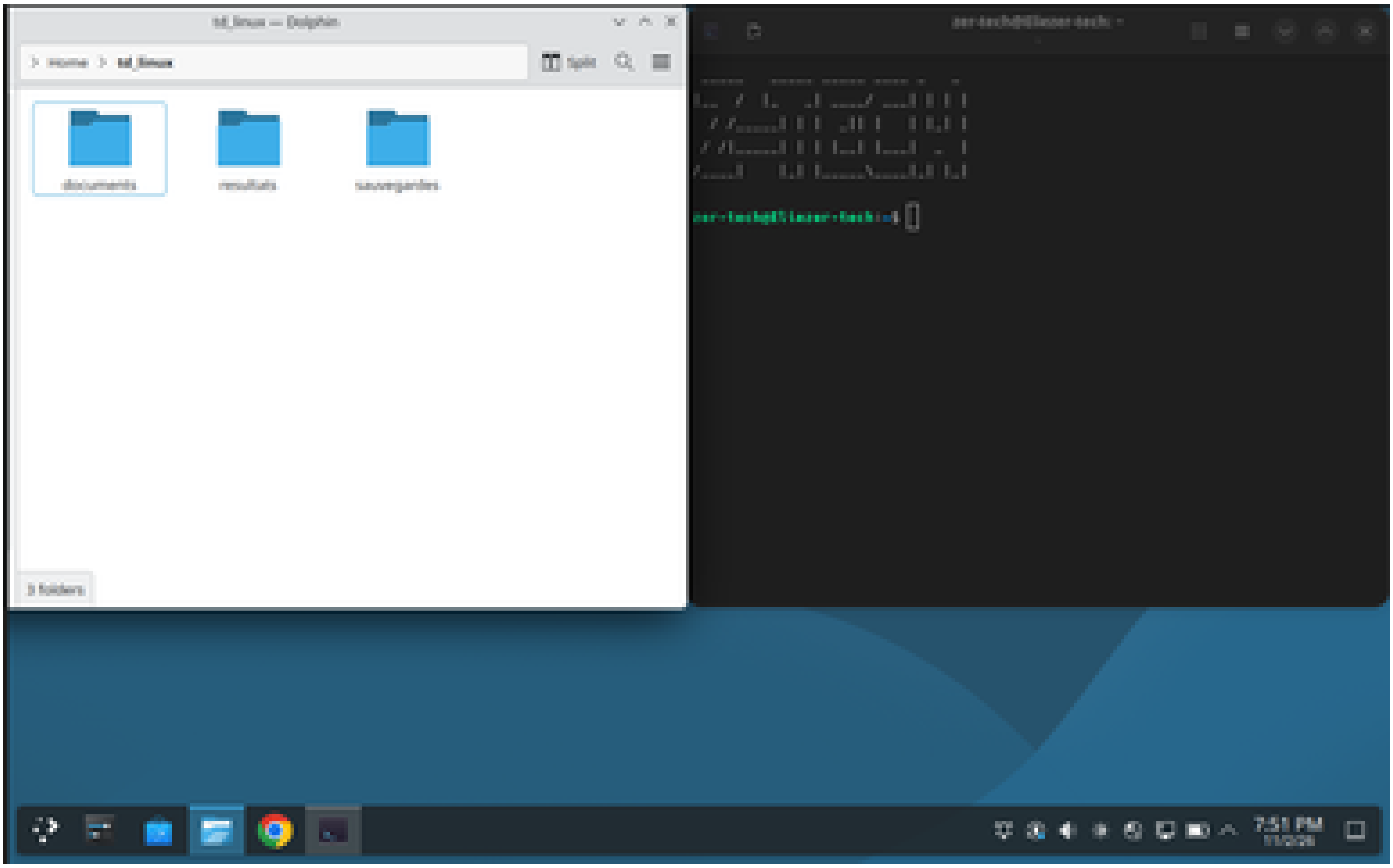
Session
- ○ GNOME
- ○ GNOME Classic
- ○ Default Xsession
- ● Plasma (Wayland)
- ○ Ubuntu
- ○ Xfce Session
- ○ Xfce Session (Wayland)

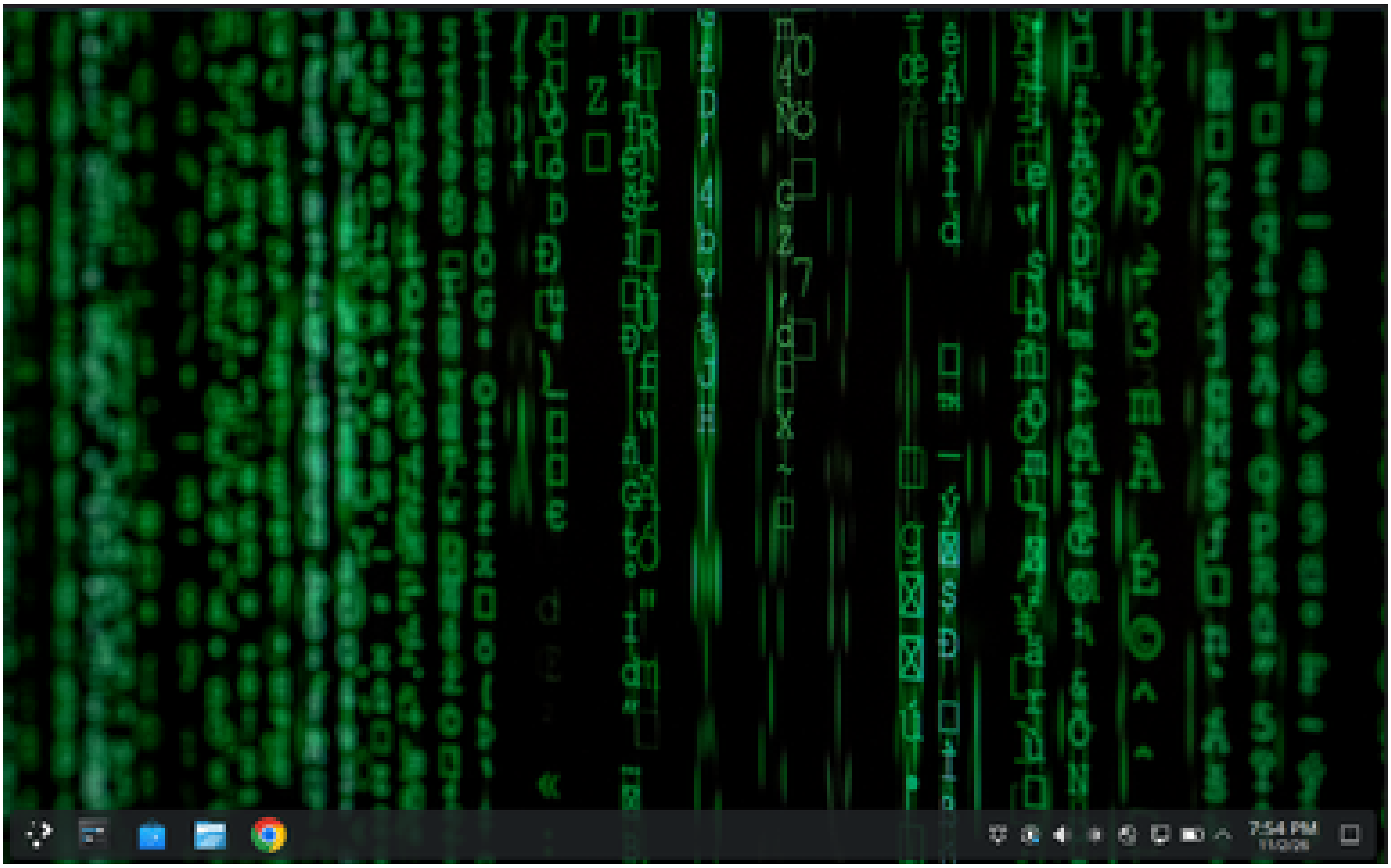

Ubuntu
Show Details
Continue
snapd-desktop-integration
Yes
No
7:31 PM
11/2/26

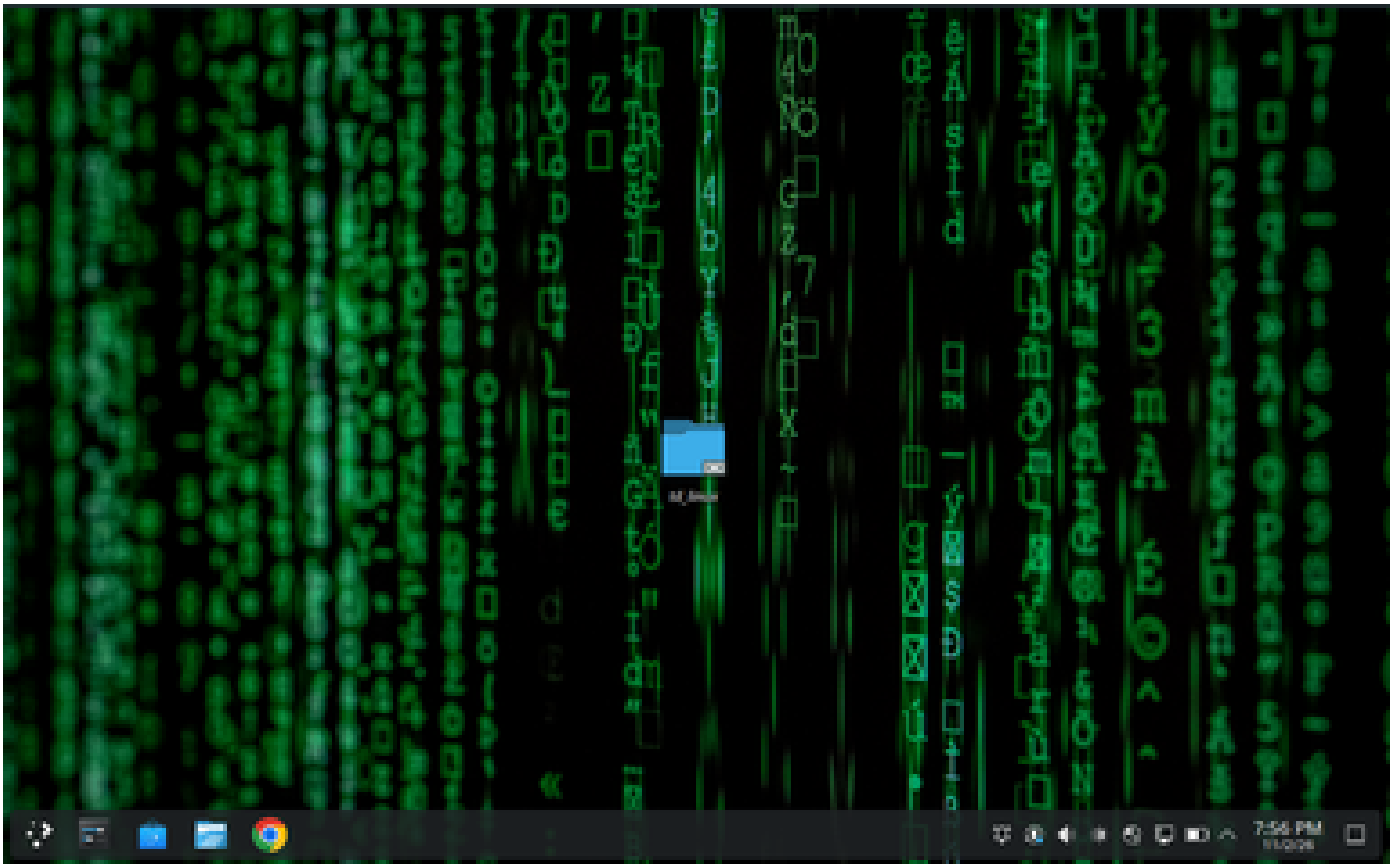
Sorry, Ubuntu 26.04 has experienced an internal error.

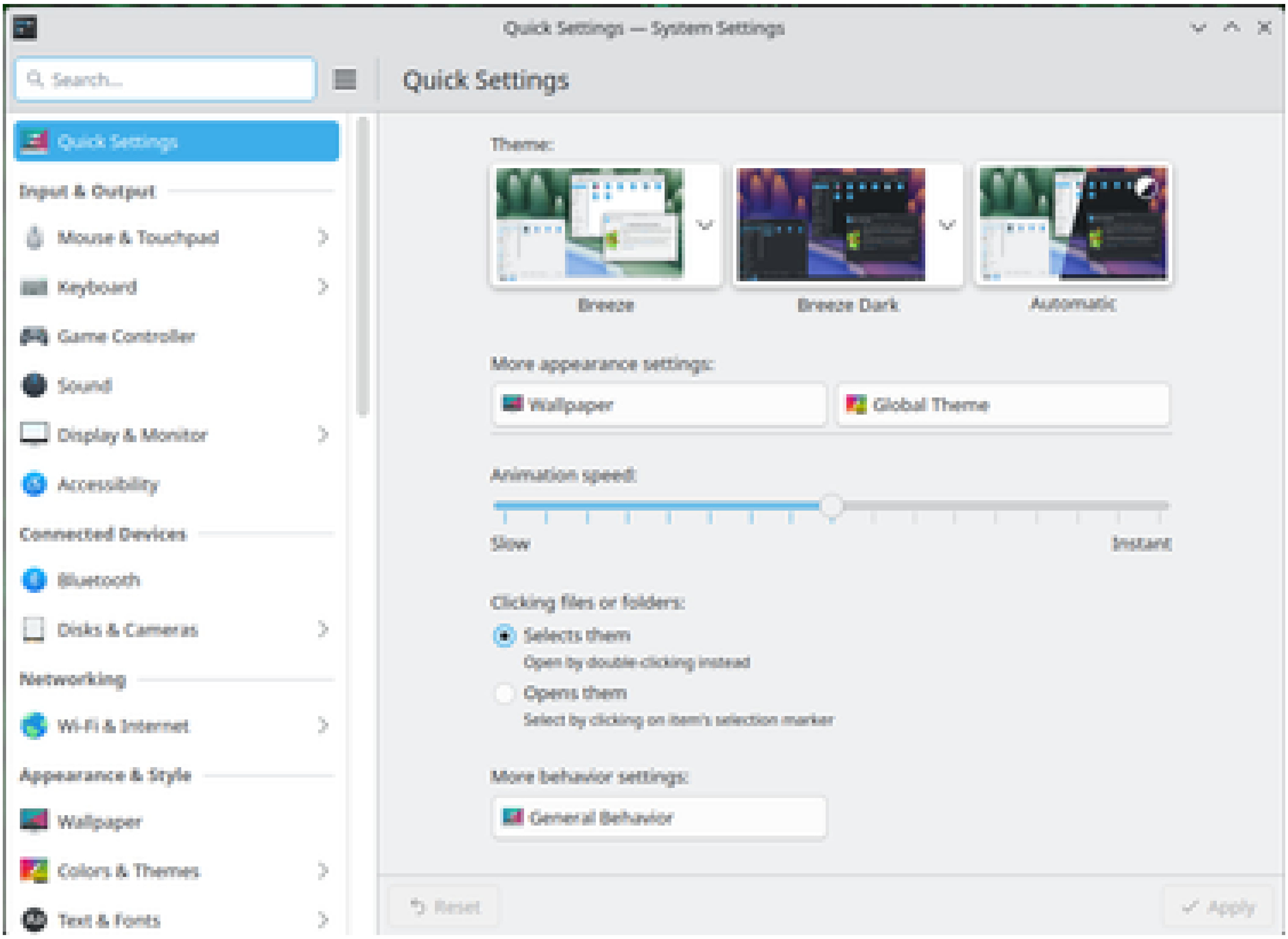
If you notice further problems, try restarting the computer.

Some required themes are missing.
Would you like to install them now?


td_linux — Dolphin
Home
td_linux
Split
documents
resultats
sauvegardes
3 folders
7:51 PM


7:54 PM




Quick Settings — System Settings
Search...
Quick Settings
Input & Output
Mouse & Touchpad
Keyboard
Game Controller
Sound
Display & Monitor
Accessibility
Connected Devices
Bluetooth
Disks & Cameras
Networking
Wi-Fi & Internet
Appearance & Style
Wallpaper
Colors & Themes
Text & Fonts
Reset
Apply

Quick Settings

Theme:

Breeze | Breeze Dark | Automatic

More appearance settings:

Wallpaper | Global Theme

Animation speed:

Slow — Instant

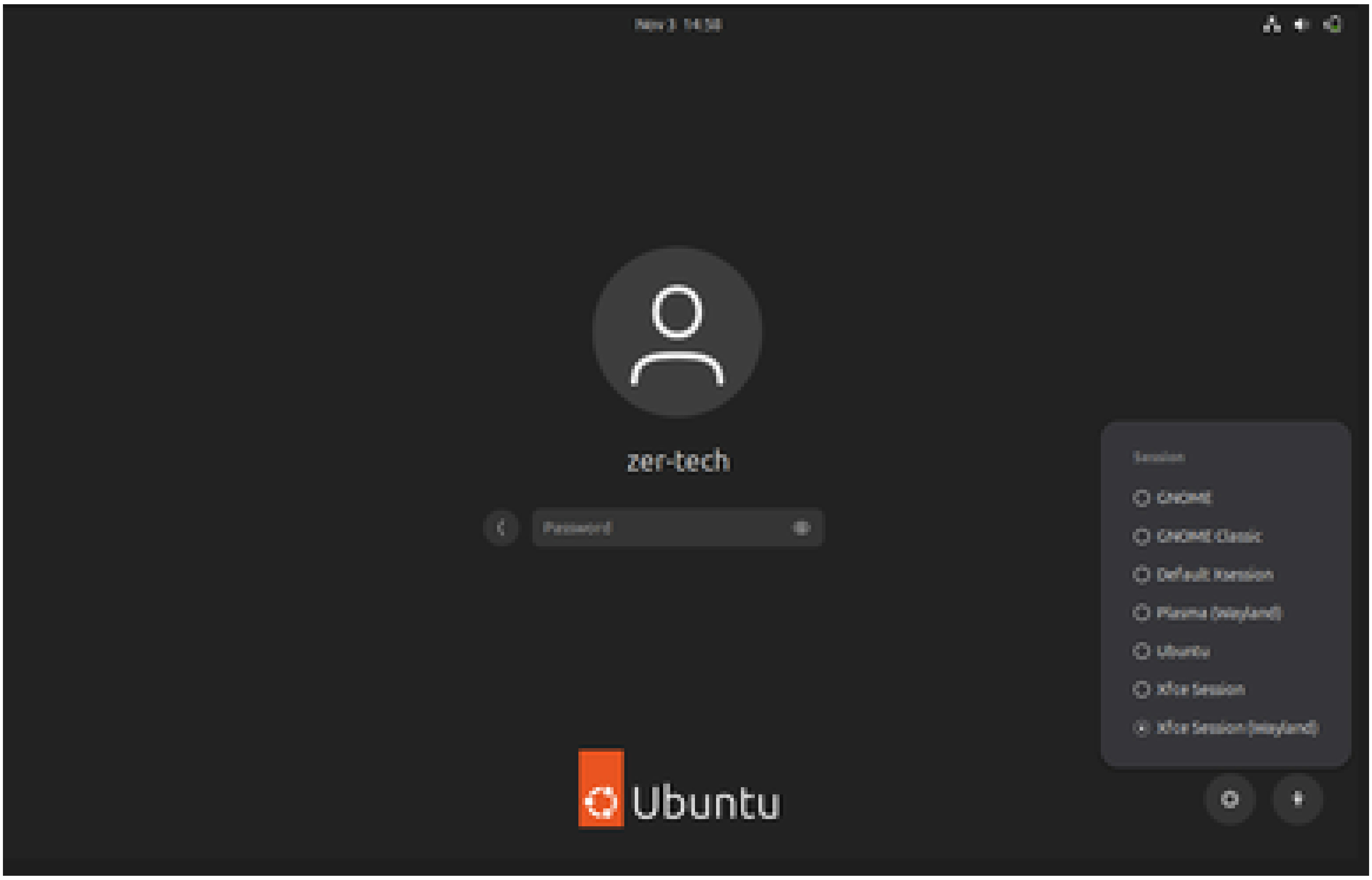
Clicking files or folders:

● Selects them
Open by double-clicking instead

○ Opens them
Select by clicking on item's selection marker

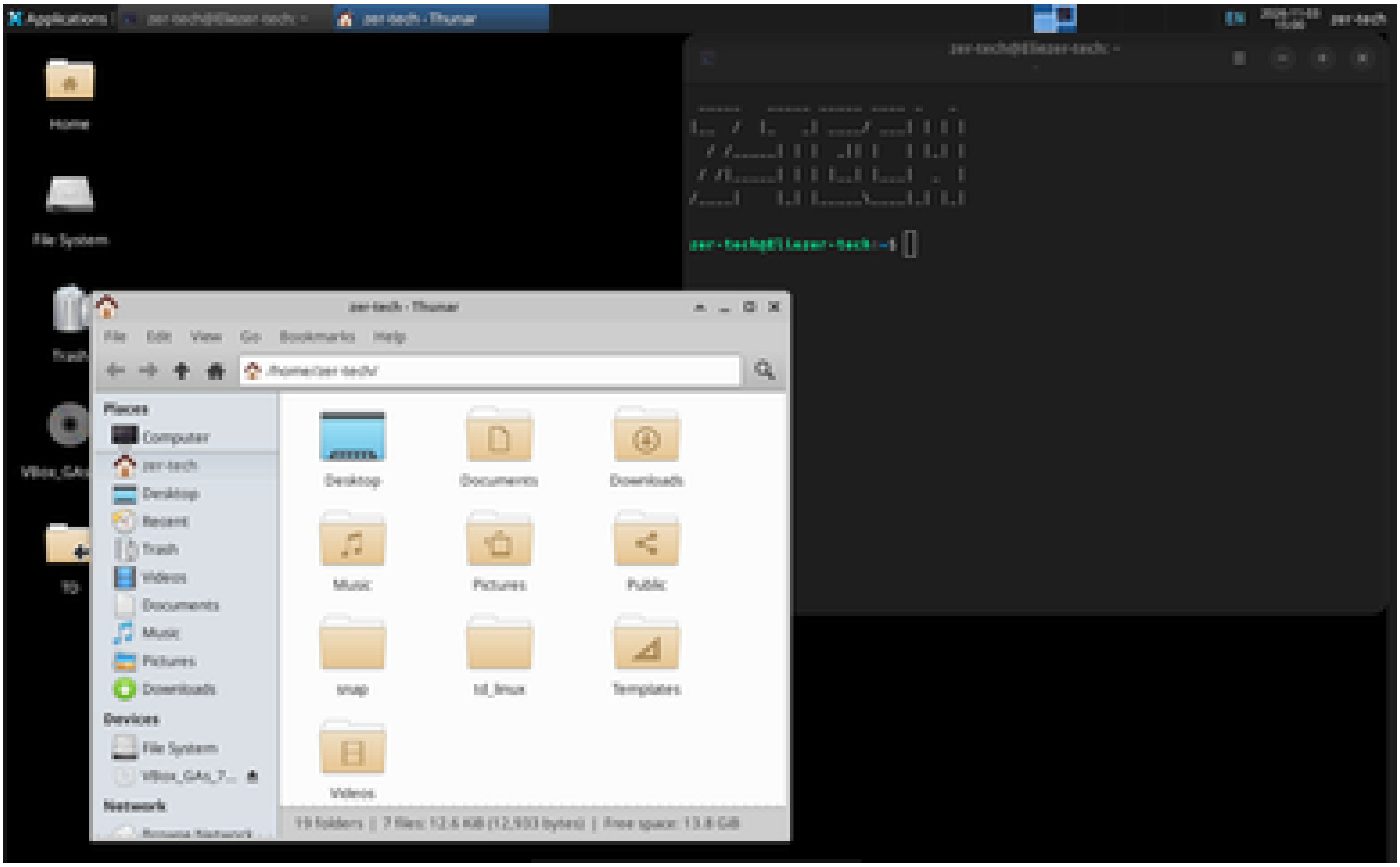
More behavior settings:

General Behavior

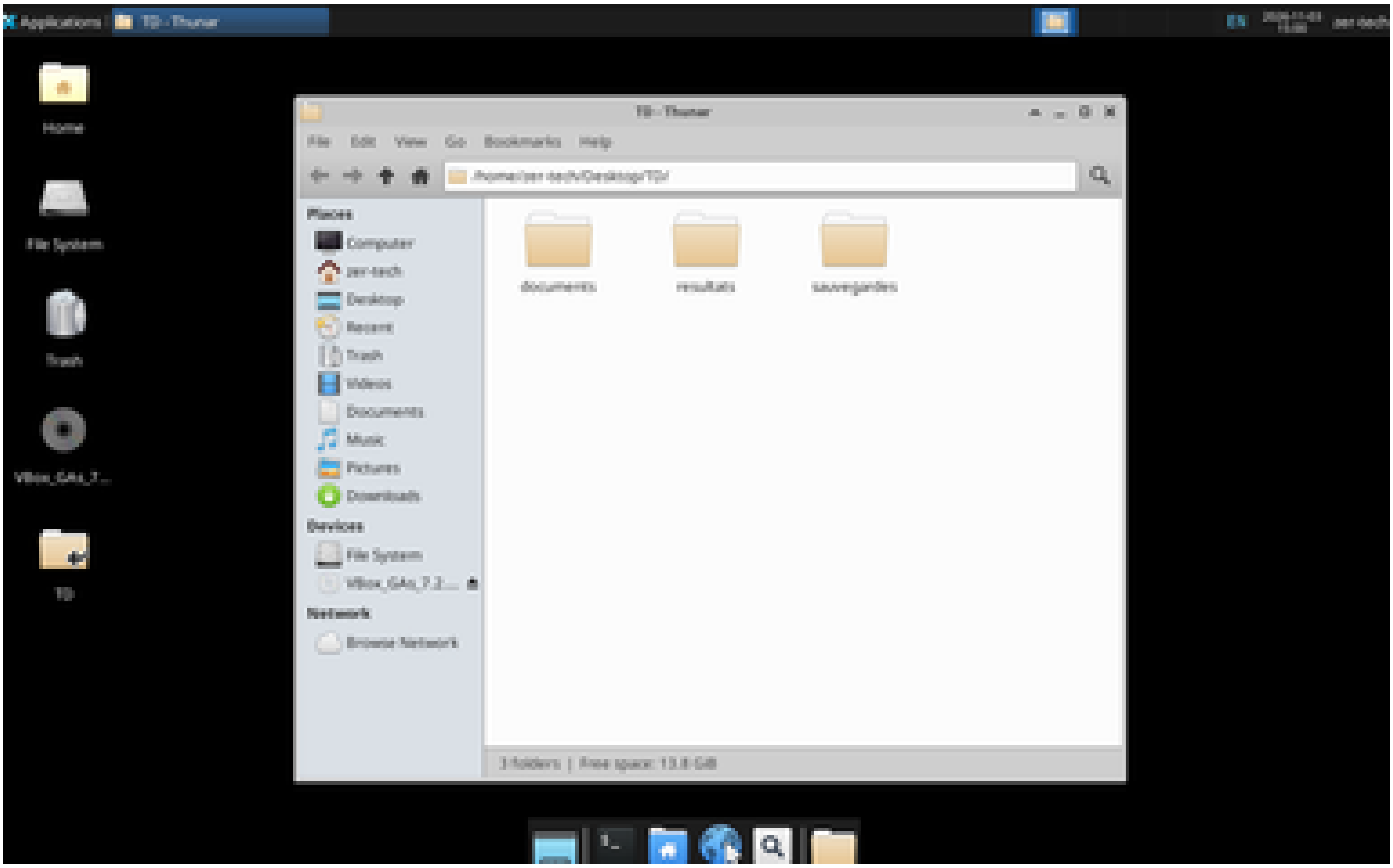

Nov 3 14:58
zer-tech
Password
Ubuntu

Session

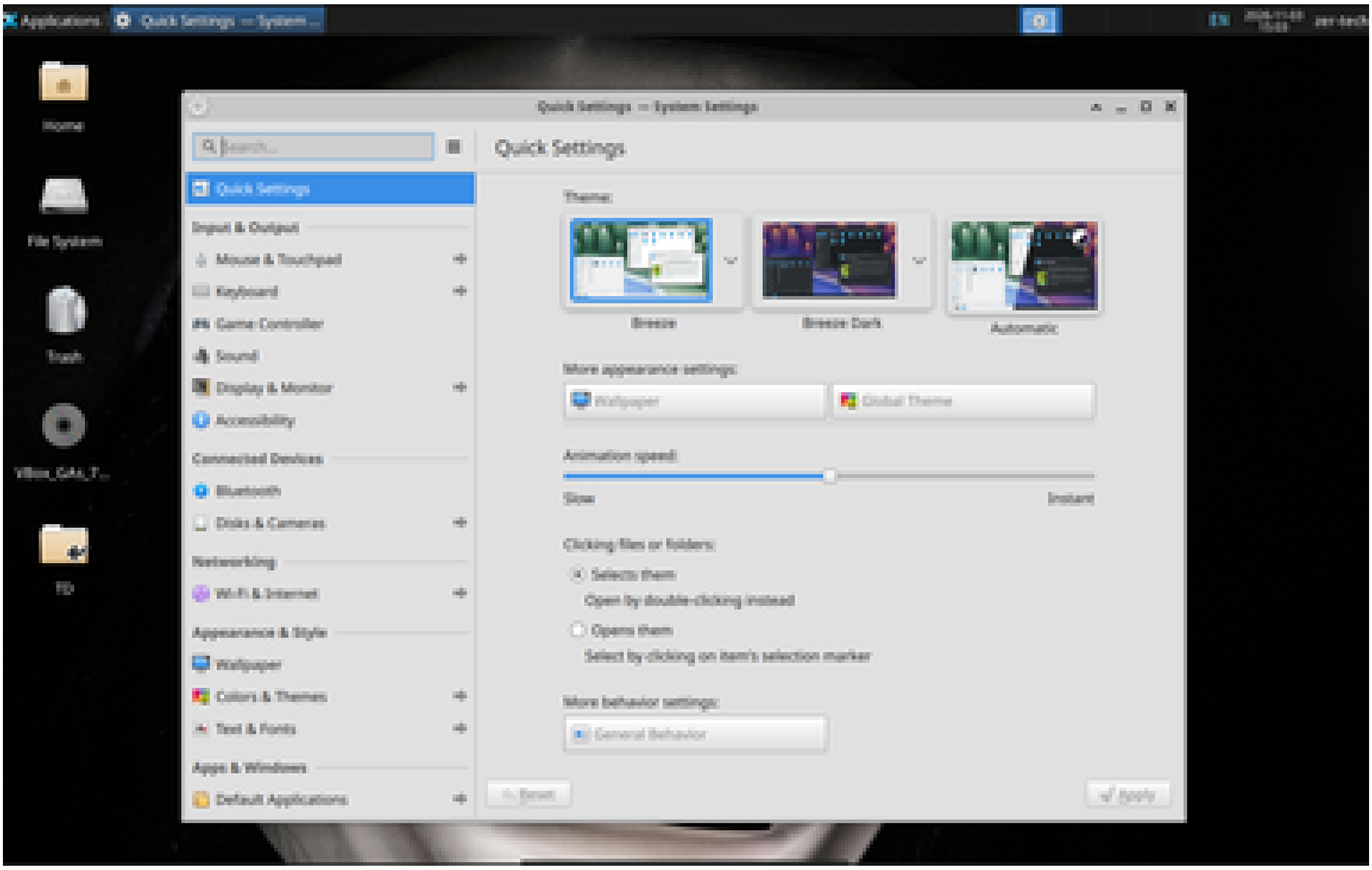
○ GNOME
○ GNOME Classic
○ Default Xsession
○ Plasma (Wayland)
○ Ubuntu
○ Xfce Session
● Xfce Session (Wayland)


Applications
zer-tech@Eliezer-tech: ~
zer-tech - Thunar
EN
zer-tech
Home
File System
Trash
VBox_GAs
TG
File
Edit
View
Go
Bookmarks
Help
/home/zer-tech/
Places
Computer
zer-tech
Desktop
Recent
Trash
Videos
Documents
Music
Pictures
Downloads
Devices
File System
VBox_GAs_7...
Network
Browse Network
Desktop
Documents
Downloads
Music
Pictures
Public
snap
td_linux
Templates
Videos
19 folders | 7 files: 12.6 KiB (12,933 bytes) | Free space: 13.8 GiB

```
zer-tech@Eliezer-tech:~$ 
```


Applications
TD - Thunar
EN
Home
File System
Trash
VBox_GAs_7...
TD
TD - Thunar
File
Edit
View
Go
Bookmarks
Help
Places
Computer
Desktop
Recent
Trash
Videos
Documents
Music
Pictures
Downloads
Devices
File System
VBox_GAs_7.2...
Network
Browse Network
documents
resultats
sauvegardes

Applications
EN
Home
File System
Trash
VBox_GAs_7...
TD


Applications
Quick Settings — System...
Home
File System
Trash
VBox_GAs_7...
TD
Quick Settings — System Settings
Search...
Quick Settings
Input & Output
Mouse & Touchpad
Keyboard
Game Controller
Sound
Display & Monitor
Accessibility
Connected Devices
Bluetooth
Disks & Cameras
Networking
Wi-Fi & Internet
Appearance & Style
Wallpaper
Colors & Themes
Text & Fonts
Apps & Windows
Default Applications
Quick Settings
Theme:
Breeze
Breeze Dark
Automatic
More appearance settings:
Wallpaper
Global Theme
Animation speed:
Slow
Instant
Clicking files or folders:
Selects them
Open by double-clicking instead
Opens them
Select by clicking on item's selection marker
More behavior settings:
General Behavior
Reset
Apply

Critère	KDE Plasma	XFCE
Terminal Utilisé	konsole	xfce4-terminal
Gestionnaire de fichiers	Dolphin	Thunar
Accès aux paramètres	systemesetting	xfce-4-settings-manager
personnalisation	Tout est modifiable depuis l'interface sans avoir besoin de faire recours à un outil tiers	personnalisation limitée, simple rapide consomme peu de ressources
mémoire disponible	1.7gi	2.2gi
choix pour un poste donné	poste de travail avec des matériels récents, avec confort visuel au max	poste de travail pour des matériels bas de gamme ou ancien, pour limiter la consommation de RAM

Conclusion

En conclusion, nous appris à manipuler le terminal sur linux et effectuer certaine tâches comme créer et manipuler des dossiers, créer des archives sans avoir besoin d'utiliser l'interface graphique. Ce qui fait la force et la maniabilité extrême de linux.

L'autre point important de ce TD c'est aussi l'initialisation aux packets, nous avons appris a simuler leur installation et suppression via *apt -s install*. la commande *sudo apt install*, collectionne les dépendances des packets et les installe d'un coup.

Difficile de passer a cote de l'étape visant à configurer les deux espaces de travail KDE plasma et XFCE, pendant l'installation nous avons dû changer l'affichage par défaut au dépend de GDN3 pour pouvoir utilisé les différentes affichages.

Les difficultés que nous avons rencontré lors de ce travail était principalement dans l'exercice 2 dans laquelle nous avons créé le fichier services.txt via la commande: *cat > documents/services.txt <<'EOF' web;actif ssh;actif impression;inactif sauvegarde;actif web;actif EOF*. mais malheureusement nous avons dû opter pour une autre option vue celle-ci était défaillante.
Nous avons aussi rencontré quelques difficultés mineurs par rapport a l'affichage de KDE plasma. et les packets cmatrix, neofetch,ncal et cawsay était absent nous les avons installé .

En somme le travail est réussi, et nous sommes en mesure de travailler sur différents espace de travail soit Ubuntu, DKE plasma ou XFCE.